ENGINEERING MATHEMATICS

# Calculus & Linear Algebra

Units I – III
From absolute beginner to exam-ready, one idea at a time

$$A\mathbf{x} = \lambda\mathbf{x} \quad \cdot \quad f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \quad \cdot \quad \int_a^b f(x)\,dx = F(b) - F(a)$$

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Unit I Matrix methods & linear systems | Unit II Differential calculus & applications | Unit III Fundamentals of integral calculus

Written around your syllabus, textbook chapters, practice decks and your professor's handwritten sheets.

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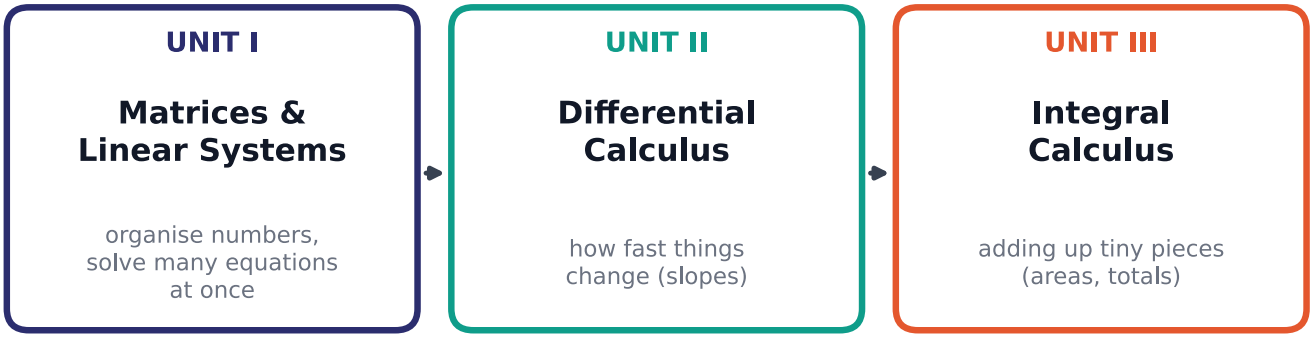
WELCOME

# Read this first

## What this book is

Calculus has a reputation for being scary. It isn't. Behind every symbol there is a simple physical idea, and once you *see* the idea, the symbols stop being a wall and become a shorthand. This book is written for someone who has never done calculus: every chapter starts from the picture, builds the idea in plain words, and only then gives the rule, the proof sketch, the exam tricks and worked examples.

It follows your syllabus exactly:




The three units and how they fit together

| Unit | Syllabus topics (as given) | Chapters |
|---|---|---|
| **I** Matrix methods and linear systems | review of matrices, elementary operations, rank, linear dependence and independence of vectors, solution of linear systems, inverse of matrices, eigenvalues and eigenvectors and their properties, Cayley–Hamilton theorem | 1 – 8 |
| **II** Differential calculus and its applications | derivatives of standard functions and general rules, parametric, implicit, logarithmic differentiation, higher-order derivatives, Rolle's theorem, mean value theorems, Taylor's and Maclaurin's theorems, indeterminate forms, L'Hôpital's rule, maxima and minima | 9 – 19 |
| **III** Fundamentals of integral calculus | general rules of integration, substitution, integration by parts, integration by partial fractions, properties of definite integrals | 20 – 26 |

Chapter 0 is a **toolkit** of school-level facts (functions, algebra, trigonometry, logarithms). If any of it is shaky, calculus will feel hard for no good reason, so spend an evening there.

## How to read each chapter

BIG IDEA **The five-step habit**

1. **Read the picture first.** Every chapter opens with a "Big idea" box and a diagram.
2. **Cover the solution.** For each *Example*, try it yourself before reading the solution.
3. **Learn the formula boxes by heart.** They are the exam currency.
4. **Do the *Try it* boxes.** They take two minutes each and expose gaps immediately.
5. **Finish with the chapter practice.** Answers are in the book; your professor's practice sets (MCQ deck, partial-fraction and definite-integral sheets, textbook exercises) are mapped to chapters in Appendix D.

## The colour code

These boxes appear throughout. Learn to recognise them; they tell you how to read what is inside.

BIG IDEA **Big idea**

The intuition in one paragraph. If you remember nothing else, remember this.

DEFINITION **Definition**

A precise statement of what a word means. Exams quote these almost verbatim.

THEOREM **Theorem**

A fact that has been proved. Usually comes with conditions that you must check before using it.

FORMULA **Formula**

Worth memorising.

WATCH OUT **Watch out**

A trap that costs marks. Every one of these has caught real students.

EXAM TIP **Exam tip**

A shortcut, a pattern the examiners love, or a way to check your answer.

TRY IT **Try it**

A quick question. Answer in the margin before reading on.

STORY TIME **Story time**

*A little history or a real-world picture, to make the idea stick.*

## A suggested 4-week plan

| Week | Focus | Goal |
|---|---|---|
| 1 | Chapter 0, Unit I (Chapters 1–8) | Be able to row-reduce a matrix without thinking; solve any 3-variable system; find eigenvalues |
| 2 | Chapters 9–14 (limits → higher derivatives → Rolle/MVT) | Differentiate anything in the table; implicit, log and parametric differentiation on autopilot |
| 3 | Chapters 15–19 (Taylor, L'Hôpital, applications, optimisation) | Handle any max/min problem with a clear 5-step method |
| 4 | Unit III (Chapters 20–26) + all appendices | Integrate by substitution, parts and partial fractions; use property tricks for definite integrals |

**EXAM TIP** **How much maths per day?**

Ninety focused minutes with a pen beats four hours of re-reading. Maths is learnt with the hand, not the eye.

CHAPTER 0 · FOUNDATIONS

# The Toolkit: everything you need before calculus

**BIG IDEA** **Why start here?**

Calculus problems are rarely *hard calculus*. They are usually easy calculus buried inside algebra, trigonometry or logarithms. A student who can simplify fluently finds calculus gentle; a student who cannot finds it painful. This chapter is the cure.

## 0.1 Numbers, intervals and absolute value

**Number families.** Each family sits inside the next: natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$, integers $\mathbb{Z}$, rationals $\mathbb{Q}$ (fractions), and the real numbers $\mathbb{R}$ (every point on the number line, including $\sqrt{2}$ and $\pi$). Calculus lives on $\mathbb{R}$.

**Intervals** are the standard way to describe "all numbers between".

| Notation | Meaning | Picture |
|---|---|---|
| $(a, b)$ | $a < x < b$ (ends **excluded**) | open circles at both ends |
| $[a, b]$ | $a \le x \le b$ (ends **included**) | filled circles |
| $[a, b)$ , $(a, b]$ | one end in, one end out | mixed |
| $(a, \infty)$ | $x > a$ | arrow to the right |
| $(-\infty, b]$ | $x \le b$ | arrow to the left |
| $\mathbb{R} = (-\infty, \infty)$ | every real number | the whole line |

Infinity is **never** included with a square bracket: $\infty$ is not a number, just a direction.

**Inequality rules.** You may add the same number to both sides, or multiply by a *positive* number. Multiplying or dividing by a **negative** number flips the sign: from $-2x < 6$ we get $x > -3$.

**DEFINITION** **Absolute value (distance from zero)**

$$|x| = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases}$$

So $|5| = 5$ and $|-5| = 5$. Geometrically, $|x - a|$ is the distance between $x$ and $a$ on the number line.

**FORMULA** **Working with absolute values**

- $|x| < a \iff -a < x < a$ (a "within $a$ of zero" statement, for $a > 0$)
- $|x| > a \iff x < -a$ or $x > a$
- $|ab| = |a|\,|b|$ and $\sqrt{x^2} = |x|$ (**not** $x$!)
- Triangle inequality: $|a + b| \le |a| + |b|$

**EXAMPLE** **Example 0.1**

Solve $|2x - 3| \le 5$.

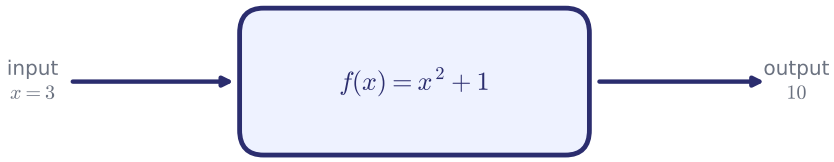
$-5 \le 2x - 3 \le 5 \Rightarrow -2 \le 2x \le 8 \Rightarrow -1 \le x \le 4$. Answer: $[-1, 4]$.

**Why it matters later.** To integrate or differentiate $|x-1|$ you will rewrite it as two separate formulas, one for $x \geq 1$ and one for $x < 1$ (Chapters 10, 12 and 25).

## 0.2 Functions: the central character

**DEFINITION** **Function**

A **function** $f$ is a rule that assigns to each input $x$ in a set (the **domain**) exactly **one** output $f(x)$. The set of outputs actually produced is the **range**.




A function is a machine: one input in, exactly one output out.

**Finding the domain.** Ask "what could go wrong?" The only three things that can go wrong at this level:

| Danger | Rule | Example |
|---|---|---|
| Division by zero | denominator $\neq 0$ | $\frac{1}{x-5}$: domain $x \neq 5$ |
| Even root of a negative | radicand $\geq 0$ | $\sqrt{x-1}$: domain $x \geq 1$ |
| Log of a non-positive | argument $> 0$ | $\ln(x+2)$: domain $x > -2$ |

Also $\sin^{-1} x$ and $\cos^{-1} x$ need $-1 \leq x \leq 1$, and $\tan x$ is undefined at odd multiples of $\pi/2$.

**Piecewise functions** use different formulas on different intervals, e.g. $|x|$ above. Almost every continuity and differentiability question in Unit II uses one.

**Composition.** $(f \circ g)(x) = f(g(x))$: feed the output of $g$ into $f$. For example if $g(x) = x^2$ and $f(x) = \sin x$ then $f(g(x)) = \sin(x^2)$, while $g(f(x)) = \sin^2 x$. Order matters! Composition is exactly what the **chain rule** (Chapter 12) differentiates.

**Even and odd functions.** These two labels are gold for definite integrals (Chapter 25).

**DEFINITION** **Even / odd**

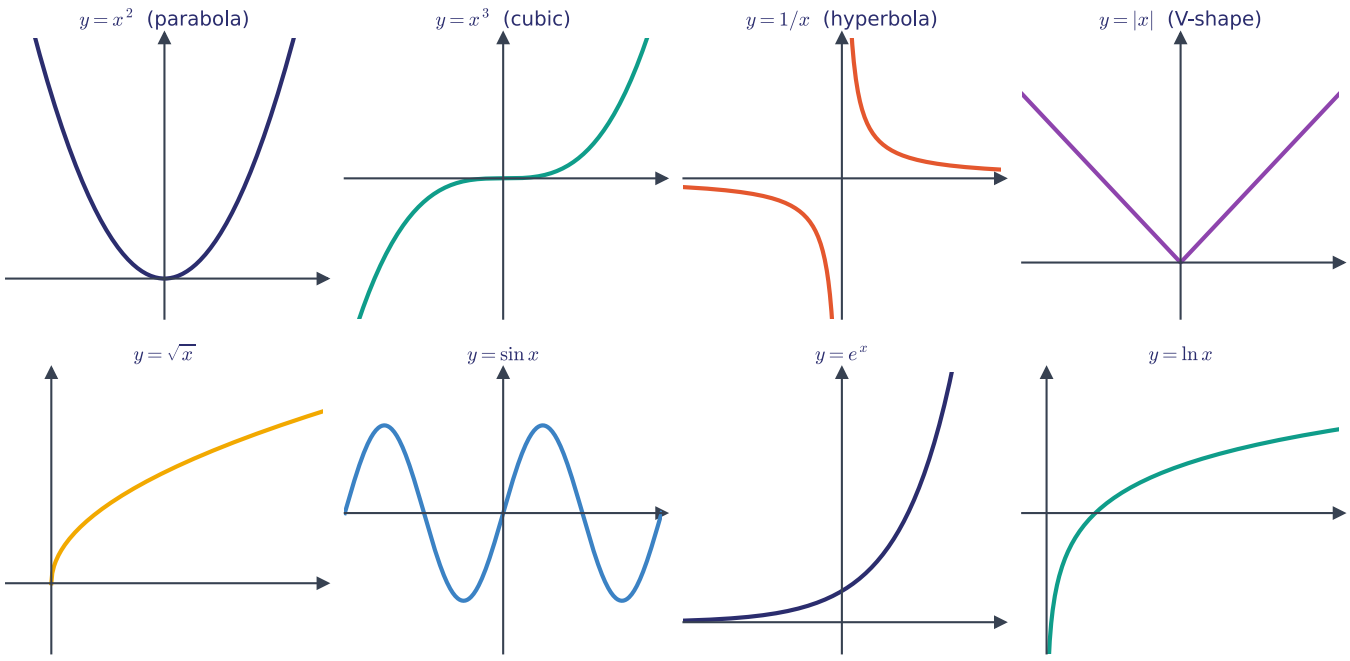
- **Even** if $f(-x) = f(x)$: the graph is a mirror image about the $y$-axis. Examples: $x^2,\ x^4,\ \cos x,\ |x|$.
- **Odd** if $f(-x) = -f(x)$: the graph is symmetric about the origin. Examples: $x,\ x^3,\ \sin x,\ \tan x$.
- Most functions are neither (e.g. $x+1$, $e^x$).

Combining: even × even = even; odd × odd = even; even × odd = odd. Sum of an odd and an even function is generally neither.

**Inverse functions.** If $f$ takes $a$ to $b$, the inverse $f^{-1}$ takes $b$ back to $a$. The graph of $f^{-1}$ is the graph of $f$ reflected in the line $y = x$. Only one-to-one functions have inverses; $\sin x$ gets one *after* restricting its domain, which is why inverse trig functions have fussy ranges (see 0.4).

**The greatest integer (floor) function** $[x]$ is the largest integer $\leq x$: $[2.7] = 2$, $[-1.2] = -2$, $[3] = 3$. Its graph is a staircase, famously discontinuous at every integer (Chapter 10).

**A gallery of the functions you must recognise on sight:**




Eight graphs to know by heart

## 0.3 Algebra you must not forget

**FORMULA** **Exponent laws (for $a, b > 0$)**

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n, \quad a^{-n} = \frac{1}{a^n}, \quad a^{1/n} = \sqrt[n]{a}, \quad a^0 = 1$$

**FORMULA** **Factorisation identities**

$$a^2 - b^2 = (a-b)(a+b), \qquad (a \pm b)^2 = a^2 \pm 2ab + b^2$$
$$a^3 - b^3 = (a-b)(a^2 + ab + b^2), \qquad a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$
$$(a+b)^3 = a^3 + 3a^2 b + 3ab^2 + b^3$$

**Quadratics.** The roots of $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. **Completing the square** rewrites $x^2 + bx + c = (x + \frac{b}{2})^2 + c - \frac{b^2}{4}$, a trick used constantly in integration (Chapter 22).

**Polynomial long division.** Dividing like long division of numbers. This is *the* tool for improper fractions in partial fractions (Chapter 24).

**EXAMPLE** **Example 0.2**

Divide $x^2 + 3x + 5$ by $x + 1$.

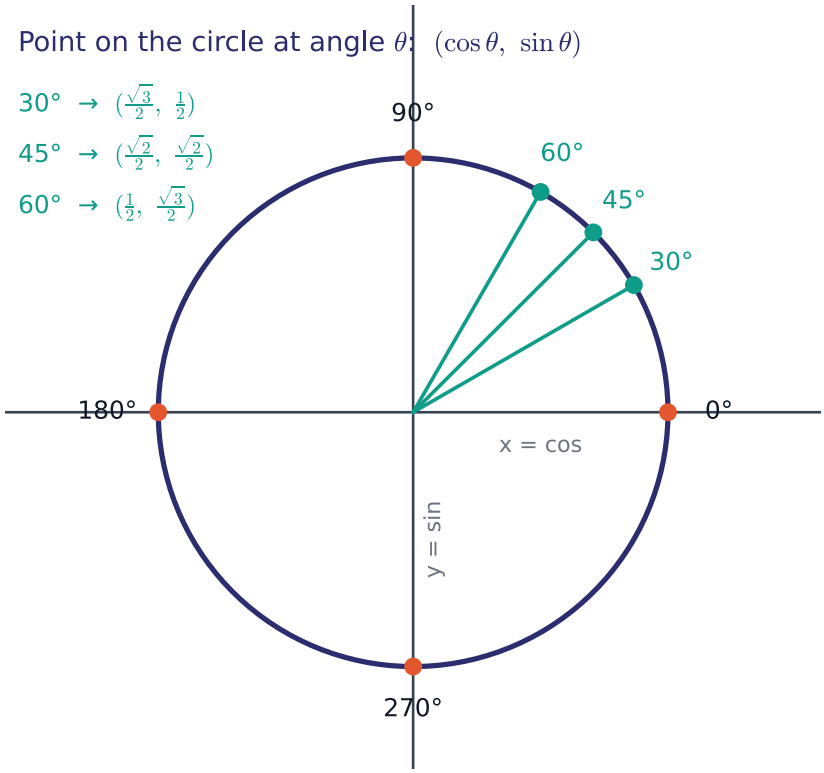
$x^2 + 3x + 5 = (x+1)(x+2) + 3$. So $\dfrac{x^2 + 3x + 5}{x+1} = x + 2 + \dfrac{3}{x+1}$.

**Binomial theorem.** $(1+x)^n = 1 + nx + \dfrac{n(n-1)}{2!}x^2 + \dfrac{n(n-1)(n-2)}{3!}x^3 + \cdots$, a fact you will meet again as a Maclaurin series.

**Sigma notation and factorials.** $n! = n(n-1)\cdots 2 \cdot 1$, with $0! = 1$. $\displaystyle\sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n$ is just shorthand for "add them up".

## 0.4 Trigonometry in one place

**Radians.** $\pi$ radians $= 180°$. Calculus **always** uses radians. In radians the limit $\lim \frac{\sin x}{x} = 1$ holds; in degrees it does not.




The unit circle: the point at angle $\theta$ has coordinates $(\cos\theta, \sin\theta)$.

**FORMULA** **The value table (memorise it)**

| $\theta$ | $0$ | $30° = \frac{\pi}{6}$ | $45° = \frac{\pi}{4}$ | $60° = \frac{\pi}{3}$ | $90° = \frac{\pi}{2}$ | $180° = \pi$ |
|---|---|---|---|---|---|---|
| $\sin\theta$ | $0$ | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | $1$ | $0$ |
| $\cos\theta$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | $0$ | $-1$ |
| $\tan\theta$ | $0$ | $\frac{1}{\sqrt{3}}$ | $1$ | $\sqrt{3}$ | undefined | $0$ |

**MEMORY TRICK** **Trick for the table**

For sin write $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$ for $0°, 30°, 45°, 60°, 90°$. For cos read the same list backwards.

**FORMULA** **Identities that appear in calculus every day**

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

$$\sin 2\theta = 2\sin\theta\cos\theta, \qquad \cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$$

$$\boxed{\cos^2\theta = \frac{1+\cos 2\theta}{2}}, \qquad \boxed{\sin^2\theta = \frac{1-\cos 2\theta}{2}} \quad \text{(power reduction: essential for integration)}$$

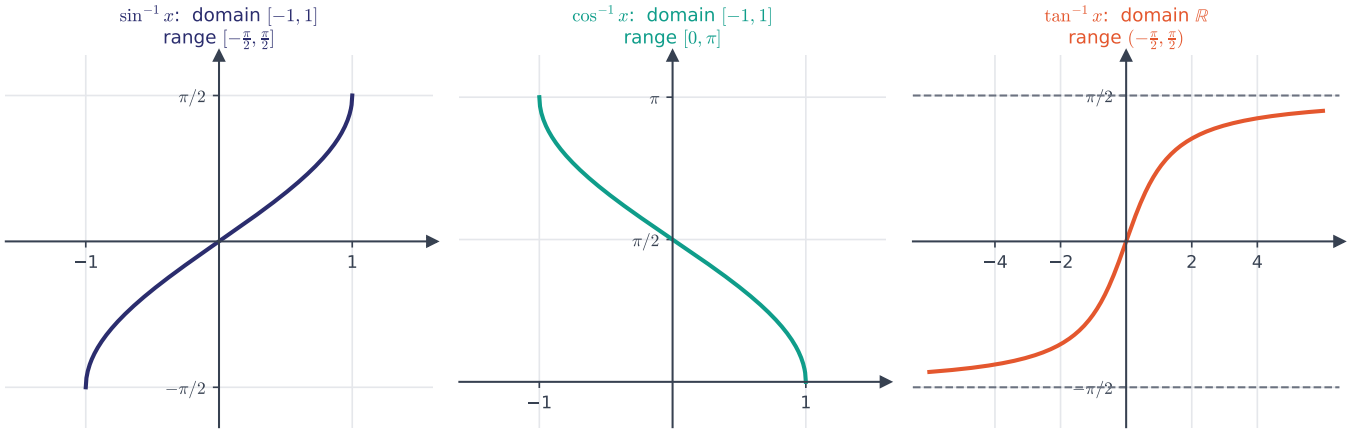
$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B, \qquad \cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

$$\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}, \qquad \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}, \qquad \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

**Inverse trigonometric functions.** "$\sin^{-1} x$" means "the angle whose sine is $x$" (it is **not** $1/\sin x$). To make it a proper function, mathematicians agree on a **principal range**:




Domains and principal ranges of the inverse trig functions

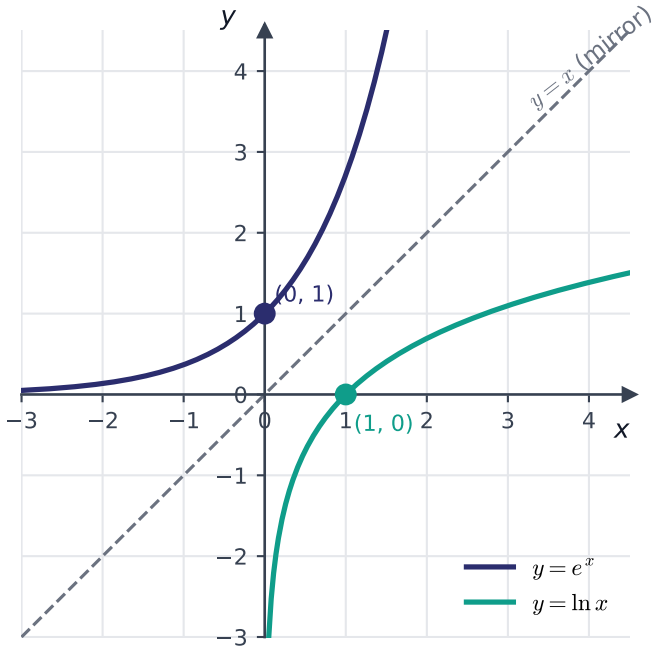
**FORMULA** **Identities for inverse trig**

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \qquad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

$$\sin^{-1}(\sin x) = x \text{ only for } x \in [-\tfrac{\pi}{2}, \tfrac{\pi}{2}], \qquad \tan^{-1} x + \tan^{-1} y = \tan^{-1}\frac{x+y}{1-xy} \quad (xy < 1)$$

## 0.5 Exponentials and logarithms

The number $e \approx 2.71828$ is defined by $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$. The function $e^x$ is the unique function that equals its own derivative. The **natural logarithm** $\ln x$ (written $\log x$ in your NCERT chapter) is its inverse: $\ln x = y \iff e^y = x$.




$e^x$ and $\ln x$ are mirror images in the line $y = x$

**FORMULA** **Log laws (all logs of positive numbers)**

$$\ln(pq) = \ln p + \ln q, \quad \ln\frac{p}{q} = \ln p - \ln q, \quad \ln p^n = n\ln p, \quad \ln 1 = 0, \quad \ln e = 1$$

$$e^{\ln x} = x \ (x > 0), \qquad \ln e^x = x, \qquad \log_a p = \frac{\ln p}{\ln a} \ \text{(change of base)}$$

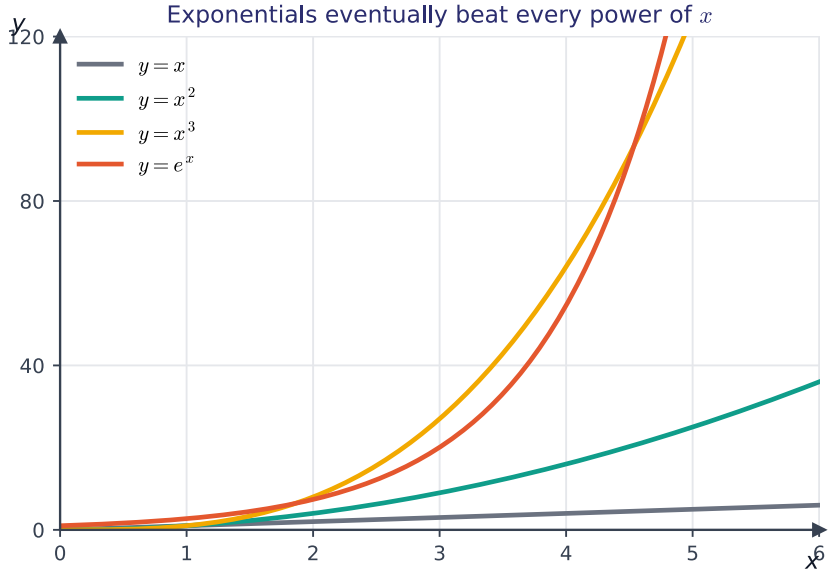
$$a^x = e^{x\ln a} \quad (a > 0)$$

**WATCH OUT** **A subtle trap**

$x = e^{\ln x}$ is true **only for** $x > 0$, because $\ln x$ does not exist otherwise (your textbook's Example 28). Domain first, algebra second.

**STORY TIME** **Where does $e$ come from?**

*In 1683 Jacob Bernoulli asked what happens if a bank pays 100% interest per year but compounds it more and more often. Compounding $n$ times a year multiplies your money by $(1 + \frac{1}{n})^n$. As $n$ grows this settles on $2.71828\ldots$, the number $e$.*




Exponentials eventually overtake every power of $x$

## 0.6 Quick self-check

**TRY IT** **Ten questions, ten minutes**

1. Write $|x+2|<3$ without absolute values.
2. Domain of $\dfrac{\sqrt{x-1}}{x-4}$?
3. Is $f(x)=x^3\cos x$ even, odd or neither?
4. Simplify $\dfrac{x^2-9}{x^2+x-12}$.
5. $\sin^2\theta = ?$ in terms of $\cos 2\theta$.
6. $\sin\frac{\pi}{6}+\cos\frac{\pi}{3}=?$
7. Simplify $\ln\dfrac{e^3}{x}$.
8. $\sin^{-1}(1)=?$ and $\tan^{-1}(1)=?$
9. Convert $\dfrac{5\pi}{6}$ to degrees.
10. Divide $x^3-1$ by $x-1$.

**Answers.** (1) $-5<x<1$. (2) $x\ge 1$ and $x\neq 4$. (3) odd (odd $\times$ even). (4) $\frac{(x-3)(x+3)}{(x+4)(x-3)}=\frac{x+3}{x+4}$, $x\neq 3$. (5) $\frac{1-\cos 2\theta}{2}$. (6) $\frac{1}{2}+\frac{1}{2}=1$. (7) $3-\ln x$. (8) $\frac{\pi}{2}$ and $\frac{\pi}{4}$. (9) $150^\circ$. (10) $x^2+x+1$.

UNIT I · CHAPTER 1

# Matrices: organising numbers

**BIG IDEA** **Big idea**

A **matrix** is just a rectangular table of numbers, like a spreadsheet. The magic is that we can *add*, *multiply* and *invert* whole tables as if they were single numbers. That lets us solve thousands of equations at once, describe rotations and stretches of space, model electrical networks, and run Google's page ranking.

## 1.1 What is a matrix?

**DEFINITION** **Matrix, order, entry**

A **matrix** of **order** $m \times n$ has $m$ rows and $n$ columns. The entry in row $i$, column $j$ is written $a_{ij}$.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad \text{is a } 2 \times 3 \text{ matrix.}$$

Read "rows first, then columns": a $2 \times 3$ matrix has 2 rows and 3 columns.

**Types you must be able to name:**

| Name | Description | Example |
|---|---|---|
| Row / column matrix | one row / one column | $[1\ 2\ 3]$, $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ |
| Square matrix | $m = n$ | $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ |
| Diagonal | square, all off-diagonal entries 0 | $\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ |
| Scalar | diagonal with equal diagonal entries | $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ |
| Identity $I_n$ | scalar with 1's: acts like the number 1 | $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ |
| Null (zero) matrix $O$ | every entry 0 | $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ |
| Upper / lower triangular | all entries below / above the diagonal are 0 | $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ |
| Symmetric | $A^T = A$ (mirror image in the diagonal) | $\begin{bmatrix} 1 & 5 \\ 5 & 2 \end{bmatrix}$ |
| Skew-symmetric | $A^T = -A$ (so the diagonal is all 0) | $\begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$ |

The **transpose** $A^T$ swaps rows and columns: $(A^T)_{ij} = A_{ji}$. The **trace** of a square matrix is the sum of its diagonal entries: $\operatorname{tr} A = a_{11} + a_{22} + \cdots + a_{nn}$. Trace will return in the eigenvalue chapter.

## 1.2 Adding and scaling

You may add matrices **only if they have the same order**, entry by entry. Scalar multiplication $kA$ multiplies every entry by $k$.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 5 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 7 \\ 2 & 5 \end{bmatrix}, \qquad 3\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$$

## 1.3 Multiplying matrices (the one that matters)

**DEFINITION** **Matrix product**

$AB$ exists only when **(number of columns of $A$) = (number of rows of $B$)**. If $A$ is $m \times n$ and $B$ is $n \times p$ then $AB$ is $m \times p$, and

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \quad \text{(row } i \text{ of } A \text{ "times" column } j \text{ of } B\text{)}.$$

**MEMORY TRICK** **Dimension check**

Write the orders side by side: $(m \times n)(n \times p)$. The two **inner** numbers must match; the two **outer** numbers give the answer's order.

**EXAMPLE** **Example 1.1: product depends on order**

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Find $AB$ and $BA$.

$$AB = \begin{bmatrix} 1\cdot 0 + 2\cdot 1 & 1\cdot 1 + 2\cdot 0 \\ 3\cdot 0 + 4\cdot 1 & 3\cdot 1 + 4\cdot 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \text{ (columns swapped).}$$

$$BA = \begin{bmatrix} 0\cdot 1 + 1\cdot 3 & 0\cdot 2 + 1\cdot 4 \\ 1\cdot 1 + 0\cdot 3 & 1\cdot 2 + 0\cdot 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \text{ (rows swapped).}$$

So $AB \neq BA$.

**WATCH OUT** **Matrices are not numbers**

- In general $AB \neq BA$ (**not commutative**).
- $AB = O$ does **not** force $A = O$ or $B = O$. Example: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = O$.
- $AB = AC$ does **not** let you cancel $A$.

**FORMULA** **Rules that do hold**

$$A(BC) = (AB)C, \quad A(B+C) = AB + AC, \quad AI = IA = A$$

$$(A^T)^T = A, \quad (A+B)^T = A^T + B^T, \quad (kA)^T = kA^T, \quad \boxed{(AB)^T = B^T A^T}$$

**EXAM TIP** **Every square matrix splits in two**

$A = \underbrace{\tfrac{1}{2}(A + A^T)}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}(A - A^T)}_{\text{skew-symmetric}}$. Exams love asking you to write this out.

**TRY IT** **Try it**

If $A$ is $3 \times 2$ and $B$ is $2 \times 4$, what is the order of $AB$? Does $BA$ exist? *Answer: $AB$ is $3 \times 4$; $BA$ would need $4 = 3$, so it does not exist.*

**STORY TIME** **Why do engineers care?**

*Rotating a point in the plane by angle $\theta$ is the matrix product $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$. Doing two rotations is multiplying two matrices. Computer graphics, robotics and signal processing are essentially fast matrix multiplication.*

UNIT I · CHAPTER 2

# Determinants and the inverse

**BIG IDEA** **Big idea**

A square matrix has a single number attached to it, its **determinant**. That number answers the biggest question you can ask: *can this matrix be undone?* If $|A| \neq 0$ the matrix is **invertible** and every system $AX = B$ has exactly one solution. If $|A| = 0$ the matrix collapses space (a plane squashed to a line) and information is lost for ever.

## 2.1 Computing determinants

$2 \times 2$: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ (down-right product minus down-left product).

$3 \times 3$ **by expansion along the first row**, with alternating signs $+ \; - \; +$:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

You may expand along **any** row or column; use the one with the most zeros. The sign of position $(i, j)$ is $(-1)^{i+j}$:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

**EXAMPLE** **Example 2.1**

Evaluate $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{vmatrix}$.

Expand along row 1: $\Delta = 1(4 \cdot 6 - 5 \cdot 0) - 2(0 \cdot 6 - 5 \cdot 1) + 3(0 \cdot 0 - 4 \cdot 1) = 24 + 10 - 12 = 22$.

**FORMULA** **Properties of determinants (use them to simplify!)**

1. $|A^T| = |A|$.
2. **Swapping** two rows (or columns) changes the sign of the determinant.
3. If two rows (or columns) are **identical** or proportional, $|A| = 0$. A row or column of zeros also gives 0.
4. Multiplying **one** row by $k$ multiplies $|A|$ by $k$. Hence for an $n \times n$ matrix $|kA| = k^n |A|$.
5. Adding a multiple of one row to another row **does not change** $|A|$ (this is the row-reduction workhorse).
6. $|AB| = |A| \; |B|$.
7. For triangular and diagonal matrices, $|A|$ is the **product of the diagonal entries**.

## 2.2 The inverse matrix

**DEFINITION** **Inverse**

$B$ is the **inverse** of $A$ (written $A^{-1}$) if $AB = BA = I$. It exists **iff** $|A| \neq 0$; such a matrix is called **non-singular** (invertible). If $|A| = 0$ the matrix is **singular**.

**Formula using the adjoint.** The **minor** $M_{ij}$ is the determinant left after deleting row $i$ and column $j$. The **cofactor** is $C_{ij} = (-1)^{i+j} M_{ij}$. The **adjoint** adj $A$ is the *transpose* of the cofactor matrix. Then

$$A\,(\operatorname{adj} A) = |A|\,I \qquad \Longrightarrow \qquad \boxed{A^{-1} = \frac{1}{|A|}\operatorname{adj} A}$$

For a $2 \times 2$ matrix this is the pleasant shortcut: swap the diagonal, negate the off-diagonal, divide by the determinant.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

**EXAMPLE** **Example 2.2:** $2 \times 2$

$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, $|A| = 6 - 5 = 1$, so $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$. *Check:* $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{bmatrix} = I$ ✓.

**EXAMPLE** **Example 2.3:** $3 \times 3$ **by the adjoint**

Find $A^{-1}$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$.

$|A| = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1$.

Cofactors: $C_{11} = -24$, $C_{12} = 20$, $C_{13} = -5$; $C_{21} = 18$, $C_{22} = -15$, $C_{23} = 4$; $C_{31} = 5$, $C_{32} = -4$, $C_{33} = 1$.

Cofactor matrix $\begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{bmatrix}$; transpose it to get the adjoint:

$$A^{-1} = \frac{1}{1}\begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}.$$

**FORMULA** **Inverse rules (these are tested directly in MCQs)**

$$\boxed{(AB)^{-1} = B^{-1}A^{-1}}, \qquad (A^T)^{-1} = (A^{-1})^T, \qquad (A^{-1})^{-1} = A, \qquad \boxed{|A^{-1}| = \frac{1}{|A|}}$$

$$|\operatorname{adj} A| = |A|^{n-1}\ (n \times n), \qquad (kA)^{-1} = \tfrac{1}{k}A^{-1}$$

**MEMORY TRICK** **Socks and shoes**

You put on socks, then shoes. To undo it you take off the **shoes first**, then socks. That is why $(AB)^{-1} = B^{-1}A^{-1}$ reverses the order.

UNIT I · CHAPTER 3

# Rank and elementary operations

**BIG IDEA** **Big idea**

The **rank** of a matrix counts how many rows carry *genuinely new information*. If one row is just a combination of others it is redundant and the rank drops. Rank is the master key to linear systems: it decides whether solutions exist, and how many.

## 3.1 Elementary row operations

There are exactly three, and none of them changes the rank or the solutions of a system:

| Operation | Symbol | Meaning |
|---|---|---|
| Swap | $R_i \leftrightarrow R_j$ | exchange two rows |
| Scale | $R_i \to kR_i \ (k \neq 0)$ | multiply a row by a non-zero number |
| Replace | $R_i \to R_i + kR_j$ | add a multiple of one row to another |

**DEFINITION** **Echelon form**

A matrix is in **row echelon form** if (i) all zero rows are at the bottom, and (ii) the first non-zero entry (the *pivot*) of each row is strictly to the right of the pivot above it, so the matrix looks like a staircase.

$$\begin{bmatrix} \boxed{2} & 1 & 3 & 0 \\ 0 & \boxed{5} & 1 & 4 \\ 0 & 0 & 0 & \boxed{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

## 3.2 Rank

**DEFINITION** **Rank**

The **rank** $\rho(A)$ is the **number of non-zero rows in an echelon form of** $A$. Equivalently, it is the largest order of a square sub-matrix (minor) with non-zero determinant.

**FORMULA** **Facts to memorise**

- $\operatorname{rank}(O) = 0$ for the **null matrix** of any order $m \times n$.
- $\operatorname{rank}(I_n) = n$.
- $\operatorname{rank}(A) \leq \min(m, n)$ for an $m \times n$ matrix.
- $\operatorname{rank}(A) = \operatorname{rank}(A^T)$.
- For a square $n \times n$ matrix: $\text{rank} = n \iff |A| \neq 0 \iff A$ invertible.
- A non-zero matrix has rank $\geq 1$.

**EXAMPLE** **Example 3.1: reduce to echelon form**

Find the rank of $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$.

$R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - 3R_1$:

$$\begin{bmatrix}1&2&3&4\\0&-1&-2&-3\\0&-2&-4&-6\end{bmatrix}\xrightarrow{R_3\to R_3-2R_2}\begin{bmatrix}1&2&3&4\\0&-1&-2&-3\\0&0&0&0\end{bmatrix}$$

Two non-zero rows, so rank $A=2$.

**EXAMPLE** **Example 3.2: rank depending on a parameter**

Find the rank of $A=\begin{bmatrix}1&1&1\\1&2&3\\1&3&k\end{bmatrix}$ for every $k$.

$|A|=1(2k-9)-1(k-3)+1(3-2)=k-5$. So for $k\neq 5$, $|A|\neq 0$ and **rank** $=3$. For $k=5$, $|A|=0$ but the first two rows are not proportional, so **rank** $=2$.

## 3.3 Finding an inverse by row operations (Gauss–Jordan)

Write $[A\mid I]$ and apply row operations until the left block becomes $I$; the right block is then $A^{-1}$.

**EXAMPLE** **Example 3.3**

Invert $A=\begin{bmatrix}2&1\\5&3\end{bmatrix}$ this way.

$$\left[\begin{array}{cc|cc}2&1&1&0\\5&3&0&1\end{array}\right]\xrightarrow{R_1\to\frac12 R_1}\left[\begin{array}{cc|cc}1&\frac12&\frac12&0\\5&3&0&1\end{array}\right]\xrightarrow{R_2\to R_2-5R_1}\left[\begin{array}{cc|cc}1&\frac12&\frac12&0\\0&\frac12&-\frac52&1\end{array}\right]$$

$$\xrightarrow{R_2\to 2R_2}\left[\begin{array}{cc|cc}1&\frac12&\frac12&0\\0&1&-5&2\end{array}\right]\xrightarrow{R_1\to R_1-\frac12 R_2}\left[\begin{array}{cc|cc}1&0&3&-1\\0&1&-5&2\end{array}\right]$$

So $A^{-1}=\begin{bmatrix}3&-1\\-5&2\end{bmatrix}$, agreeing with Example 2.2.

**EXAM TIP** **Which method?**

For a $2\times 2$ use the shortcut. For $3\times 3$ the adjoint formula is fine but error-prone; Gauss–Jordan is more reliable and generalises to any size.

UNIT I · CHAPTER 4

# Linear dependence and independence

**BIG IDEA** **Big idea**

A set of vectors is **linearly dependent** if at least one of them is *redundant*, i.e. can be built from the others. It is **independent** if every vector adds a new direction. In the plane, two vectors pointing along the same line are dependent; two vectors pointing in different directions are independent. Three vectors in the plane are *always* dependent, since there are only two directions to go.

**DEFINITION** **Linear independence**

Vectors $v_1, \dots, v_k$ are **linearly independent** if

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0 \quad \Longrightarrow \quad c_1 = c_2 = \cdots = c_k = 0.$$

They are **linearly dependent** if the equation has a solution with **at least one** $c_i \neq 0$.

**FORMULA** **Quick facts**

- Any set **containing the zero vector** is dependent (give the zero vector coefficient 1 and all others 0).
- **Two non-zero vectors** are dependent iff one is a scalar multiple of the other.
- More than $n$ vectors in $\mathbb{R}^n$ are always dependent.
- A single non-zero vector is independent.
- Any subset of an independent set is independent; any superset of a dependent set is dependent.

**The practical test.** Put the vectors as the rows of a matrix and find its rank.

$$\text{rank} = \text{number of vectors} \;\Rightarrow\; \text{independent}; \qquad \text{rank} < \text{number of vectors} \;\Rightarrow\; \text{dependent}.$$

For $n$ vectors in $\mathbb{R}^n$ this simplifies to: **independent** $\iff \det \neq 0$.

**EXAMPLE** **Example 4.1**

Are $v_1 = (1,0,1),\ v_2 = (0,1,1),\ v_3 = (1,1,2)$ independent?

Notice $v_3 = v_1 + v_2$, i.e. $1 \cdot v_1 + 1 \cdot v_2 - 1 \cdot v_3 = 0$ with non-zero coefficients. **Dependent.** Check: $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 1(2-1) - 0 + 1(0-1) = 0$ ✓.

**EXAMPLE** **Example 4.2: find the parameter**

For which $\lambda$ are $(1,2,3),\ (3,-1,2),\ (5,3,\lambda)$ linearly dependent?

Dependent $\iff$ determinant $= 0$:

$$\begin{vmatrix} 1 & 2 & 3 \\ 3 & -1 & 2 \\ 5 & 3 & \lambda \end{vmatrix} = 1(-\lambda - 6) - 2(3\lambda - 10) + 3(9+5) = -7\lambda + 56 = 0 \;\Rightarrow\; \lambda = 8.$$

**TRY IT** **Try it**

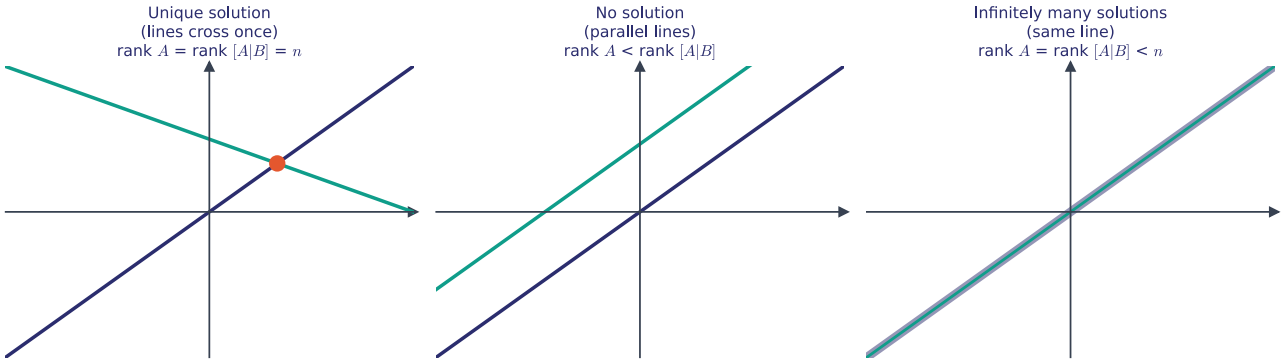
Are $(1,2,-1), (2,-1,3), (0,1,1)$ independent? *Answer: the determinant is $-10 \neq 0$, so yes.*

UNIT I · CHAPTER 5

# Solving systems of linear equations

**BIG IDEA** **Big idea**

Every system of linear equations can be packaged as a single matrix equation $AX = B$. Each equation is a line (or plane); solving the system means finding where they all meet. Three things can happen, and **rank** tells us which.




Three possible pictures for two equations in two unknowns. The same trichotomy holds in every dimension.

## 5.1 Matrix form

The system $a_{11}x_1 + \cdots + a_{1n}x_n = b_1,\ \ldots,\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m$ becomes

$$\underbrace{\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}}_{B}$$

The **augmented matrix** is $[A \mid B]$: the coefficient matrix with the right-hand side glued on as an extra column.

## 5.2 The Rouché–Capelli theorem

**THEOREM** **Consistency theorem (Rouché–Capelli)**

Let $r = \operatorname{rank}(A)$, $r' = \operatorname{rank}[A \mid B]$ and $n =$ number of unknowns.

| Condition | Conclusion |
|---|---|
| $r \neq r'$ (i.e. $r < r'$) | **No solution** (inconsistent) |
| $r = r' = n$ | **Unique solution** |
| $r = r' < n$ | **Infinitely many solutions**, with $n - r$ free parameters |

**MEMORY TRICK** **Why?**

If $r < r'$ the row-reduced augmented matrix contains a row $[0\ 0\ \cdots\ 0 \mid c]$ with $c \neq 0$, which says $0 = c$: an impossibility. If $r = r' < n$, some variables are never pinned down and can be chosen freely.

**Homogeneous systems ($AX = O$).** These are always consistent, because $X = O$ (the **trivial solution**) always works. So only two cases arise:

- rank $A = n$: only the trivial solution.

- rank $A < n$: **non-trivial solutions** exist (infinitely many).
- For a **square** homogeneous system: non-trivial solutions exist $\iff |A| = 0$.

## 5.3 Gauss elimination in practice

Row-reduce $[A \mid B]$ to echelon form, then back-substitute.

**EXAMPLE** **Example 5.1: unique solution**

Solve $x + y + z = 6, \quad x - y + z = 2, \quad 2x + y - z = 1.$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array}\right] \xrightarrow{R_3 \to R_3 - \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & -3 & -9 \end{array}\right]$$

Here $r = r' = 3 = n$. Back-substitute: $z = 3$, $y = 2$, $x = 6 - 2 - 3 = 1$. **Solution** $(1, 2, 3)$.

**EXAMPLE** **Example 5.2: the classic 'find $\lambda, \mu$' question**

Discuss the solutions of $x + y + z = 6, \;\; x + 2y + 3z = 10, \;\; x + 2y + \lambda z = \mu.$

$R_2 \to R_2 - R_1$ gives $y + 2z = 4$. $R_3 \to R_3 - R_1$ gives $y + (\lambda - 1)z = \mu - 6$. Now $R_3 \to R_3 - R_2$ gives $(\lambda - 3)z = \mu - 10$. So in echelon form the last row is $[0 \;\; 0 \;\; \lambda - 3 \mid \mu - 10]$.

- $\lambda \neq 3$: $r = r' = 3$. **Unique solution** (for any $\mu$).
- $\lambda = 3, \;\; \mu \neq 10$: last row $[0\,0\,0 \mid \mu - 10]$ with $\mu - 10 \neq 0$. $r = 2 < r' = 3$. **No solution.**
- $\lambda = 3, \;\; \mu = 10$: $r = r' = 2 < 3$. **Infinitely many solutions.** Put $z = t$: $y = 4 - 2t$, $x = 6 - y - z = 2 + t$. The solutions are $(2 + t, \; 4 - 2t, \; t)$.

**EXAMPLE** **Example 5.3: homogeneous system**

For what $k$ does $x + 2y + z = 0, \;\; 2x + y + 3z = 0, \;\; 3x + 3y + kz = 0$ have a non-trivial solution?

Need $|A| = 0$: $\begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 3 & 3 & k \end{vmatrix} = 1(k - 9) - 2(2k - 9) + 1(6 - 3) = 12 - 3k = 0 \Rightarrow k = 4.$

## 5.4 Two other methods (know they exist)

- **Inverse method:** if $|A| \neq 0$, then $X = A^{-1}B$.
- **Cramer's rule** (square, $|A| \neq 0$): $x_i = \dfrac{|A_i|}{|A|}$, where $A_i$ is $A$ with column $i$ replaced by $B$.

**WATCH OUT** **Common slip**

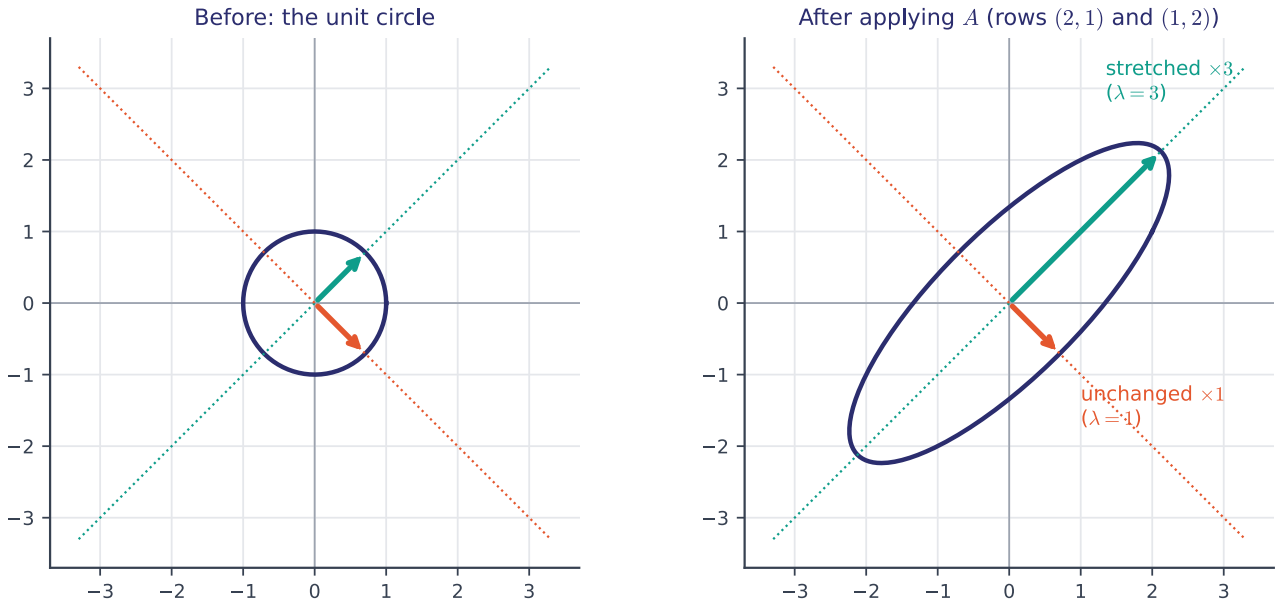
Do **not** compare rank $A$ with the *number of equations*. Compare it with the number of **unknowns** $n$ and with rank$[A \mid B]$.

UNIT I · CHAPTER 6

# Eigenvalues and eigenvectors

**BIG IDEA** **Big idea**

A matrix $A$ transforms vectors: it rotates, stretches and shears them. Most vectors get turned to point in a *new* direction. But some special directions are only **stretched or shrunk**, never turned. Those directions are the **eigenvectors**, and the stretch factors are the **eigenvalues**. (*Eigen* is German for "own": these are the matrix's own special directions.)




The matrix $A$ stretches the diagonal direction by 3 and leaves the anti-diagonal direction unchanged; the whole circle becomes an ellipse whose axes are the eigen-directions.

**DEFINITION** **Eigenvalue and eigenvector**

For a square matrix $A$, a number $\lambda$ is an **eigenvalue** and a **non-zero** vector $x$ is a corresponding **eigenvector** if

$$Ax = \lambda x.$$

## 6.1 How to find them

Rewrite $Ax = \lambda x$ as $(A - \lambda I)x = 0$. A non-zero solution $x$ exists only if $A - \lambda I$ is singular:

$$\boxed{|A - \lambda I| = 0} \qquad \text{(the characteristic equation)}$$

Expanding the determinant gives a polynomial in $\lambda$ of degree $n$ (the characteristic polynomial); its roots are the eigenvalues. For each eigenvalue, solve $(A - \lambda I)x = 0$ to get the eigenvectors.

**FORMULA** **Shortcuts for the characteristic equation**

$2 \times 2$: $\lambda^2 - T\lambda + D = 0$, where $T = \operatorname{tr} A$ and $D = |A|$.

$3 \times 3$: $\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$, where $S_1 = \operatorname{tr} A$ and $S_2 =$ sum of the three principal $2 \times 2$ minors (the minors of the diagonal entries).

**EXAMPLE** **Example 6.1: a $2 \times 2$ from start to finish**

Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

$T = 4$, $D = 3$, so $\lambda^2 - 4\lambda + 3 = 0 \Rightarrow (\lambda - 1)(\lambda - 3) = 0$: $\lambda = 1,\ 3$.

$\lambda = 3$: $(A - 3I)x = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0 \Rightarrow x_1 = x_2$. Eigenvector $(1, 1)$ (any non-zero multiple).

$\lambda = 1$: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0 \Rightarrow x_1 = -x_2$. Eigenvector $(1, -1)$.

Matches the picture above: the diagonal $(1, 1)$ is stretched by 3, the anti-diagonal $(1, -1)$ is unchanged.

**EXAMPLE** **Example 6.2: a $3 \times 3$ symmetric matrix**

Find the eigenvalues of $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and one eigenvector.

$S_1 = 8 + 7 + 3 = 18$. $S_2 = (56 - 36) + (24 - 4) + (21 - 16) = 20 + 20 + 5 = 45$. $|A| = 8(21 - 16) + 6(-18 + 8) + 2(24 - 14) = 40 - 60 + 20 = 0$.

Characteristic equation: $\lambda^3 - 18\lambda^2 + 45\lambda = 0 \Rightarrow \lambda(\lambda - 3)(\lambda - 15) = 0$.

**Eigenvalues:** $0,\ 3,\ 15$. (Check: sum $= 18 =$ trace ✓, product $= 0 = |A|$ ✓.)

For $\lambda = 3$: $(A - 3I)x = 0$ gives $x = 2y,\ z = -2y$, so an eigenvector is $(2, 1, -2)$. Similarly $(1, 2, 2)$ for $\lambda = 0$ and $(2, -2, 1)$ for $\lambda = 15$. These three eigenvectors are mutually perpendicular, as the theory of symmetric matrices predicts.

## 6.2 Properties of eigenvalues

**FORMULA** **The big list (examiners adore these)**

Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of an $n \times n$ matrix $A$.

1. **Sum = trace:** $\lambda_1 + \cdots + \lambda_n = \operatorname{tr} A$.
2. **Product = determinant:** $\lambda_1 \lambda_2 \cdots \lambda_n = |A|$.
3. $A$ is **singular** ($|A| = 0$) $\iff$ $0$ is an eigenvalue.
4. $A$ and $A^T$ have the **same** eigenvalues.
5. $kA$ has eigenvalues $k\lambda_i$.
6. $A^m$ has eigenvalues $\lambda_i^m$.
7. If $A$ is invertible, $A^{-1}$ has eigenvalues $\dfrac{1}{\lambda_i}$ (same eigenvectors).
8. $A + kI$ has eigenvalues $\lambda_i + k$.
9. For any polynomial $p$, $p(A)$ has eigenvalues $p(\lambda_i)$.
10. For a **triangular or diagonal** matrix the eigenvalues are the **diagonal entries**.
11. Eigenvectors for **distinct** eigenvalues are linearly independent.
12. For a **real symmetric** matrix: all eigenvalues are real, and eigenvectors of different eigenvalues are orthogonal.

**PROOF** **Why is (7) true?**

$Ax = \lambda x$. Multiply both sides by $A^{-1}$: $x = \lambda A^{-1} x$, so $A^{-1} x = \frac{1}{\lambda} x$. (We need $\lambda \neq 0$, guaranteed since $A$ is invertible.)

**EXAMPLE** **Example 6.3: using the properties**

A $3 \times 3$ matrix has eigenvalues $1, 2, 3$. Find $|A|$, $\operatorname{tr} A$, and the eigenvalues of $A^{-1}$, $A^2 + I$ and $2A$.

$|A| = 1 \cdot 2 \cdot 3 = 6$; $\operatorname{tr} A = 6$. $A^{-1}$: $1, \tfrac{1}{2}, \tfrac{1}{3}$. $A^2 + I$: $2, 5, 10$. $2A$: $2, 4, 6$.

**TRY IT** **Try it**

If the trace of a $2 \times 2$ matrix is $5$ and its determinant is $6$, what are its eigenvalues? *Answer:* $\lambda^2 - 5\lambda + 6 = 0$, so $2$ *and* $3$.

UNIT I · CHAPTER 7

# The Cayley–Hamilton theorem

**BIG IDEA** **Big idea**

We know every matrix has a characteristic polynomial $p(\lambda)$. The astonishing theorem says: if you plug the **matrix itself** into its own characteristic polynomial, you get the zero matrix. This lets you compute inverses and high powers of a matrix with nothing but addition and multiplication.

**THEOREM** **Cayley–Hamilton**

Every square matrix satisfies its own characteristic equation. If $\lambda^n + c_1\lambda^{n-1} + \cdots + c_n = 0$ is the characteristic equation of $A$, then

$$A^n + c_1A^{n-1} + \cdots + c_nI = O.$$

(The constant term $c_n$ is multiplied by the **identity matrix** $I$.)

For a $2 \times 2$ matrix this reads $\boxed{A^2 - TA + DI = O}$, with $T = \operatorname{tr} A$ and $D = |A|$.

**EXAMPLE** **Example 7.1: verify and find $A^{-1}$**

For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ($T = 5$, $D = -2$) the characteristic equation is $\lambda^2 - 5\lambda - 2 = 0$.

*Verify:* $A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$, $5A = \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix}$, so $A^2 - 5A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$, i.e. $A^2 - 5A - 2I = O$ ✓.

*Inverse:* multiply $A^2 - 5A - 2I = O$ by $A^{-1}$: $A - 5I - 2A^{-1} = O$, so

$$A^{-1} = \tfrac{1}{2}(A - 5I) = \tfrac{1}{2}\begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.$$

**EXAM TIP** **The standard exam recipe**

1. Write $|A - \lambda I| = 0$ (use the $S_1, S_2$ shortcut).
2. Replace $\lambda$ by $A$ and the constant by $\times I$.
3. To find $A^{-1}$: multiply by $A^{-1}$ and solve for it. (The constant term must be non-zero, i.e. $|A| \neq 0$.)
4. To find a high power, e.g. $A^4$: reduce it with $A^3 = \cdots$ repeatedly.

**EXAMPLE** **Example 7.2: a $3 \times 3$ inverse**

Use Cayley–Hamilton to find $A^{-1}$ for $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$.

$S_1 = 6$; principal minors: $6,\ -1,\ 2$ so $S_2 = 7$; $|A| = 1(6 - 0) + 2(0 - 4) = -2$. Characteristic equation: $\lambda^3 - 6\lambda^2 + 7\lambda + 2 = 0$.

By Cayley–Hamilton $A^3 - 6A^2 + 7A + 2I = O$. Multiply by $A^{-1}$: $A^2 - 6A + 7I + 2A^{-1} = O$, so $A^{-1} = -\frac{1}{2}(A^2 - 6A + 7I)$.

With $A^2 = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$: $A^2 - 6A + 7I = \begin{bmatrix} 6 & 0 & -4 \\ 2 & -1 & -1 \\ -4 & 0 & 2 \end{bmatrix}$, hence

$$A^{-1} = \begin{bmatrix} -3 & 0 & 2 \\ -1 & \frac{1}{2} & \frac{1}{2} \\ 2 & 0 & -1 \end{bmatrix}.$$

**EXAMPLE** **Example 7.3: the MCQ pattern**

If $A^2 - 3A + 2I = O$, express $A^{-1}$.

Multiply by $A^{-1}$: $A - 3I + 2A^{-1} = O \Rightarrow A^{-1} = \frac{1}{2}(3I - A)$. (This exact question is in your practice set.) Also, the eigenvalues must satisfy $\lambda^2 - 3\lambda + 2 = 0$, so they can only be 1 or 2.

**STORY TIME** **A theorem with a surprising history**

*Arthur Cayley stated it for small matrices in 1858 and William Rowan Hamilton had met a special case in his work on quaternions a few years earlier. The general proof came later, from Frobenius in 1878.*

UNIT I · CHAPTER 8

# Unit I: practice and answers

## 8.1 Your professor's practice questions (with reasons)

These are the Unit I questions from the mixed practice set. Cover the answer column and try each first.

| Q | Question (short) | Answer | Why |
|---|---|---|---|
| 2 | Rank of the null matrix $m \times n$ | **(A) 0** | no non-zero row |
| 3 | Two non-zero vectors dependent iff | **(B) one is a scalar multiple of the other** | parallel vectors |
| 5 | Characteristic equation | **(B)** $\lvert A - \lambda I\rvert = 0$ | from $(A - \lambda I)x = 0$ |
| 6 | Cayley–Hamilton: satisfies its own | **(B) characteristic equation** | the theorem |
| 7 | Invertible iff | **(B) non-singular,** $\lvert A\rvert \neq 0$ | $A^{-1} = \operatorname{adj} A / \lvert A\rvert$ |
| 8 | Rank of $I_n$ | **(D)** $n$ | $n$ pivots |
| 9 | Set containing the zero vector | **(A) linearly dependent** | $1 \cdot 0 = 0$ |
| 10 | Unique solution of $AX = B$ | **(C) rank** $A =$ **rank**$[A \mid B] = n$ | Rouché–Capelli |
| 11 | Sum of eigenvalues | **(B) the trace of** $A$ | property 1 |
| 13 | $(AB)^{-1}$ | **(B)** $B^{-1}A^{-1}$ | socks and shoes |
| 16 | Infinitely many solutions | **(C) rank** $A =$ **rank**$[A \mid B] < n$ | free parameters |
| 17 | Product of eigenvalues | **(B) the determinant** | property 2 |
| 18 | $2 \times 2$ with trace $T$, det $D$ | **(B)** $\lambda^2 - T\lambda + D = 0$ | shortcut |
| 19 | $\lvert A^{-1}\rvert$ | **(B)** $1/\lvert A\rvert$ | $\lvert A\rvert\,\lvert A^{-1}\rvert = 1$ |
| 21 | Independent if $\sum c_i v_i = 0$ implies | **(C) all** $c_i = 0$ | definition |
| 22 | No solution if | **(B) rank** $A <$ **rank**$[A \mid B]$ | inconsistent |
| 23 | Eigenvalue of $A^{-1}$ | **(C)** $1/\lambda$ | property 7 |
| 24 | $A^2 - 3A + 2I = O$, $A^{-1} =$ | **(A)** $\frac{1}{2}(3I - A)$ | Example 7.3 |

## 8.2 Extra practice

**TRY IT** **Questions**

1. Find the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$.
2. Find $A^{-1}$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$.
3. Solve $x + y + z = 3,\ 2x - y + z = 2,\ x + 2y - z = 2$.
4. For what $k$ is $x + y + z = 1,\ x + 2y + 4z = k,\ x + 4y + 10z = k^2$ consistent?
5. Eigenvalues and eigenvectors of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.
6. $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Find the eigenvalues of $A^{-1}$, $A^2 + I$ and $2A$.
7. Use Cayley–Hamilton to find $A^{-1}$ for $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.
8. Are $(1, 2, 3), (2, 3, 4), (3, 4, 5)$ independent?
9. A $3 \times 3$ matrix has eigenvalues $1, 2, 3$. Find the eigenvalues of $A^3$ and $|\text{adj } A|$.

**Answers.** (1) Rank 2 ($R_2 - 2R_1 = [0, 0, 1]$, $R_3 - 3R_1 = [0, 0, 1]$). (2) $\begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$. (3) $|A| = 7 \neq 0$; unique solution $(1, 1, 1)$. (4) Row reduce: the last row becomes $[0\,0\,0 \mid k^2 - 3k + 2]$, so consistent for $k = 1$ or $k = 2$ (then infinitely many solutions). (5) $\lambda^2 - 7\lambda + 10 = 0$: $\lambda = 2$ (eigenvector $(1, -2)$), $\lambda = 5$ (eigenvector $(1, 1)$). (6) Triangular, so eigenvalues $1, 3$. $A^{-1}$: $1, \frac{1}{3}$. $A^2 + I$: $2, 10$. $2A$: $2, 6$. (7) $A^2 - 3A + I = O$, so $A^{-1} = 3I - A = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$. (8) Dependent: $(3, 4, 5) = 2(2, 3, 4) - (1, 2, 3)$. (9) $A^3$: $1, 8, 27$. $|A| = 6$, so $|\text{adj } A| = 6^2 = 36$.

UNIT II · CHAPTER 9

# Limits: where is it heading?

**BIG IDEA** **Big idea**

A **limit** describes what value a function is *heading towards* as its input gets close to some point. It does **not** care what happens *at* the point itself. This one idea is the foundation under both derivatives and integrals: a derivative is a limit of slopes, an integral is a limit of sums.

## 9.1 The idea in a table

Let $f(x) = \dfrac{x^2 - 1}{x - 1}$. At $x = 1$ it is $\frac{0}{0}$, undefined. But look at nearby values:

| $x$ | 0.9 | 0.99 | 0.999 | 1 | 1.001 | 1.01 | 1.1 |
|---|---|---|---|---|---|---|---|
| $f(x)$ | 1.9 | 1.99 | 1.999 | **?** | 2.001 | 2.01 | 2.1 |

From both sides the outputs close in on 2. We write

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

Algebra confirms it: for $x \neq 1$, $\frac{x^2-1}{x-1} = x + 1 \to 2$.

**DEFINITION** **Limit (informal definition)**

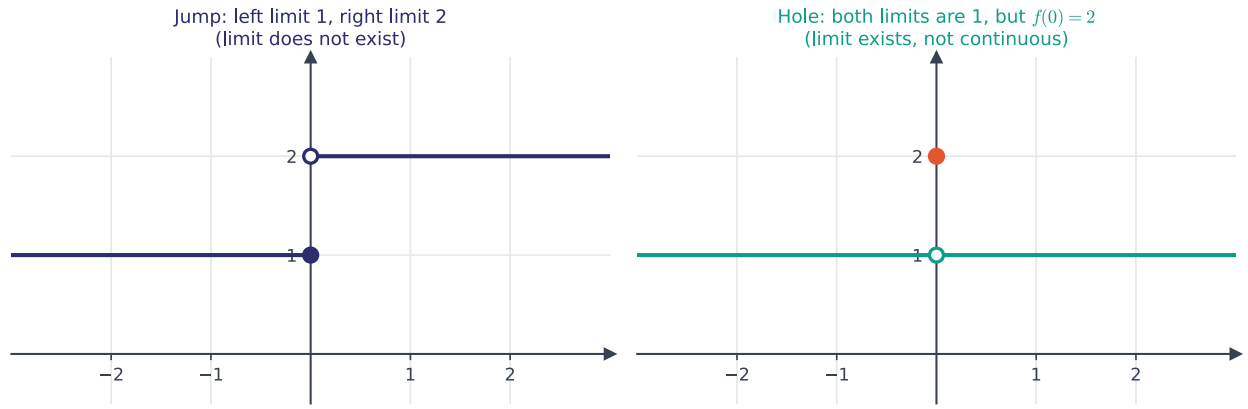
$\lim\limits_{x \to c} f(x) = L$ means: $f(x)$ can be made as close to $L$ as we like by taking $x$ close enough to $c$ (but $x \neq c$).

## 9.2 One-sided limits

Approach $c$ from the **left** ($x \to c^-$) or from the **right** ($x \to c^+$). These can differ.

**THEOREM** **Existence of a limit**

$\lim\limits_{x \to c} f(x)$ exists **if and only if** the left-hand limit and the right-hand limit both exist and are **equal**.




Left and right limits. In the left picture they disagree, so the limit does not exist. In the right picture both are 1 even though $f(0) = 2$.

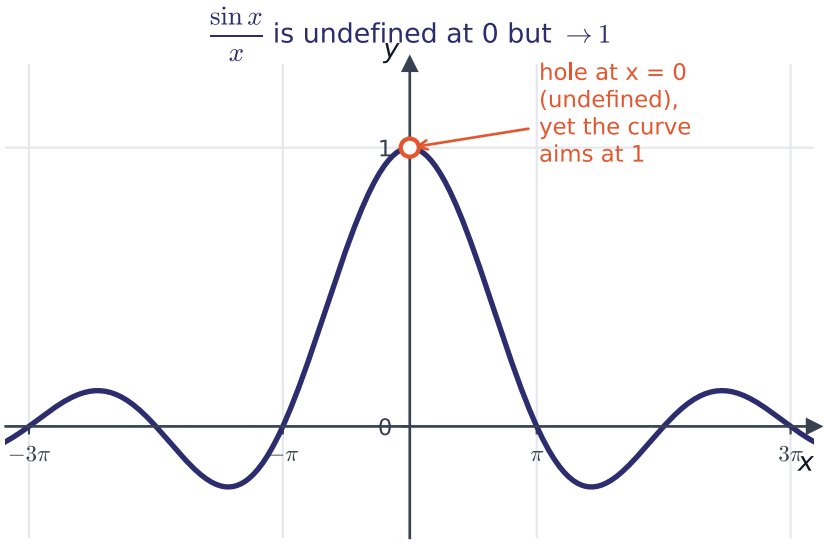
## 9.3 Limit laws

If $\lim f = L$ and $\lim g = M$ (as $x \to c$), then:

**FORMULA** **Limit laws**

$$\lim(f \pm g) = L \pm M, \quad \lim(fg) = LM, \quad \lim(kf) = kL, \quad \lim \frac{f}{g} = \frac{L}{M}\ (M \neq 0), \quad \lim f^n = L^n$$

**Squeeze (sandwich) theorem.** If $g(x) \le f(x) \le h(x)$ near $c$ and $g, h$ both tend to $L$, then $f \to L$ too. This is how we prove the most important limit in the course:




$\frac{\sin x}{x}$ is not defined at $0$ but tends to $1$.

**PROOF** **Why $\sin x/x \to 1$**

For small $x > 0$ (radians!) geometry gives $\sin x < x < \tan x$. Dividing by $\sin x$: $1 < \frac{x}{\sin x} < \frac{1}{\cos x}$, i.e. $\cos x < \frac{\sin x}{x} < 1$. As $x \to 0$, $\cos x \to 1$, so $\frac{\sin x}{x}$ is squeezed to 1.

## 9.4 Techniques for finding limits

**EXAM TIP** **Always try these in order**

1. **Substitute.** If the function is continuous at $c$ (polynomials, sin, $e^x$, ...) the answer is $f(c)$. Done.
2. If you get $\frac{0}{0}$: **factorise and cancel**, or **rationalise**.
3. For $x \to \pm\infty$: **divide numerator and denominator by the highest power of $x$**.
4. Recognise a **standard limit** (next table), possibly after a substitution.
5. As a last resort use **L'Hôpital's rule** (Chapter 16) or a **series** (Chapter 15).

**EXAMPLE** **Example 9.1: factorise**

$$\lim_{x\to 2} \frac{x^2 - 4}{x - 2} = \lim_{x\to 2}(x + 2) = 4.$$

**EXAMPLE** **Example 9.2: rationalise**

$$\lim_{x\to 0} \frac{\sqrt{1+x} - 1}{x} = \lim_{x\to 0} \frac{(1+x) - 1}{x(\sqrt{1+x} + 1)} = \lim_{x\to 0} \frac{1}{\sqrt{1+x} + 1} = \frac{1}{2}.$$

**EXAMPLE** **Example 9.3: limits at infinity**

$$\lim_{x\to\infty} \frac{3x^2 + 2x}{5x^2 - 1} = \lim_{x\to\infty} \frac{3 + 2/x}{5 - 1/x^2} = \frac{3}{5}.$$

$$\lim_{x\to\infty}\left(\sqrt{x^2+x}-x\right)=\lim\frac{x}{\sqrt{x^2+x}+x}=\lim\frac{1}{\sqrt{1+1/x}+1}=\frac{1}{2}.$$

## 9.5 Standard limits (learn these cold)

**FORMULA** **The standard limit table**

| Limit | Value |
|---|---|
| $\lim_{x\to0}\frac{\sin x}{x}$, $\lim_{x\to0}\frac{\tan x}{x}$, $\lim_{x\to0}\frac{\sin^{-1}x}{x}$, $\lim_{x\to0}\frac{\tan^{-1}x}{x}$ | $1$ |
| $\lim_{x\to0}\frac{1-\cos x}{x^2}$ | $\frac{1}{2}$ |
| $\lim_{x\to0}\frac{1-\cos x}{x}$ | $0$ |
| $\lim_{x\to0}\frac{e^x-1}{x}$ | $1$ |
| $\lim_{x\to0}\frac{a^x-1}{x}$ | $\ln a$ |
| $\lim_{x\to0}\frac{\ln(1+x)}{x}$ | $1$ |
| $\lim_{x\to0}(1+x)^{1/x}$, $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$ | $e$ |
| $\lim_{x\to a}\frac{x^n-a^n}{x-a}$ | $na^{n-1}$ |

**WATCH OUT** **Radians only**

$\lim\frac{\sin x}{x}=1$ is true only when $x$ is in **radians**. In degrees the limit is $\pi/180$.

## 9.6 Infinite limits

Look at $f(x)=\dfrac{1}{x}$ near 0. As $x\to0^+$ the values $10,\ 100,\ 1000,\ldots$ grow without bound; as $x\to0^-$ they fall to $-10,\ -100,\ -1000,\ldots$ We write $\lim_{x\to0^+}\frac{1}{x}=+\infty$ and $\lim_{x\to0^-}\frac{1}{x}=-\infty$.

**WATCH OUT** **$\infty$ is not a number**

"$=\infty$" describes **how** a limit fails to exist (it grows without bound). It never means the limit is a real number. You cannot do ordinary arithmetic like $\infty-\infty$ (that is an **indeterminate form**, Chapter 16).

The line $x=0$ is a **vertical asymptote**; $y=L$ is a **horizontal asymptote** if $\lim_{x\to\pm\infty}f(x)=L$.

**TRY IT** **Try it**

Find: (a) $\lim_{x\to0}\dfrac{\sin 5x}{x}$ (b) $\lim_{x\to0}\dfrac{1-\cos x}{x^2}$ (c) $\lim_{x\to\infty}\dfrac{2x+1}{x^2+3}$. *Answers:* (*a*) 5 (*write* $5\cdot\frac{\sin 5x}{5x}$). (*b*) $\frac{1}{2}$. (*c*) 0.

UNIT II · CHAPTER 10

# Continuity

**BIG IDEA** **Big idea**

**Pen test.** A function is continuous if you can draw its graph without lifting your pen. A discontinuity is a place where you *must* lift the pen: a hole, a jump, or a blow-up. Mathematically: *no surprises*, meaning the value of the function at a point equals the value it was heading towards.

## 10.1 Definition

**DEFINITION** **Continuity at a point**

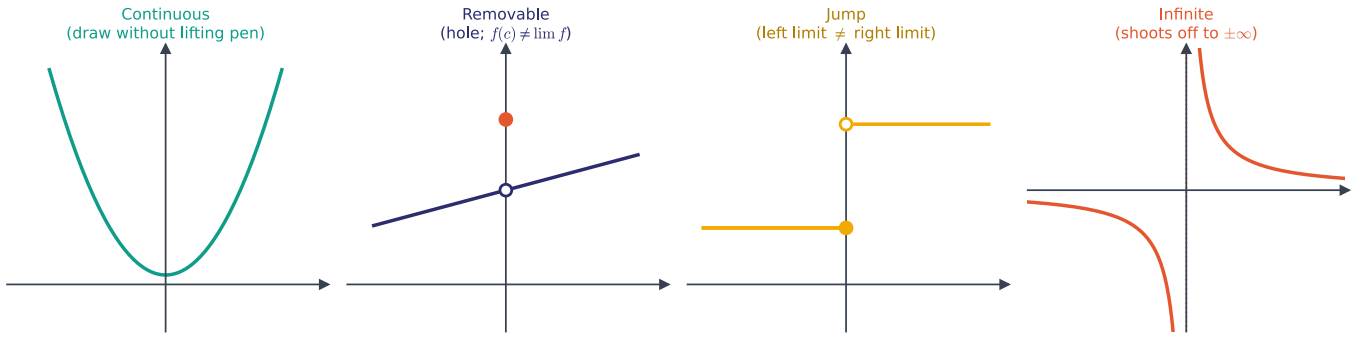
A function $f$ is **continuous at** $x = c$ if all three hold:

1. $f(c)$ is defined;
2. $\lim_{x \to c} f(x)$ exists (left limit = right limit);
3. $\lim_{x \to c} f(x) = f(c)$.

In one line: $\text{LHL} = \text{RHL} = f(c)$. If this fails, $f$ is **discontinuous** at $c$.

A function is **continuous on an interval** if it is continuous at every point of it. At an end point $a$ of $[a, b]$ we use the one-sided limit: $\lim_{x \to a^+} f(x) = f(a)$, and $\lim_{x \to b^-} f(x) = f(b)$.

## 10.2 The kinds of discontinuity




Four behaviours at a point. The first is the only one that passes the pen test.

| Type | What happens | Fixable? |
|---|---|---|
| **Removable** | limit exists but $\neq f(c)$, or $f(c)$ missing | Yes: redefine $f(c)$ to equal the limit |
| **Jump** | left limit $\neq$ right limit (both finite) | No |
| **Infinite** | a one-sided limit is $\pm\infty$ | No |

## 10.3 Examples from your textbook (worked)

**EXAMPLE** **Example 10.1: continuous at a point**

$f(x) = 2x + 3$ at $x = 1$: $f(1) = 5$ and $\lim_{x \to 1}(2x + 3) = 5$. Equal, so continuous. Similarly $x^2$ is continuous at 0.

**EXAMPLE** **Example 10.2: $|x|$ at $0$**

$|x| = -x$ for $x < 0$ and $x$ for $x \geq 0$. LHL $= \lim_{x\to 0^-}(-x) = 0$, RHL $= \lim_{x\to 0^+} x = 0$, and $f(0) = 0$. All three agree: **continuous**.

**EXAMPLE** **Example 10.3: a removable discontinuity**

$f(x) = x^3 + 3$ for $x \neq 0$, and $f(0) = 1$. Then $\lim_{x\to 0} f(x) = 3 \neq f(0) = 1$. **Discontinuous** at $0$ (and only there).

**EXAMPLE** **Example 10.4: a jump**

$$f(x) = \begin{cases} x + 2, & x \leq 1 \\ x - 2, & x > 1 \end{cases}.$$

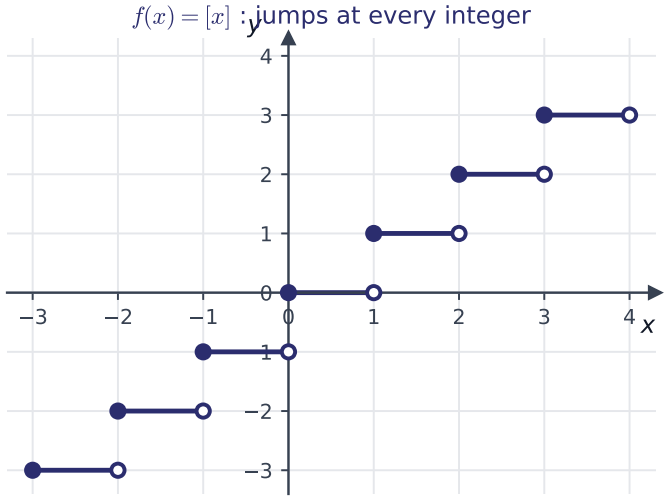
Away from $x = 1$ both pieces are polynomials, hence continuous. At $x = 1$: LHL $= 1 + 2 = 3$, RHL $= 1 - 2 = -1$. They differ, so $f$ has a **jump discontinuity at** $x = 1$ (the only one). Changing the value of $f(1)$ to $0$ (as in your textbook's next example) does not repair it.

**EXAMPLE** **Example 10.5: stitched nicely**

$f(x) = \begin{cases} x, & x \geq 0 \\ x^2, & x < 0 \end{cases}$. At 0: RHL $= 0$, LHL $= 0^2 = 0$, $f(0) = 0$. **Continuous everywhere.**

**WATCH OUT** **Is $f(x) = 1/x$ a continuous function?**

Yes! At every point of its **domain** ($x \neq 0$) it is continuous. $x = 0$ is not in the domain, so there is nothing to be continuous *at*. The graph has a break, but not a *point of discontinuity* in the technical sense. (Your textbook, Example 9, makes exactly this point.) By contrast, the **greatest integer function** below is defined at every real number and *is* discontinuous at every integer.




The greatest-integer (floor) function $[x]$ is discontinuous at every integer. Filled dot: included; open dot: excluded.

## 10.4 Which functions are continuous?

**THEOREM** **Algebra of continuous functions**

If $f$ and $g$ are continuous at $c$, so are $f + g$, $f - g$, $f \cdot g$, $kf$, and $f/g$ (provided $g(c) \neq 0$). If $g$ is continuous at $c$ and $f$ is continuous at $g(c)$, then the **composition** $f \circ g$ is continuous at $c$.

Using this, the following are continuous **wherever they are defined**: polynomials, rational functions, $\sin x$, $\cos x$, $\tan x$ (except odd multiples of $\pi/2$), $e^x$, $\ln x$ ($x>0$), $|x|$, $x^{1/n}$, inverse trigonometric functions, and any sum, product, quotient or composition of these. So $\sin(x^2)$ and $|1-x+|x||$ are continuous everywhere with no further work.

## 10.5 Finding unknown constants

The standard exam question: "*find $k$ so that $f$ is continuous at $c$.*" The method never changes:

**EXAM TIP** **Three-line method**

1. Compute the LHL and RHL at the break point (they will contain $k$).
2. Compute $f(c)$.
3. Set LHL $=$ RHL $=f(c)$ and solve.

**EXAMPLE** **Worked set (textbook Exercise 5.1)**

**(a)** $f(x)=\begin{cases} ax+1, & x\le 3 \\ bx+3, & x>3 \end{cases}$ continuous at 3: $3a+1=3b+3 \Rightarrow \boxed{3a-3b=2}$.

**(b)** $f(x)=\begin{cases} \dfrac{k\cos x}{\pi-2x}, & x\neq\frac{\pi}{2} \\ 3, & x=\frac{\pi}{2} \end{cases}$. Put $t=\frac{\pi}{2}-x\to 0$: $\cos x=\sin t$, $\pi-2x=2t$, so the limit is $\frac{k}{2}\cdot\frac{\sin t}{t}\to\frac{k}{2}$. Set $\frac{k}{2}=3\Rightarrow$ $\boxed{k=6}$.

**(c)** $f(x)=\begin{cases} kx^2, & x\le 2 \\ 3, & x>2 \end{cases}$: $4k=3\Rightarrow \boxed{k=\frac{3}{4}}$.

**(d)** $f(x)=\begin{cases} kx+1, & x\le \pi \\ \cos x, & x>\pi \end{cases}$: $k\pi+1=-1\Rightarrow \boxed{k=-\frac{2}{\pi}}$.

**(e)** $f(x)=\begin{cases} kx+1, & x\le 5 \\ 3x-5, & x>5 \end{cases}$: $5k+1=10\Rightarrow \boxed{k=\frac{9}{5}}$.

**(f)** $f=\begin{cases} 5, & x\le 2 \\ ax+b, & 2<x<10 \\ 21, & x\ge 10 \end{cases}$ continuous: $2a+b=5$ and $10a+b=21\Rightarrow \boxed{a=2,\ b=1}$.

**(g)** $f(x)=\begin{cases} \lambda(x^2-2x), & x\le 0 \\ 4x+1, & x>0 \end{cases}$. At 0: LHL $=0$, RHL $=1$: **never** equal, so no $\lambda$ makes $f$ continuous at 0. At $x=1$: $f$ is just $4x+1$ near 1, so it is continuous there for every $\lambda$.

**EXAMPLE** **A famous one: $x^2\sin(1/x)$**

$f(x)=\begin{cases} x^2\sin\frac{1}{x}, & x\neq 0 \\ 0, & x=0 \end{cases}$. Since $-x^2\le x^2\sin\frac{1}{x}\le x^2$ and both bounds $\to 0$, the **squeeze theorem** gives $\lim_{x\to 0}f(x)=0=f(0)$. So $f$ is continuous at 0.

## 10.6 Two big theorems about continuous functions

**THEOREM** **Intermediate Value Theorem**

If $f$ is continuous on $[a,b]$ and $N$ lies between $f(a)$ and $f(b)$, then $f(c)=N$ for some $c\in(a,b)$. (*A continuous path from below sea level to above sea level must cross sea level.*) Use: $f(x)=x^3-x-1$ has a root in $(1,2)$ because $f(1)=-1<0<5=f(2)$.

**THEOREM** **Extreme Value Theorem**

A function continuous on a **closed interval** $[a, b]$ attains an absolute maximum and an absolute minimum on it. This guarantee is what makes the max/min method of Chapter 18 work.

**TRY IT** **Try it**

Is $f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$ continuous at 3? If not, what value of $f(3)$ would fix it? *Answer: the limit is $6 \neq 5$, so no. Define $f(3) = 6$.*

UNIT II · CHAPTER 11

# The derivative from scratch

**BIG IDEA** **Big idea**

The derivative measures **how fast something is changing at an instant**. A speedometer shows your derivative of distance with respect to time. On a graph, the derivative is the **slope of the tangent line**. To get it we do something clever: compute the average rate over a short interval, then shrink the interval to nothing.

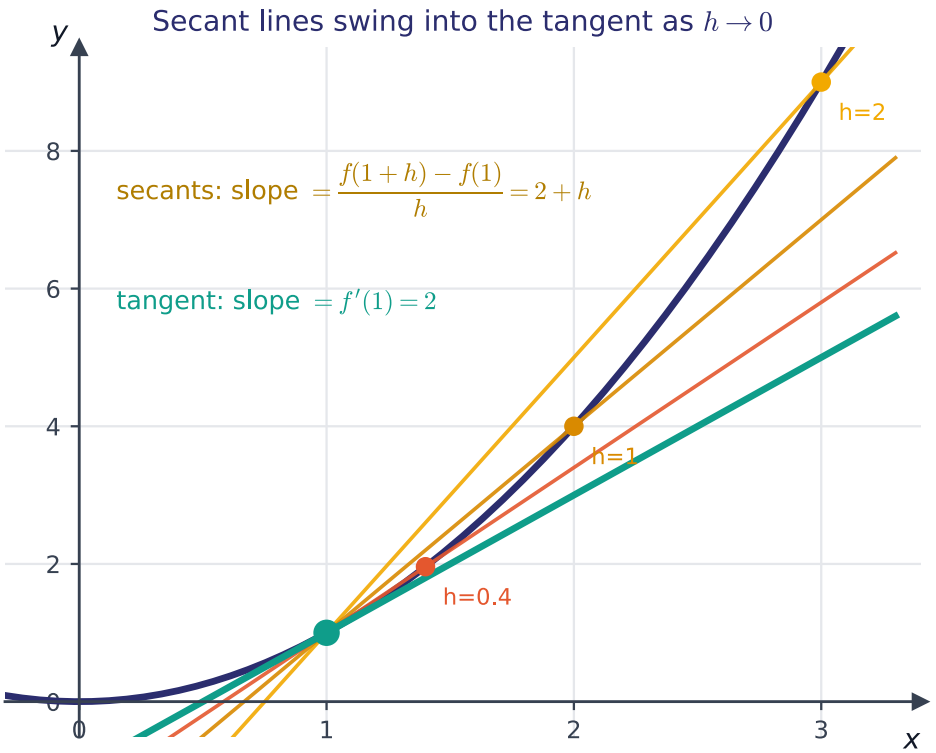
## 11.1 From average speed to instant speed

A ball falls so that its distance after $t$ seconds is $s(t) = t^2$ metres. Its *average* speed between $t = 1$ and $t = 1 + h$ is

$$\frac{s(1+h) - s(1)}{h} = \frac{(1+h)^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h.$$

Make the time gap $h$ tinier and tinier: $h = 0.1$ gives 2.1; $h = 0.001$ gives 2.001. The speed *at the instant* $t = 1$ is the limit: **2 m/s**.

Geometrically, the average speed is the slope of a **secant** line through two points on the graph, and the instant speed is the slope of the **tangent**.




As $h$ shrinks, the secant lines (amber/orange) rotate into the tangent line (teal).

## 11.2 The definition

**DEFINITION** **Derivative**

The derivative of $f$ at $x = c$ is

$$f'(c) = \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} \qquad \left( = \lim_{x\to c} \frac{f(x) - f(c)}{x - c} \right)$$

provided the limit exists (is finite). As a function: $f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$.

**Notation.** $f'(x)$, $\dfrac{dy}{dx}$, $\dfrac{d}{dx}f(x)$, $y'$, $Df$ all mean the same thing. $\left.\dfrac{dy}{dx}\right|_{x=c}$ means "evaluate at $x = c$". **Meaning:** slope of the tangent, rate of change of $y$ with respect to $x$, instantaneous velocity.

## 11.3 Computing derivatives from first principles

**EXAMPLE** **Example 11.1:** $f(x) = x^2$

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \;\to\; 2x.$$

**EXAMPLE** **Example 11.2:** $f(x) = 1/x$

$$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{x - (x+h)}{hx(x+h)} = \frac{-1}{x(x+h)} \to -\frac{1}{x^2}.$$

**EXAMPLE** **Example 11.3:** $f(x) = \sqrt{x}$

Multiply by the conjugate: $\dfrac{\sqrt{x+h} - \sqrt{x}}{h} = \dfrac{1}{\sqrt{x+h} + \sqrt{x}} \to \dfrac{1}{2\sqrt{x}}$.

**EXAMPLE** **Example 11.4:** $f(x) = \sin x$

$$\frac{\sin(x+h) - \sin x}{h} = \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h} \to \sin x \cdot 0 + \cos x \cdot 1 = \cos x.$$

## 11.4 The rules of differentiation

**FORMULA** **General rules**

For differentiable $u(x)$, $v(x)$ and constant $k$:

$$(k)' = 0, \quad (ku)' = ku', \quad (u \pm v)' = u' \pm v'$$

$$\boxed{(uv)' = u'v + uv'} \text{ (product rule)}, \qquad \boxed{\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}} \; (v \neq 0) \text{ (quotient rule)}$$

**MEMORY TRICK** **Quotient rule chant**

"**Low d-high minus high d-low, over the square of what's below.**" (Low = $v$, high = $u$.) In symbols: $\dfrac{vu' - uv'}{v^2}$. The order of subtraction matters.

**PROOF** **Why the product rule is true**

$\dfrac{u(x+h)v(x+h) - u(x)v(x)}{h} = \dfrac{u(x+h) - u(x)}{h} v(x+h) + u(x)\dfrac{v(x+h) - v(x)}{h}$ (add and subtract $u(x)v(x+h)$ in the numerator). Let $h \to 0$: $u'v + uv'$.

**FORMULA** **Table of standard derivatives (memorise all of it)**

| $f(x)$ | $f'(x)$ | $f(x)$ | $f'(x)$ |
|---|---|---|---|
| $x^n$ | $nx^{n-1}$ | $\sin x$ | $\cos x$ |
| $e^x$ | $e^x$ | $\cos x$ | $-\sin x$ |
| $a^x$ | $a^x \ln a$ | $\tan x$ | $\sec^2 x$ |
| $\ln x$ | $\dfrac{1}{x}$ | $\cot x$ | $-\operatorname{cosec}^2 x$ |
| $\log_a x$ | $\dfrac{1}{x \ln a}$ | $\sec x$ | $\sec x \tan x$ |
| $\sqrt{x}$ | $\dfrac{1}{2\sqrt{x}}$ | $\operatorname{cosec} x$ | $-\operatorname{cosec} x \cot x$ |
| $\sin^{-1} x$ | $\dfrac{1}{\sqrt{1-x^2}}$ | $\cos^{-1} x$ | $-\dfrac{1}{\sqrt{1-x^2}}$ |
| $\tan^{-1} x$ | $\dfrac{1}{1+x^2}$ | $\cot^{-1} x$ | $-\dfrac{1}{1+x^2}$ |
| $\sec^{-1} x$ | $\dfrac{1}{\lvert x\rvert \sqrt{x^2-1}}$ | $\operatorname{cosec}^{-1} x$ | $-\dfrac{1}{\lvert x\rvert \sqrt{x^2-1}}$ |

**MEMORY TRICK** **The co- pattern**

Every "co-" function ($\cos, \cot, \operatorname{cosec}, \cos^{-1}, \cot^{-1}, \operatorname{cosec}^{-1}$) has a **minus sign** in its derivative. (Your textbook writes $x$ instead of $|x|$ in the $\sec^{-1}$ and $\operatorname{cosec}^{-1}$ formulas; they agree for $x > 1$.)

**EXAMPLE** **Example 11.5: derive $\tan x$ with the quotient rule**

$$(\tan x)' = \left(\frac{\sin x}{\cos x}\right)' = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$ (Practice MCQ Q1.)

## 11.5 Differentiability

**DEFINITION** **Differentiable**

$f$ is **differentiable at** $c$ if $f'(c)$ exists, i.e. the **left-hand derivative** $\lim_{h \to 0^-} \dfrac{f(c+h)-f(c)}{h}$ and the **right-hand derivative** $\lim_{h \to 0^+} \dfrac{f(c+h)-f(c)}{h}$ both exist, are finite, and are **equal**.

**THEOREM** **Differentiable ⇒ continuous**

If $f$ is differentiable at $c$, then $f$ is continuous at $c$.

**PROOF** **Proof**

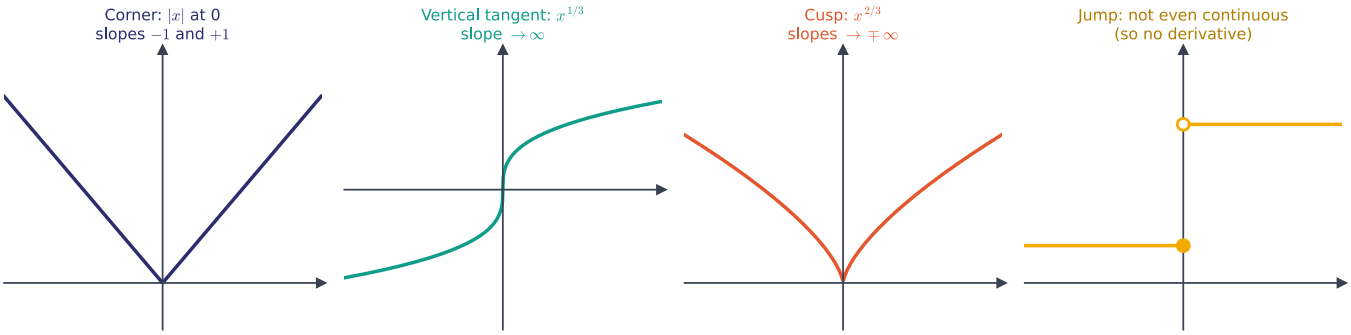
For $x \neq c$: $f(x) - f(c) = \dfrac{f(x)-f(c)}{x-c} \cdot (x-c)$. Let $x \to c$: the first factor tends to $f'(c)$ and the second to $0$, so $f(x) - f(c) \to f'(c) \cdot 0 = 0$. Hence $\lim f(x) = f(c)$. ■

**WATCH OUT** **The converse is FALSE**

Continuous does **not** imply differentiable. $f(x) = |x|$ is continuous at $0$, but

$$\text{left derivative} = \lim_{h \to 0^-} \frac{|h|}{h} = \frac{-h}{h} = -1, \qquad \text{right derivative} = \lim_{h \to 0^+} \frac{h}{h} = +1.$$

They differ, so $|x|$ has a **corner** at $0$ and no derivative there. *Differentiable* is the stronger property; continuity is a necessary but not sufficient condition.




Four ways a function can fail to be differentiable at a point.

**EXAMPLE** **Example 11.6: gluing two pieces**

For $f(x) = \begin{cases} x^2, & x \le 1 \\ ax + b, & x > 1 \end{cases}$ to be differentiable at 1, find $a, b$.

*Continuity*: $1 = a + b$. *Equal slopes*: left derivative $2x|_{x=1} = 2$, right derivative $a$, so $a = 2$ and then $b = -1$. A function is **smooth** (differentiable) at a join only if the values *and* the slopes match.

**EXAMPLE** **Example 11.7: smooth despite $|\,|$**

$f(x) = x\,|x|$ at 0: $\dfrac{f(h) - f(0)}{h} = \dfrac{h\,|h|}{h} = |h| \to 0$. So $f'(0) = 0$: differentiable even though it contains an absolute value. (Absolute value does not automatically kill differentiability; the function must have a corner.)

**EXAMPLE** **Example 11.8 (textbook Ex. 5.2 Q9, Q10)**

$f(x) = |x - 1|$ is not differentiable at $x = 1$ (a corner, same argument as $|x|$). The greatest integer function $[x]$ is not even continuous at $x = 1, 2$, so it cannot be differentiable there.

**TRY IT** **Try it**

Is $f(x) = |x|^3$ differentiable at 0? *Answer: yes.* $\dfrac{f(h) - f(0)}{h} = \dfrac{|h|^3}{h} = h\,|h| \to 0$, so $f'(0) = 0$. *In fact* $f'(x) = 3x\,|x|$ *and* $f''(x) = 6\,|x|$ *exist everywhere, but the third derivative at 0 does not (textbook Misc. Ex. 18).*

UNIT II · CHAPTER 12

# The differentiation toolkit

**BIG IDEA** **Big idea**

With the table of basic derivatives and four rules (sum, product, quotient, **chain**) you can differentiate anything built from familiar functions. This chapter adds the remaining techniques the syllabus names: **implicit**, **inverse trigonometric**, **exponential/logarithmic**, **logarithmic differentiation** and **parametric** differentiation.

## 12.1 The chain rule (functions inside functions)

**THEOREM** **Chain rule**

If $y = f(u)$ and $u = g(x)$ then

$$\boxed{\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}} \qquad \text{i.e.} \qquad \big(f(g(x))\big)' = f'(g(x))\cdot g'(x).$$

**MEMORY TRICK** **The gears picture**

Three gears turn together: gear $x$ drives gear $u$, which drives gear $y$. If $y$ turns $3\times$ as fast as $u$, and $u$ turns $2\times$ as fast as $x$, then $y$ turns $3\times 2 = 6\times$ as fast as $x$. Rates **multiply** along the chain. (*Pretend $du$ cancels in $\frac{dy}{du}\cdot\frac{du}{dx}$: it is a good memory aid, not a proof.*)

**Working outside-in.** Differentiate the **outer** function, leaving the inside alone, then multiply by the derivative of the inside.

**EXAMPLE** **Worked examples**

- $\dfrac{d}{dx}(2x+1)^3 = 3(2x+1)^2\cdot 2 = 6(2x+1)^2.$
- $\dfrac{d}{dx}\sin(x^2) = \cos(x^2)\cdot 2x = 2x\cos x^2.$ (Not $\cos x^2$ alone!)
- $\dfrac{d}{dx}\tan(2x+3) = 2\sec^2(2x+3).$
- $\dfrac{d}{dx}\cos 3x = -3\sin 3x.$
- **Three links:** $\dfrac{d}{dx}\sin\big(\cos(x^2)\big) = \cos\big(\cos x^2\big)\cdot\big(-\sin x^2\big)\cdot 2x = -2x\sin x^2\cos(\cos x^2).$
- $\dfrac{d}{dx}\sqrt{3x+2} = \dfrac{3}{2\sqrt{3x+2}}.$

**WATCH OUT** **Never forget the inside**

The most common lost mark in calculus is forgetting the "$\times g'(x)$". Ask: "*is the inside just $x$?*" If not, multiply by its derivative.

## 12.2 Implicit differentiation

Sometimes $y$ is tangled up with $x$ in an equation like $x^2 + y^2 = r^2$ and solving for $y$ is awkward or impossible. We can still find $\dfrac{dy}{dx}$.

**BIG IDEA** **The trick**

**Differentiate both sides with respect to $x$, treating $y$ as a function of $x$.** Every time you differentiate an expression in $y$, the chain rule leaves a factor $\frac{dy}{dx}$: $\frac{d}{dx}y^n = ny^{n-1}\frac{dy}{dx}$, $\frac{d}{dx}\sin y = \cos y\,\frac{dy}{dx}$. Then collect $\frac{dy}{dx}$ terms and solve.

**EXAMPLE** **Example 12.1: circle**

$x^2 + y^2 = r^2 \Rightarrow 2x + 2y\frac{dy}{dx} = 0 \Rightarrow \boxed{\frac{dy}{dx} = -\frac{x}{y}}$. (Geometrically: the tangent to a circle is perpendicular to the radius, whose slope is $y/x$.) Also $xy = c^2 \Rightarrow y + x\,y' = 0 \Rightarrow y' = -\frac{y}{x}$.

**EXAMPLE** **Example 12.2: $y + \sin y = \cos x$**

$y' + \cos y \cdot y' = -\sin x \Rightarrow y'(1 + \cos y) = -\sin x \Rightarrow y' = -\frac{\sin x}{1 + \cos y}$.

**EXAMPLE** **Example 12.3: $x^2 + xy + y^2 = 100$**

$2x + (y + x\,y') + 2y\,y' = 0 \Rightarrow y'(x + 2y) = -(2x + y) \Rightarrow y' = -\frac{2x + y}{x + 2y}$.

**EXAMPLE** **Example 12.4: the folium $x^3 + y^3 = 3axy$**

$3x^2 + 3y^2y' = 3a(y + xy') \Rightarrow y'(y^2 - ax) = ay - x^2 \Rightarrow y' = \frac{ay - x^2}{y^2 - ax}$.

## 12.3 Derivatives of inverse trigonometric functions

**Derivation (this is the whole trick for any inverse function).** Let $y = \sin^{-1} x$, so $x = \sin y$ with $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Differentiate implicitly: $1 = \cos y \cdot y'$, so $y' = \frac{1}{\cos y}$. In that range $\cos y \geq 0$, so $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$. Hence

$$\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1 - x^2}}, \qquad |x| < 1.$$

Likewise $y = \tan^{-1} x \Rightarrow x = \tan y \Rightarrow 1 = \sec^2 y \cdot y' \Rightarrow y' = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$. The full table is in Chapter 11 (and your textbook's Table 5.4).

**MEMORY TRICK** **Why $\cos^{-1} x$ has a minus**

Since $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ is a constant, the two derivatives must cancel: $(\cos^{-1} x)' = -(\sin^{-1} x)'$.

**Simplify first, differentiate second.** Expressions like $\sin^{-1}\frac{2x}{1+x^2}$ look terrifying but collapse with a trig substitution, turning a nightmare into a one-liner.

**FORMULA** **Substitution cheat-sheet**

| Expression | Substitute | Simplifies to | Derivative |
|---|---|---|---|
| $\sin^{-1}\dfrac{2x}{1+x^2}$ $(\lvert x\rvert \le 1)$ | $x=\tan\theta$ | $2\tan^{-1}x$ | $\dfrac{2}{1+x^2}$ |
| $\cos^{-1}\dfrac{1-x^2}{1+x^2}$ $(x\ge 0)$ | $x=\tan\theta$ | $2\tan^{-1}x$ | $\dfrac{2}{1+x^2}$ |
| $\tan^{-1}\dfrac{2x}{1-x^2}$ $(\lvert x\rvert < 1)$ | $x=\tan\theta$ | $2\tan^{-1}x$ | $\dfrac{2}{1+x^2}$ |
| $\tan^{-1}\dfrac{3x-x^3}{1-3x^2}$ $(\lvert x\rvert < \frac{1}{\sqrt{3}})$ | $x=\tan\theta$ | $3\tan^{-1}x$ | $\dfrac{3}{1+x^2}$ |
| $\sin^{-1}\left(2x\sqrt{1-x^2}\right)$ $(\lvert x\rvert \le \frac{1}{\sqrt{2}})$ | $x=\sin\theta$ | $2\sin^{-1}x$ | $\dfrac{2}{\sqrt{1-x^2}}$ |
| $\cos^{-1}(2x^2-1)$ $(0\le x\le 1)$ | $x=\cos\theta$ | $2\cos^{-1}x$ | $-\dfrac{2}{\sqrt{1-x^2}}$ |
| $\sin^{-1}(3x-4x^3)$ $(\lvert x\rvert \le \frac{1}{2})$ | $x=\sin\theta$ | $3\sin^{-1}x$ | $\dfrac{3}{\sqrt{1-x^2}}$ |

**MEMORY TRICK** **Which substitution?**

See $1+x^2$ or $1-x^2$ in a fraction: try $x=\tan\theta$ (matches $\sin 2\theta, \cos 2\theta, \tan 2\theta$ identities from Chapter 0). See $\sqrt{1-x^2}$: try $x=\sin\theta$ or $\cos\theta$.

**EXAMPLE** **Example 12.5 (textbook Ex. 45)**

i. $\cos^{-1}(\sin x)=\cos^{-1}\cos(\frac{\pi}{2}-x)=\frac{\pi}{2}-x$ (for $x\in[-\frac{\pi}{2},\frac{\pi}{2}]$), so the derivative is $-1$.

ii. $\tan^{-1}\dfrac{\sin x}{1+\cos x}=\tan^{-1}\dfrac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}=\tan^{-1}\tan\frac{x}{2}=\dfrac{x}{2}$, so the derivative is $\dfrac{1}{2}$.

## 12.4 Exponential and logarithmic functions

$$\frac{d}{dx}e^x=e^x,\qquad \frac{d}{dx}\ln x=\frac{1}{x}\ (x>0),\qquad \frac{d}{dx}\ln|x|=\frac{1}{x}\ (x\neq 0),\qquad \frac{d}{dx}a^x=a^x\ln a.$$

With the chain rule: $\dfrac{d}{dx}e^{g(x)}=g'(x)e^{g(x)}$ and $\dfrac{d}{dx}\ln g(x)=\dfrac{g'(x)}{g(x)}$.

**EXAMPLE** **Example 12.6 (textbook Ex. 29)**

- $\dfrac{d}{dx}e^{-x}=-e^{-x}$
- $\dfrac{d}{dx}e^{\cos x}=-\sin x\,e^{\cos x}$
- $\dfrac{d}{dx}\sin(\ln x)=\dfrac{\cos(\ln x)}{x}$
- $\dfrac{d}{dx}\cos^{-1}(e^x)=\dfrac{-e^x}{\sqrt{1-e^{2x}}}$
- $\dfrac{d}{dx}\ln(x^3)=\dfrac{3}{x}$ (simplify first: $\ln x^3=3\ln x$)
- $\dfrac{d}{dx}\ln(\sin x)=\cot x$

$\dfrac{d}{dx}\log_7(\ln x)$: change base to get $\dfrac{\ln(\ln x)}{\ln 7}$, whose derivative is $\dfrac{1}{x\ln x\ln 7}$.

## 12.5 Logarithmic differentiation

Use it for **(a)** a variable raised to a variable power, like $x^x$ or $(\sin x)^{\cos x}$, or **(b)** a huge product/quotient/power. Taking ln turns products into sums, quotients into differences and powers into multiples.

**BIG IDEA** **Recipe**

1. Take ln of both sides and simplify with log laws.
2. Differentiate both sides **implicitly** ($\ln y \to \frac{1}{y}\frac{dy}{dx}$).
3. Multiply by $y$ and substitute the original $y$ back. (The function and all bases must be **positive**, otherwise the logs do not exist.)

**FORMULA** **General power-exponential formula**

For $y = [u(x)]^{v(x)}$ with $u > 0$:

$$\frac{dy}{dx} = y\left[\frac{v\,u'}{u} + v'\ln u\right]$$

(= "treat the exponent as constant" **plus** "treat the base as constant".)

**EXAMPLE** **Example 12.7:** $y = x^x$

$\ln y = x\ln x \Rightarrow \frac{y'}{y} = \ln x + 1 \Rightarrow \boxed{y' = x^x(1+\ln x)}$. (Practice MCQ: option (C).) The tempting answer $x \cdot x^{x-1}$ treats the exponent as constant and is **wrong**.

**EXAMPLE** **Example 12.8:** $y = x^{\sin x}$

$\ln y = \sin x \ln x \Rightarrow \frac{y'}{y} = \cos x \ln x + \frac{\sin x}{x}$, so $y' = x^{\sin x}\left[\frac{\sin x}{x} + \cos x \ln x\right]$.

**EXAMPLE** **Example 12.9: a big quotient**

$y = \sqrt{\dfrac{(x-3)(x^2+4)}{3x^2+4x+5}}$. Then $\ln y = \frac{1}{2}\big[\ln(x-3) + \ln(x^2+4) - \ln(3x^2+4x+5)\big]$ and

$$y' = \frac{y}{2}\left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5}\right].$$

**EXAMPLE** **Example 12.10:** $y^x = x^y$

$x\ln y = y\ln x \Rightarrow \ln y + \dfrac{x}{y}y' = y'\ln x + \dfrac{y}{x} \Rightarrow y'\left(\dfrac{x}{y} - \ln x\right) = \dfrac{y}{x} - \ln y$, so $y' = \dfrac{y(y - x\ln y)}{x(x - y\ln x)}$.

**TRY IT** **Try it**

Differentiate $y = (\sin x)^{\sin x}$. *Answer:* $y' = (\sin x)^{\sin x}\cos x\,(1 + \ln\sin x)$.

## 12.6 Parametric differentiation

Sometimes $x$ and $y$ are both given in terms of a third variable $t$ (the **parameter**), e.g. the path of a moving point. Then by the chain rule $\dfrac{dy}{dt} = \dfrac{dy}{dx}\cdot\dfrac{dx}{dt}$, giving

**FORMULA** **Parametric derivative**

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)} \qquad \text{for } x = f(t),\ y = g(t),\ f'(t) \neq 0.$$

The answer is naturally in terms of $t$. You do **not** need to eliminate $t$.

**EXAMPLE** **Worked examples (textbook Ex. 34–37)**

- $x = at^2,\ y = 2at$: $\dfrac{dy}{dx} = \dfrac{2a}{2at} = \dfrac{1}{t}$.

- $x = a\cos\theta,\ y = a\sin\theta$: $\dfrac{dy}{dx} = \dfrac{a\cos\theta}{-a\sin\theta} = -\cot\theta$.
- $x = a(\theta + \sin\theta),\ y = a(1 - \cos\theta)$: $\dfrac{dy}{dx} = \dfrac{a\sin\theta}{a(1+\cos\theta)} = \tan\dfrac{\theta}{2}$.
- Astroid $x = a\cos^3\theta,\ y = a\sin^3\theta$: $\dfrac{dy}{dx} = \dfrac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta} = -\tan\theta$ (which equals $-\sqrt[3]{y/x}$, since $x^{2/3} + y^{2/3} = a^{2/3}$ ). Eliminating $\theta$ first would have been very tedious.

**Derivative with respect to another function.** To find $\dfrac{du}{dv}$ where both are functions of $x$: $\dfrac{du}{dv} = \dfrac{du/dx}{dv/dx}$. Example: differentiate $\sin^2 x$ w.r.t. $e^{\cos x}$: $\dfrac{2\sin x\cos x}{-\sin x\, e^{\cos x}} = -\dfrac{2\cos x}{e^{\cos x}}$.

**EXAM TIP** **Pick the right tool**

| You see… | Use |
|---|---|
| a function of a function | chain rule |
| $x$ and $y$ mixed in one equation | implicit differentiation |
| variable$^{\text{variable}}$, or a big product/quotient/power | logarithmic differentiation |
| $x = f(t), y = g(t)$ | parametric formula |
| an inverse trig of a messy fraction | trig substitution first |

UNIT II · CHAPTER 13

# Higher-order derivatives

**BIG IDEA** **Big idea**

Differentiate again and you get the **rate of change of the rate of change**. If $s(t)$ is position, $s'$ is velocity and $s''$ is **acceleration**. On a graph, $f''$ tells you how the slope itself is changing: whether the curve bends up (like a cup) or down (like a cap).

## 13.1 Second-order derivatives

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = f''(x) = y'' = y_2.$$

Higher orders: $y_3,\ y_4,\ \ldots,\ y_n = \dfrac{d^n y}{dx^n} = f^{(n)}(x)$.

**Quick values (all appear in your practice set):** $(x^4)'' = 12x^2,\ (e^{3x})'' = 9e^{3x},\ (\ln x)'' = -\dfrac{1}{x^2},\ (a^x)'' = a^x(\ln a)^2$.

**EXAMPLE** **Example 13.1: prove a differential equation**

If $y = A\sin x + B\cos x$ then $y' = A\cos x - B\sin x$, $y'' = -A\sin x - B\cos x = -y$. Hence $y'' + y = 0$. (Pattern: the second derivative of sin and cos returns the function with a minus sign.)

**EXAMPLE** **Example 13.2**

If $y = 3e^{2x} + 2e^{3x}$ then $y' = 6e^{2x} + 6e^{3x}$, $y'' = 12e^{2x} + 18e^{3x}$, and $y'' - 5y' + 6y = (12 - 30 + 18)e^{2x} + (18 - 30 + 12)e^{3x} = 0$.

**EXAMPLE** **Example 13.3: inverse trig**

If $y = \sin^{-1} x$, show $(1 - x^2)y_2 - xy_1 = 0$.

$y_1 = \dfrac{1}{\sqrt{1 - x^2}}$, so $(1 - x^2)y_1^2 = 1$. Differentiate: $(1 - x^2) \cdot 2y_1y_2 + y_1^2(-2x) = 0$. Divide by $2y_1$: $(1 - x^2)y_2 - xy_1 = 0$. ✓ *(Squaring first avoids differentiating a square root.)*

**TRY IT** **Try it**

If $y = (\tan^{-1} x)^2$, show $(x^2 + 1)^2 y_2 + 2x(x^2 + 1)y_1 = 2$. (*Hint:* $y_1 = \dfrac{2\tan^{-1} x}{1 + x^2}$, so $(1 + x^2)y_1 = 2\tan^{-1} x$; *differentiate again.*)

**Parametric second derivative.** $\dfrac{d^2y}{dx^2} = \dfrac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$. (Note: it is **not** $\frac{d^2y/dt^2}{d^2x/dt^2}$.)

**EXAMPLE** **Example 13.4: the cycloid**

$x = a(t - \sin t),\ y = a(1 - \cos t)$. Then $\dfrac{dy}{dx} = \dfrac{\sin t}{1 - \cos t} = \cot\dfrac{t}{2}$, so $\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) = -\dfrac{1}{2}\operatorname{cosec}^2\dfrac{t}{2}$, and dividing by $\dfrac{dx}{dt} = 2a\sin^2\dfrac{t}{2}$:

$$\frac{d^2y}{dx^2} = -\frac{1}{4a\sin^4(t/2)}.$$

## 13.2 The $n$-th derivative: patterns

For a few families the $n$-th derivative has a closed formula. Memorise these; each is checked by differentiating once or twice and spotting the pattern.

**FORMULA** **Standard $n$-th derivatives ($D = \frac{d}{dx}$)**

$$D^n x^m = m(m-1)\cdots(m-n+1)\,x^{m-n} \quad \Big(\text{so } D^n x^n = n!\Big), \qquad D^n e^{ax} = a^n e^{ax}, \qquad D^n a^x = (\ln a)^n a^x$$

$$D^n \sin(ax+b) = a^n \sin\Big(ax+b+\frac{n\pi}{2}\Big), \qquad D^n \cos(ax+b) = a^n \cos\Big(ax+b+\frac{n\pi}{2}\Big)$$

$$D^n \frac{1}{ax+b} = \frac{(-1)^n n!\,a^n}{(ax+b)^{n+1}}, \qquad D^n \ln(ax+b) = \frac{(-1)^{n-1}(n-1)!\,a^n}{(ax+b)^n}$$

$$D^n\big[e^{ax}\sin(bx+c)\big] = (a^2+b^2)^{n/2} e^{ax}\sin(bx+c+n\varphi), \qquad \varphi = \tan^{-1}\frac{b}{a}$$

**Rational functions:** split into partial fractions first, then use the $\frac{1}{ax+b}$ formula.

**EXAMPLE** **Example 13.5**

$$y = \frac{1}{x^2-1} = \frac{1}{2}\left[\frac{1}{x-1} - \frac{1}{x+1}\right], \text{ so } y_n = \frac{(-1)^n n!}{2}\left[\frac{1}{(x-1)^{n+1}} - \frac{1}{(x+1)^{n+1}}\right].$$

## 13.3 Leibniz's formula for a product

**THEOREM** **Leibniz formula**

$$(uv)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} u^{(n-k)} v^{(k)} = u^{(n)}v + n u^{(n-1)}v' + \binom{n}{2} u^{(n-2)}v'' + \cdots + uv^{(n)}$$

It is the binomial theorem in disguise. Choose $v$ to be the factor that **dies quickly** (a polynomial), so only a few terms survive.

**EXAMPLE** **Example 13.6**

$D^n(x^2e^x)$: take $v = x^2$ (derivatives $2x,\ 2,\ 0, \ldots$) and $u = e^x$ ($u^{(k)} = e^x$):

$$D^n(x^2e^x) = e^x\big[x^2 + 2nx + n(n-1)\big].$$

**EXAMPLE** **Example 13.7 (Jain, Ex. 1.15)**

The 4th derivative of $e^{ax}\sin bx$ at $x = 0$. By the $n$-th derivative formula, $D^4 = (a^2+b^2)^2 e^{ax}\sin(bx+4\varphi)$; at 0 this is $(a^2+b^2)^2\sin 4\varphi = \mathbf{4ab(a^2-b^2)}$, using $\sin 4\varphi = 2\sin 2\varphi\cos 2\varphi$ with $\sin\varphi = \frac{b}{\sqrt{a^2+b^2}}$.

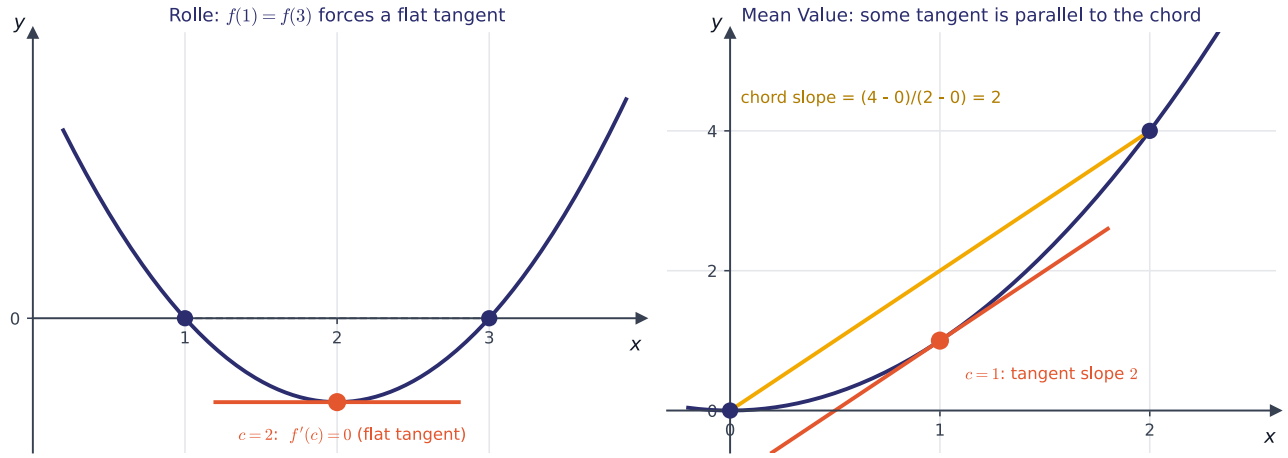
UNIT II · CHAPTER 14

# Rolle's theorem and the Mean Value theorems

**BIG IDEA** **Big idea**

Two of the most useful sentences in calculus, each with a one-line everyday meaning:

- **Rolle:** if you leave home and later return to the same spot, then at some moment your velocity was exactly **zero** (when you turned around).
- **Mean Value:** if your *average* speed over a trip was 80 km/h, then at some moment your *speedometer* read exactly 80 km/h.




Left: Rolle's theorem. Right: the Mean Value Theorem. In both, a tangent line is parallel to a chord (flat in the Rolle case).

**STORY TIME** **The speed-camera theorem**

*Two cameras 10 km apart on a motorway clock a car at 12:00 and 12:05. That is 10 km in 5 minutes: an average of 120 km/h. The driver cannot argue "I was never at 120": by the Mean Value Theorem there was an instant when the speedometer read exactly 120. (Real speed cameras use exactly this logic.)*

## 14.1 Rolle's theorem

**THEOREM** **Rolle's theorem**

Let $f : [a, b] \to \mathbb{R}$ satisfy

1. $f$ is **continuous** on the closed interval $[a, b]$,
2. $f$ is **differentiable** on the open interval $(a, b)$,
3. $f(a) = f(b)$.

Then there is **at least one** $c \in (a, b)$ with $f'(c) = 0$.

**Why is it true?** A continuous function on $[a, b]$ has a highest and a lowest point (Extreme Value Theorem). If both occur at the end points, the function is constant ($f' = 0$ everywhere). Otherwise one occurs *inside*, at some $c$, where the graph has a hilltop or valley, so the tangent is flat: $f'(c) = 0$.

**WATCH OUT** **Each condition matters**

- **Not continuous:** $f(x) = x$ for $0 \le x < 1$ and $f(1) = 0$ has $f(0) = f(1)$ and is differentiable on $(0,1)$, but $f'(x) = 1$ never vanishes. The jump at $x = 1$ breaks Rolle.
- **Not differentiable:** $f(x) = |x|$ on $[-1,1]$ has $f(-1) = f(1)$ but a corner at 0 and $f'(x) = \pm 1$ never 0. Rolle **fails** (practice MCQ: the reason is that $f$ is *not differentiable at* $x = 0$).
- $f(a) \neq f(b)$**:** $f(x) = x^2 - 1$ on $[1,2]$ has $f'(x) = 2x \neq 0$ in $(1,2)$.

Also, Rolle gives a **sufficient** condition. Its converse is false: $f'(c) = 0$ can happen when the hypotheses fail.

**EXAMPLE** **Example 14.1: verifying Rolle**

$f(x) = x^2 - 4x + 3$ on $[1,3]$: polynomial, so continuous and differentiable; $f(1) = 0 = f(3)$. $f'(x) = 2x - 4 = 0 \Rightarrow c = 2 \in (1,3)$ ✓. Similarly $f(x) = x^2 + 2x - 8$ on $[-4,2]$ ($f(-4) = f(2) = 0$): $f'(c) = 2c + 2 = 0 \Rightarrow c = -1$ ✓.

**EXAMPLE** **Example 14.2: using Rolle to locate roots**

If $p(x) = (x-1)(x-2)(x-3)$ then $p(1) = p(2) = p(3) = 0$, so Rolle on $[1,2]$ and $[2,3]$ shows that $p'(x) = 0$ has a root in $(1,2)$ **and** one in $(2,3)$. Since $p'$ is a quadratic, these are *all* its roots.

*Contrapositive form:* if $f$ is differentiable and $f'$ **never vanishes** on $[-5,5]$, then $f(-5) \neq f(5)$ (textbook Ex. 5.8 Q3).

## 14.2 Lagrange's Mean Value Theorem

**THEOREM** **Mean Value Theorem (MVT)**

If $f$ is continuous on $[a,b]$ and differentiable on $(a,b)$, there exists $c \in (a,b)$ with

$$\boxed{f'(c) = \frac{f(b) - f(a)}{b - a}}.$$

Geometrically: some tangent is **parallel to the chord** joining $(a, f(a))$ and $(b, f(b))$. When $f(a) = f(b)$ the chord is horizontal and we recover Rolle's theorem.

**PROOF** **Proof (by reducing to Rolle)**

Define $F(x) = f(x) - \dfrac{f(b) - f(a)}{b - a}\,x$. Then $F$ is continuous on $[a,b]$, differentiable on $(a,b)$, and one checks $F(a) = F(b)$. By Rolle there is $c$ with $F'(c) = 0$, i.e. $f'(c) = \dfrac{f(b) - f(a)}{b - a}$. ■

**EXAMPLE** **Example 14.3: finding $c$**

- $f(x) = x^2$ on $[0,2]$: slope of chord $= \frac{4-0}{2} = 2 = 2c \Rightarrow c = 1$. On $[2,4]$: $\frac{16-4}{2} = 6 = 2c \Rightarrow c = 3$.
- $f(x) = x^2 - 4x - 3$ on $[1,4]$: chord slope $= \frac{-3-(-6)}{3} = 1$; $f'(c) = 2c - 4 = 1 \Rightarrow c = \frac{5}{2}$.
- $f(x) = x^3 - 5x^2 - 3x$ on $[1,3]$: chord slope $= \frac{-27-(-7)}{2} = -10$; $3c^2 - 10c - 3 = -10 \Rightarrow 3c^2 - 10c + 7 = 0 \Rightarrow c = 1$ or $\frac{7}{3}$. Only $c = \frac{7}{3}$ lies in the **open** interval $(1,3)$.
- $f(x) = \ln x$ on $[1,e]$: $\frac{1-0}{e-1} = \frac{1}{c} \Rightarrow c = e - 1 \approx 1.72$.

**What MVT is used for.** MVT is the engine behind many results:

- If $f'(x) = 0$ on an interval then $f$ is **constant** there (every chord has slope 0).
- If $f' > 0$ then $f$ is **increasing**; if $f' < 0$, decreasing (the basis of Chapter 17).
- **Inequalities and error bounds.**

**EXAMPLE** **Example 14.4: proving inequalities with MVT**

i. $|\cos b - \cos a| \le |b - a|$: $\dfrac{\cos b - \cos a}{b - a} = -\sin c$ and $|\sin c| \le 1$.

ii. $e^x > 1 + x$ for $x > 0$: $\dfrac{e^x - 1}{x} = e^c > 1$ for some $c \in (0, x)$.

iii. $\ln(1 + x) < x$ for $x > 0$: $\dfrac{\ln(1+x) - 0}{x} = \dfrac{1}{1+c} < 1$.

iv. *Bounds on a value* (Jain Ex. 1.5): if $f'(x) = \dfrac{1}{3 - x^2}$ and $f(0) = 1$, then $f(1) - 1 = f'(c) \in \left[\frac{1}{3}, \frac{1}{2}\right]$ because $\frac{1}{3-x^2}$ ranges over $\left[\frac{1}{3}, \frac{1}{2}\right]$ on $[0, 1]$. Hence $\frac{4}{3} \le f(1) \le \frac{3}{2}$.

## 14.3 Cauchy's Mean Value Theorem

**THEOREM** **Cauchy MVT**

If $f, g$ are continuous on $[a, b]$, differentiable on $(a, b)$, and $g'(x) \neq 0$ on $(a, b)$, then for some $c \in (a, b)$

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

This is the MVT for a curve given *parametrically* $(x, y) = (g(t), f(t))$: some tangent is parallel to the chord. With $g(x) = x$ it becomes Lagrange's MVT. **Do not** try to prove it by applying Lagrange separately to numerator and denominator: that gives two *different* points $c_1 \neq c_2$.

**EXAMPLE** **Example 14.5**

$f(x) = x^2$, $g(x) = x^3$ on $[1, 2]$: $\dfrac{4 - 1}{8 - 1} = \dfrac{3}{7} = \dfrac{2c}{3c^2} = \dfrac{2}{3c} \Rightarrow c = \dfrac{14}{9} \in (1, 2)$ ✓.

**EXAM TIP** **Exam checklist for any Rolle/MVT question**

1. State and **verify the three conditions** (continuity on $[a, b]$, differentiability on $(a, b)$, plus $f(a) = f(b)$ for Rolle).
2. Compute $f'$ and solve for $c$ (Rolle: $f'(c) = 0$; MVT: $f'(c) = \frac{f(b)-f(a)}{b-a}$).
3. **Reject** any $c$ outside the *open* interval $(a, b)$.

UNIT II · CHAPTER 15

# Taylor's and Maclaurin's theorems

**BIG IDEA** **Big idea**

Polynomials are the easiest functions in the world: you can evaluate them with just $+, -, \times$. Taylor's theorem says that **any smooth function behaves like a polynomial near a point**, and tells you exactly which polynomial and how big the error is. That is how calculators compute sin, cos, $e^x$ and ln.

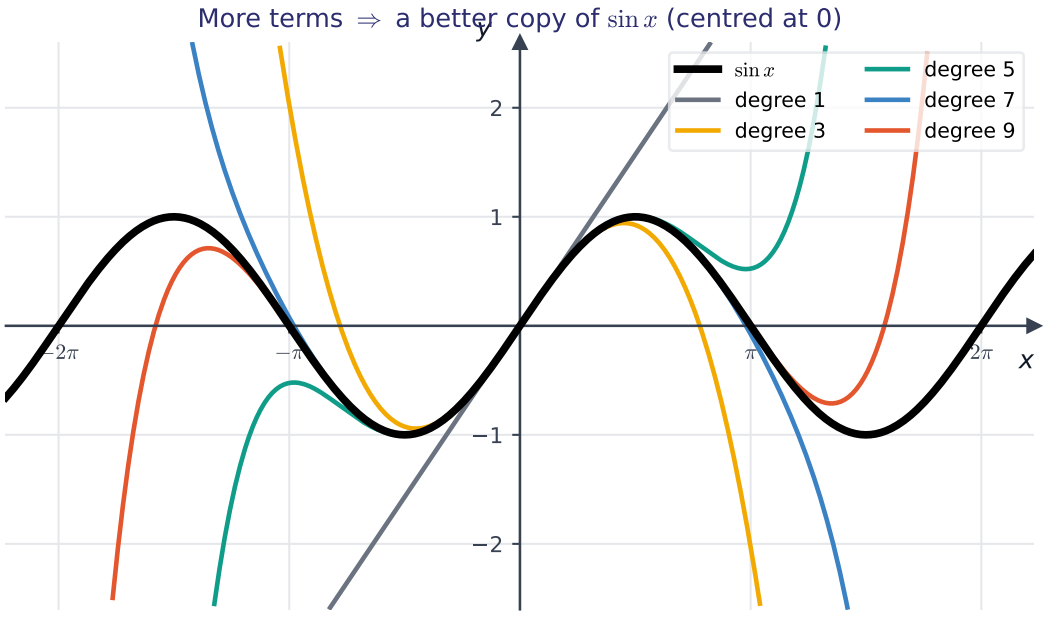
## 15.1 Building the polynomial

Near $x = a$ the tangent line approximates $f$: $f(x) \approx f(a) + f'(a)(x - a)$. It matches the **value** and the **slope** at $a$. To hug the curve better, also match the **curvature** ($f''$), then $f'''$, and so on. Demand that a degree-$n$ polynomial

$$P_n(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots + c_n(x - a)^n$$

has the same value and the first $n$ derivatives as $f$ at $x = a$. Differentiating $k$ times and putting $x = a$ gives $P_n^{(k)}(a) = k!\, c_k$. Setting this equal to $f^{(k)}(a)$:

$$\boxed{c_k = \frac{f^{(k)}(a)}{k!}}$$




Taylor polynomials of $\sin x$ about $0$. Each extra term adds a wiggle and the copy stays accurate for longer.

## 15.2 Taylor's theorem

**THEOREM** **Taylor's theorem with Lagrange's remainder (your professor's sheet)**

Let $f$ and its first $(n-1)$ derivatives be continuous on $[a, a+h]$, and let $f^{(n)}$ exist on $(a, a+h)$. Then there is a number $\theta$ with $0 < \theta < 1$ such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \underbrace{\frac{h^n}{n!}f^{(n)}(a+\theta h)}_{\text{Lagrange remainder } R_n}$$

Writing $a + h = x$, this reads

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a+\theta h).$$

**FORMULA** **The same theorem in Jain's notation (remainder after the degree-$n$ term)**

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x), \qquad R_n(x) = \frac{(x-a)^{n+1}}{(n+1)!}f^{(n+1)}(c), \quad c \text{ between } a \text{ and } x.$$

The two statements are identical; they only differ in whether $n$ counts the terms or the degree. Know both so that you are never confused by a textbook.

**Special cases worth knowing.**

- $n = 1$ gives $f(a+h) = f(a) + hf'(a+\theta h)$: **the Mean Value Theorem**. Taylor's theorem is MVT for higher derivatives.
- **Maclaurin's theorem** is Taylor with $a = 0$, $h = x$:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(\theta x).$$

- If the remainder $R_n \to 0$ as $n \to \infty$, we get the infinite **Taylor series** $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$ and, for $a = 0$, the **Maclaurin series** $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.

**MEMORY TRICK** **The recipe**

**Taylor coefficient of $(x-a)^n$ is $\frac{f^{(n)}(a)}{n!}$** (practice MCQ answer B). Not $f^{(n)}(a)$ alone: the $n!$ cancels the factors that appear when you differentiate $(x-a)^n$ repeatedly.

**Error bound.** Since $c$ is unknown, we *bound* the remainder:

$$|R_n(x)| \le \frac{M\,|x-a|^{n+1}}{(n+1)!}, \qquad M = \max\left|f^{(n+1)}\right| \text{ on the interval.}$$

## 15.3 The standard Maclaurin series

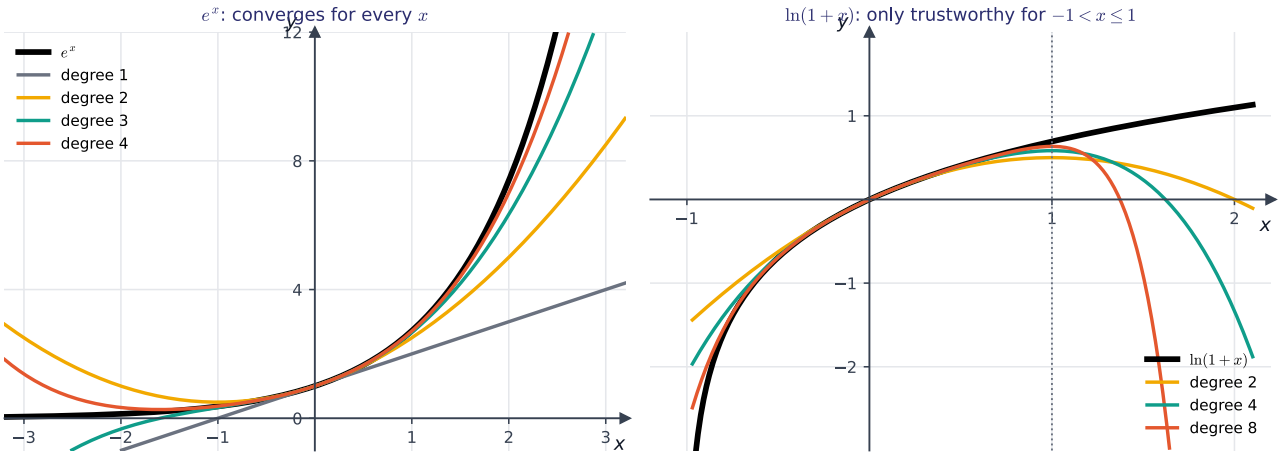
**FORMULA** **Memorise (these are on your professor's sheet)**

| Function | Series | Valid for |
|---|---|---|
| $e^x$ | $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n\geq 0} \frac{x^n}{n!}$ | all $x$ |
| $e^{-x}$ | $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$ | all $x$ |
| $\sin x$ | $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n\geq 0} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ | all $x$ |
| $\cos x$ | $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n\geq 0} (-1)^n \frac{x^{2n}}{(2n)!}$ | all $x$ |
| $\sinh x$ | $x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$ | all $x$ |
| $\cosh x$ | $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ | all $x$ |
| $\tan^{-1} x$ | $x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ | $\lvert x\rvert \leq 1$ |
| $\ln(1+x)$ | $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$ | $-1 < x \leq 1$ |
| $\ln(1-x)$ | $-x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$ | $-1 \leq x < 1$ |
| $(1+x)^n$ | $1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$ | $\lvert x\rvert < 1$ (any $n$) |
| $\frac{1}{1-x}$ | $1 + x + x^2 + x^3 + \cdots$ | $\lvert x\rvert < 1$ |
| $\tan x$ | $x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$ | $\lvert x\rvert < \frac{\pi}{2}$ |

**MEMORY TRICK** **Patterns that answer MCQs instantly**

- $\sin x$ contains only **odd** powers (it is an odd function); $\cos x$ only **even** powers (an even function).
- Signs **alternate** for $\sin, \cos, \ln(1+x), \tan^{-1} x$; they do **not** for $e^x,\ \sinh,\ \cosh,\ \frac{1}{1-x}$.
- $\frac{1}{1-x}$: first three terms $1 + x + x^2$ (practice MCQ answer B). Replace $x$ by $-x$ for $\frac{1}{1+x} = 1 - x + x^2 - \cdots$.

**Derivation of** $\sin x$**.** The derivatives of $\sin x$ at 0 cycle: $0,\ 1,\ 0,\ -1,\ 0,\ 1, \ldots$ Divide by $n!$ and you get $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$. For $e^x$ every derivative is $e^x$, which is 1 at 0, so $c_n = \frac{1}{n!}$.




Left: $e^x$ is captured perfectly by its polynomials for every $x$. Right: $\ln(1+x)$ is only reproduced for $-1 < x \le 1$; outside that range the polynomials fly away.

**STORY TIME** **Euler's masterpiece**

*Put $x = i\theta$ (with $i^2 = -1$) into the series for $e^x$ and separate real and imaginary parts: the real part is exactly the series for $\cos\theta$, the imaginary part is the series for $\sin\theta$. So $e^{i\theta} = \cos\theta + i\sin\theta$, and at $\theta = \pi$ we get the famous $e^{i\pi} + 1 = 0$, which ties together $e,\ i,\ \pi,\ 1,\ 0$.*

## 15.4 Worked examples

**EXAMPLE** **Example 15.1: Taylor polynomial about $a \neq 0$**

Find the Taylor polynomial of degree 3 for $f(x) = \sqrt{x}$ about $a = 1$.

$f = x^{1/2},\ f' = \frac{1}{2}x^{-1/2},\ f'' = -\frac{1}{4}x^{-3/2},\ f''' = \frac{3}{8}x^{-5/2}$. At $x = 1$: $1,\ \frac{1}{2},\ -\frac{1}{4},\ \frac{3}{8}$. Divide by $k!$:

$$P_3(x) = 1 + \tfrac{1}{2}(x-1) - \tfrac{1}{8}(x-1)^2 + \tfrac{1}{16}(x-1)^3.$$

Test: $P_3(1.2) = 1 + 0.1 - 0.005 + 0.0005 = 1.0955$, and $\sqrt{1.2} = 1.09545$ ✓.

**EXAMPLE** **Example 15.2: compute $\sin 29^\circ$**

Convert: $29^\circ = \dfrac{29\pi}{180} \approx 0.506145$ rad. Then $\sin x \approx x - \dfrac{x^3}{6} + \dfrac{x^5}{120} = 0.506145 - 0.021611 + 0.000277 \approx 0.48481$. The true value is $0.48481$. The first omitted term is about $x^7/5040 \approx 1.7 \times 10^{-6}$, so the answer is right to 5 decimal places.

**EXAMPLE** **Example 15.3: how many terms for $e$?**

Use $e^x$ at $x = 1$: $R_n = \dfrac{e^c}{(n+1)!} < \dfrac{3}{(n+1)!}$ since $e^c < e < 3$. For error $< 0.001$ we need $(n+1)! > 3000$, i.e. $n + 1 = 7$, so $n = 6$ (since $7! = 5040$). Indeed $1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \frac{1}{720} = 2.71806$, and $e = 2.71828$.

**EXAMPLE** **Example 15.4: build new series from old ones**

- Replace $x$ by $-x^2$ in $e^x$: $e^{-x^2} = 1 - x^2 + \dfrac{x^4}{2} - \dfrac{x^6}{6} + \cdots$
- **Multiply** series: $e^x \sin x = (1 + x + \frac{x^2}{2} + \frac{x^3}{6})(x - \frac{x^3}{6}) = x + x^2 + \frac{x^3}{3} + \cdots$ (the $x^3$ terms: $\frac{1}{2} - \frac{1}{6} = \frac{1}{3}$).
- Likewise $e^x \cos x = 1 + x - \dfrac{x^3}{3} - \dfrac{x^4}{6} + \cdots$

**EXAMPLE** **Example 15.5: series make limits easy**

$$\lim_{x\to 0}\frac{\sin x - x}{x^3} = \lim \frac{-\frac{x^3}{6} + \frac{x^5}{120} - \cdots}{x^3} = -\frac{1}{6}.$$

$$\lim_{x\to 0}\frac{e^x - 1 - x}{x^2} = \frac{1}{2}, \qquad \lim_{x\to 0}\frac{x - \ln(1+x)}{x^2} = \frac{1}{2}, \qquad \lim_{x\to 0}\frac{\tan x - x}{x^3} = \frac{1}{3}.$$

## 15.5 Summary

**QUICK RECALL** **Taylor in a nutshell**

- **Idea:** match $f$ and its first $n$ derivatives at $a$.
- **Coefficient:** $\dfrac{f^{(k)}(a)}{k!}(x-a)^k$.
- **Error:** $R_n = \dfrac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$; bound it with the largest $\left|f^{(n+1)}\right|$.
- **Maclaurin** = Taylor at $a = 0$. **Taylor** with $n = 1$ = **MVT**.
- **Validity:** the infinite series converges only inside its interval (e.g. $|x| < 1$ for $\ln(1+x)$).

UNIT II · CHAPTER 16

# Indeterminate forms and L'Hôpital's rule

**BIG IDEA** **Big idea**

Some limits look like a tug-of-war with no winner: $\frac{0}{0}$ ("zero over zero") or $\frac{\infty}{\infty}$. Each could be anything: $\frac{x}{x} \to 1$, $\frac{x^2}{x} \to 0$, $\frac{x}{x^2} \to \infty$. They are called **indeterminate forms**. L'Hôpital's rule resolves them by comparing **how fast** numerator and denominator are changing, i.e. by comparing their derivatives.

## 16.1 The seven indeterminate forms

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 1^{\infty}, \quad 0^{0}, \quad \infty^{0}.$$

**WATCH OUT** **Forms that are NOT indeterminate**

$\infty + \infty = \infty$, $\infty \cdot \infty = \infty$, $\infty^{\infty} = \infty$, $0^{\infty} = 0$, $\infty^{-\infty} = 0$, and $\frac{c}{0}$ $(c \neq 0)$ is infinite. L'Hôpital's rule must **not** be used on these (Jain, Remark 4).

## 16.2 L'Hôpital's rule

**THEOREM** **L'Hôpital's rule**

Let $f, g$ be differentiable near $a$ (except possibly at $a$), with $g'(x) \neq 0$ nearby. If $\lim_{x \to a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ **or** $\frac{\pm\infty}{\pm\infty}$, then

$$\boxed{\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}}$$

provided the right-hand limit exists (or is $\pm\infty$). If $\frac{f'}{g'}$ is again indeterminate, apply the rule again. It also works as $x \to \pm\infty$ and for one-sided limits.

**Why it works ($\frac{0}{0}$ case).** If $f(a) = g(a) = 0$ then $\frac{f(x)}{g(x)} = \frac{[f(x) - f(a)]/(x - a)}{[g(x) - g(a)]/(x - a)} \to \frac{f'(a)}{g'(a)}$.

**WATCH OUT** **Three big warnings**

1. **Check the form first.** Apply the rule only to $\frac{0}{0}$ or $\frac{\infty}{\infty}$. (Applying it to a limit like $\frac{1-\cos x}{x}$ is fine, since it is $\frac{0}{0}$.)
2. **Differentiate top and bottom separately.** Do **not** use the quotient rule!
3. If $\lim \frac{f'}{g'}$ does not exist, L'Hôpital says **nothing**. Example: $\lim_{x \to \infty} \frac{x + \sin x}{x}$ is $\frac{\infty}{\infty}$, and $\frac{1 + \cos x}{1}$ has no limit, yet the original limit equals 1 (divide by $x$).

## 16.3 Worked examples (including your practice questions)

**EXAMPLE** **Example 16.1: the practice-set limits**

- $\lim_{x \to 0} \frac{\sin x}{x} = \lim \frac{\cos x}{1} = 1.$ $\quad \lim_{x \to 0} \frac{\tan x}{x} = \lim \frac{\sec^2 x}{1} = 1.$
- $\lim_{x \to 0} \frac{1 - \cos x}{x} = \lim \frac{\sin x}{1} = 0.$
- $\lim_{x \to 1} \frac{\ln x}{x - 1} = \lim \frac{1/x}{1} = 1.$
- $\lim_{x \to \infty} \frac{e^x}{x} = \lim \frac{e^x}{1} = \infty.$

- $\lim_{x\to\infty} xe^{-x} = \lim \dfrac{x}{e^x} = \lim \dfrac{1}{e^x} = 0$ (a $0\cdot\infty$ form rewritten as $\frac{\infty}{\infty}$).

**EXAMPLE** **Example 16.2: repeated application**

$\lim_{x\to 0} \dfrac{e^x - e^{-x} - 2x}{x - \sin x}$ is $\frac{0}{0}$. Differentiate: $\frac{e^x+e^{-x}-2}{1-\cos x}$, still $\frac{0}{0}$. Again: $\frac{e^x-e^{-x}}{\sin x}$, still $\frac{0}{0}$. Again: $\frac{e^x+e^{-x}}{\cos x} \to \frac{2}{1} = \mathbf{2}$.

## 16.4 Reducing the other forms

**FORMULA** **Conversion table**

| Form | What to do |
|---|---|
| $0\cdot\infty$ | write $f\cdot g = \dfrac{f}{1/g}$ to get $\frac{0}{0}$ or $\frac{\infty}{\infty}$ |
| $\infty - \infty$ | combine into one fraction (common denominator) or rationalise |
| $1^\infty,\ 0^0,\ \infty^0$ | let $y = f^g$, take $\ln y = g\ln f$ (a $0\cdot\infty$ form), find its limit $L$, then the answer is $e^L$ |

**EXAMPLE** **Example 16.3: $0^0$**

$\lim_{x\to 0^+} x^x$: $\ln y = x\ln x = \dfrac{\ln x}{1/x} \to \dfrac{1/x}{-1/x^2} = -x \to 0$. So $y \to e^0 = \mathbf{1}$.

**EXAMPLE** **Example 16.4: $\infty - \infty$**

$$\lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{\sin x}\right) = \lim \frac{\sin x - x}{x\sin x} = \lim \frac{\cos x - 1}{\sin x + x\cos x} = \lim \frac{-\sin x}{2\cos x - x\sin x} = \frac{0}{2} = \mathbf{0}.$$

**EXAMPLE** **Example 16.5: $1^\infty$**

$\lim_{x\to\infty}\left(1+\dfrac{1}{x}\right)^x = e$. More generally, $\lim_{x\to\infty}\left(\dfrac{x+4}{x+2}\right)^{x+3}$: write $\frac{x+4}{x+2} = 1 + \frac{2}{x+2}$, so $\ln y = (x+3)\ln\left(1+\frac{2}{x+2}\right) \approx (x+3)\cdot\frac{2}{x+2} \to 2$ and the limit is $\mathbf{e^2}$.

**EXAMPLE** **Example 16.6: more $\infty^0$ and $1^\infty$ type limits**

- $\lim_{x\to\pi/2} (\sin x)^{\tan x}$: $\ln y = \dfrac{\ln\sin x}{\cot x} \to \dfrac{\cot x}{-\operatorname{cosec}^2 x} = -\sin x\cos x \to 0$, so the limit is $e^0 = \mathbf{1}$.
- $\lim_{x\to 0}(\cos x)^{1/x^2} = e^{-1/2}$ because $\dfrac{\ln\cos x}{x^2} \to \dfrac{-\tan x}{2x} \to -\dfrac{1}{2}$.
- $\lim_{x\to\infty} x\tan\dfrac{1}{x} = \lim \dfrac{\tan(1/x)}{1/x} = 1$.

## 16.5 The growth ladder

**FORMULA** **Who wins at infinity?**

$$\ln x \ll x^p\ (p>0) \ll a^x\ (a>1) \ll x! \ll x^x$$

Meaning: $\dfrac{\ln x}{x^p} \to 0$ and $\dfrac{x^p}{a^x} \to 0$ as $x\to\infty$. So any exponential beats any power, and any power beats any logarithm. This is why $\lim \frac{e^x}{x} = \infty$ and $\lim xe^{-x} = 0$. Near 0: $x^n \ln x \to 0$ for every $n > 0$.

**EXAM TIP** **L'Hôpital or Taylor?**

For limits with $\sin$, $\cos$, $e^x$, $\ln(1+x)$ at $0$, expanding each in its series often finishes in one line and avoids repeated differentiation. For limits at infinity or with messy quotients, L'Hôpital is the cleaner tool.

UNIT II · CHAPTER 17

# Applications of derivatives (I)

**BIG IDEA** **Big idea**

Now the derivative earns its keep. In this chapter it answers four practical questions: **How fast is it changing?** (rates), **Is it going up or down?** (monotonicity), **What does the curve look like at this point?** (tangents and normals), and **What is a quick estimate?** (approximations).

## 17.1 Rate of change and related rates

$\frac{dy}{dx}$ is the **rate of change of $y$ with respect to $x$**. If $x$ and $y$ both depend on time $t$, the chain rule links their rates:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \left(\frac{dx}{dt} \neq 0\right), \qquad \frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}.$$

**Sign:** $\frac{dy}{dx} > 0$ if $y$ increases as $x$ increases; $< 0$ if $y$ decreases.

**BIG IDEA** **The 5-step recipe for related-rates problems**

1. **Draw** a picture and name every changing quantity ($x, y, r, V, \ldots$). Treat each as a function of time $t$.
2. **Write one equation** linking them (a formula from geometry, Pythagoras, ...).
3. **Differentiate both sides with respect to** $t$ (chain rule!).
4. **Only now substitute** the numbers for the instant asked about. (Substituting before differentiating freezes the variables and gives nonsense.)
5. Solve and state the answer **with units and sign** (negative = decreasing).

**EXAMPLE** **Example 17.1: a growing ripple**

A stone makes circular ripples whose radius grows at 4 cm/s. How fast is the area increasing when $r = 10$ cm?

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(10)(4) = \mathbf{80\pi\ cm^2/s}$.

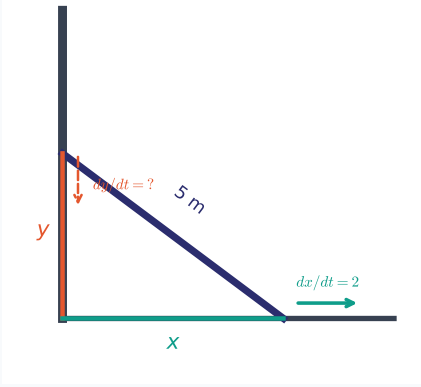
**EXAMPLE** **Example 17.2: the cube**

A cube's volume grows at 9 $\text{cm}^3/\text{s}$. How fast is its surface area growing when the edge is 10 cm?

$V = x^3 \Rightarrow 9 = 3x^2\frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{3}{x^2}$. $S = 6x^2 \Rightarrow \frac{dS}{dt} = 12x\frac{dx}{dt} = \frac{36}{x} = \mathbf{3.6\ cm^2/s}$ at $x = 10$.

**EXAMPLE** **Example 17.3: the sliding ladder**

A 5 m ladder leans on a wall. Its foot is pulled away at 2 cm/s. How fast does its top slide down when the foot is 4 m from the wall?




The ladder: $x^2 + y^2 = 25$.

$x^2 + y^2 = 25 \Rightarrow 2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0 \Rightarrow \dfrac{dy}{dt} = -\dfrac{x}{y}\dfrac{dx}{dt}$. When $x = 4$, $y = 3$: $\dfrac{dy}{dt} = -\dfrac{4}{3}(2) = -\dfrac{8}{3}$. The top **slides down at $\frac{8}{3}$ cm/s**.

**EXAMPLE** **Example 17.4: two more classics**

**Balloon.** A spherical balloon is inflated at 900 $\text{cm}^3/\text{s}$. At $r = 15$ cm: $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$, so $\dfrac{dr}{dt} = \dfrac{900}{4\pi \cdot 225} = \dfrac{1}{\pi}$ cm/s.

**Sand cone.** Sand pours at 12 $\text{cm}^3/\text{s}$ into a cone whose height is always $\frac{1}{6}$ of the base radius, so $r = 6h$. $V = \frac{1}{3}\pi r^2 h = 12\pi h^3$, $\frac{dV}{dt} = 36\pi h^2 \frac{dh}{dt} = 12$. At $h = 4$: $\dfrac{dh}{dt} = \dfrac{12}{36\pi \cdot 16} = \dfrac{1}{48\pi}$ cm/s.

**EXAMPLE** **Example 17.5: rectangle with two changing sides**

Length $x$ decreases at 3 cm/min, width $y$ increases at 2 cm/min. At $x = 10,\ y = 6$: perimeter $P = 2(x + y)$ gives $\dfrac{dP}{dt} = 2(-3 + 2) = -2$ cm/min. Area $A = xy$ gives $\dfrac{dA}{dt} = y\dfrac{dx}{dt} + x\dfrac{dy}{dt} = 6(-3) + 10(2) = \mathbf{2}$ $\text{cm}^2/\text{min}$.

**Marginal cost and revenue (economics).** If $C(x)$ is the total cost of producing $x$ items, the **marginal cost** $MC = \dfrac{dC}{dx}$ is the instantaneous rate of change of cost, approximately the cost of making **one more** item. Likewise the marginal revenue $MR = \dfrac{dR}{dx}$.

**EXAMPLE** **Example 17.6**

$C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$. Then $MC = 0.015x^2 - 0.04x + 30$; at $x = 3$ this is $0.135 - 0.12 + 30 =$ Rs $30.015 \approx 30.02$. For $R(x) = 3x^2 + 36x + 5$: $MR = 6x + 36$, so at $x = 5$, $MR =$ Rs 66.

## 17.2 Increasing and decreasing functions

**DEFINITION** **Monotonic functions**

On an interval $I$, $f$ is **increasing** if $x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)$; **strictly increasing** if $f(x_1) < f(x_2)$. Reversing the inequality gives **decreasing** and **strictly decreasing**. "Increasing" means *the graph rises as you move left to right*.

**THEOREM** **First-derivative test for monotonicity**

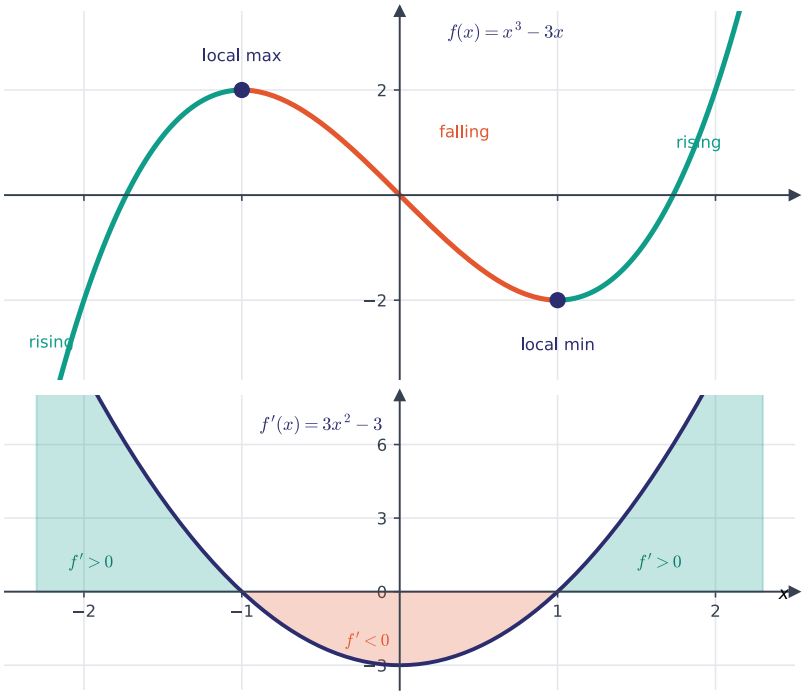
If $f$ is continuous on $[a, b]$ and differentiable on $(a, b)$, then

$$f'(x) > 0\ \forall x \in (a,b) \Rightarrow f \text{ increasing}, \qquad f'(x) < 0 \Rightarrow f \text{ decreasing}, \qquad f'(x) = 0 \Rightarrow f \text{ constant}.$$

(This is MVT: $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$ has the sign of $f'(c)$.)

**BIG IDEA** **The sign-chart method**

1. Compute $f'(x)$.
2. Solve $f'(x) = 0$ (and note where $f'$ is undefined): these **critical points** cut the line into intervals.
3. Pick a test point in each interval and find the **sign of $f'$** (or factorise and multiply signs).
4. Conclude: $+$ means increasing, $-$ means decreasing.
5. Because $f$ is continuous, you may include the endpoint critical points in both neighbouring intervals.




The sign of $f'$ (bottom) controls the direction of $f$ (top).

**EXAMPLE** **Example 17.7: a cubic**

$f(x) = 4x^3 - 6x^2 - 72x + 30$. $f'(x) = 12x^2 - 12x - 72 = 12(x-3)(x+2)$, zero at $x = -2, 3$.

| Interval | $(-\infty, -2)$ | $(-2, 3)$ | $(3, \infty)$ |
|---|---|---|---|
| sign of $f' = 12(x-3)(x+2)$ | $(-)(-) = +$ | $(-)(+) = -$ | $(+)(+) = +$ |
| behaviour of $f$ | increasing | decreasing | increasing |

So $f$ is increasing on $(-\infty, -2]$ and $[3, \infty)$, decreasing on $[-2, 3]$.

**EXAMPLE** **Example 17.8: more from the textbook**

- $f(x) = x^2 - 4x + 6$: $f' = 2x - 4$; decreasing on $(-\infty, 2]$, increasing on $[2, \infty)$.
- $f(x) = x^3 - 3x^2 + 4x$: $f' = 3(x-1)^2 + 1 > 0$ for all $x$, so **strictly increasing on** $\mathbb{R}$.
- $f(x) = \sin 3x$ on $[0, \frac{\pi}{2}]$: $f' = 3\cos 3x$ vanishes at $x = \frac{\pi}{6}$; increasing on $[0, \frac{\pi}{6}]$, decreasing on $[\frac{\pi}{6}, \frac{\pi}{2}]$.
- $y = x^2 e^{-x}$: $y' = x(2-x)e^{-x} > 0$ exactly for $0 < x < 2$, so increasing on $(0, 2)$ (a practice MCQ).
- $f(x) = \cos x$: strictly decreasing on $(0, \pi)$, strictly increasing on $(\pi, 2\pi)$.

**EXAM TIP** **Proving inequalities with monotonicity**

To show $e^x > 1 + x$ for $x > 0$: let $g(x) = e^x - 1 - x$. Then $g(0) = 0$ and $g'(x) = e^x - 1 > 0$ for $x > 0$, so $g$ increases from 0 and stays positive.

## 17.3 Tangents and normals

The **tangent** at a point touches the curve with the same direction; the **normal** is the perpendicular to the tangent at that point.

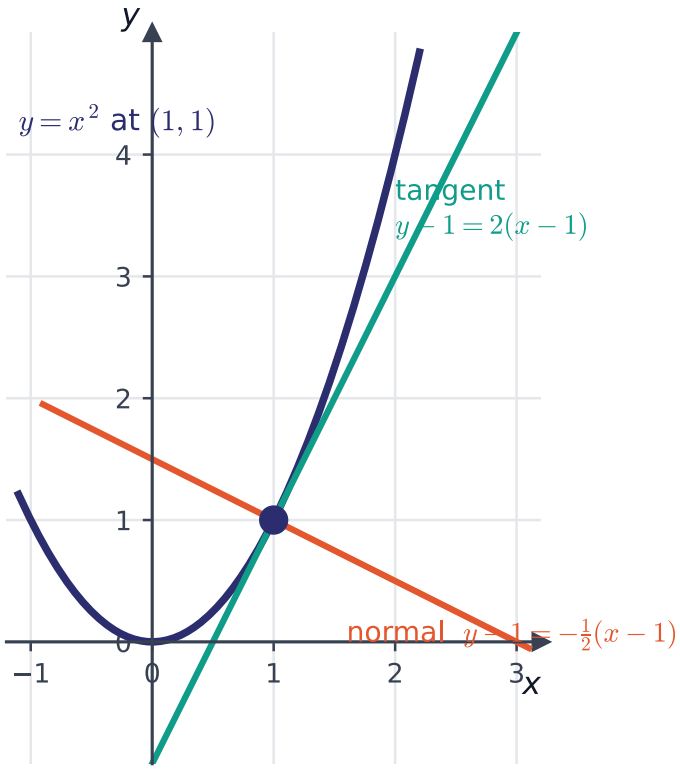
**FORMULA** **Tangent and normal at $(x_0, y_0)$ on $y = f(x)$**

$$\text{slope of tangent} = f'(x_0) = \left.\frac{dy}{dx}\right|_{(x_0,y_0)}$$

$$\text{Tangent:}\quad y - y_0 = f'(x_0)\,(x - x_0)$$

$$\text{Normal:}\quad y - y_0 = -\frac{1}{f'(x_0)}(x - x_0) \quad\Longleftrightarrow\quad (y - y_0)f'(x_0) + (x - x_0) = 0$$

**Special cases.** If $f'(x_0) = 0$ the tangent is **horizontal**: $y = y_0$ (and the normal is $x = x_0$). If $f'(x_0) \to \infty$ the tangent is **vertical**: $x = x_0$ (and the normal is $y = y_0$). If the tangent makes angle $\theta$ with the positive $x$-axis, $f'(x_0) = \tan\theta$.




Tangent (teal) and normal (coral) to $y = x^2$ at $(1, 1)$ are perpendicular.

**EXAMPLE** **Example 17.9: the basics**

- Slope of the tangent to $y = x^3 - x$ at $x = 2$: $y' = 3x^2 - 1 = 11$.
- Tangent and normal to $y = x^2$ at $(1, 1)$: slope 2, so tangent $y - 1 = 2(x - 1)$, i.e. $y = 2x - 1$; normal $y - 1 = -\frac{1}{2}(x - 1)$.
- Point where the tangent to $y = \sqrt{4x - 3} - 1$ has slope $\frac{2}{3}$: $y' = \dfrac{2}{\sqrt{4x-3}} = \dfrac{2}{3} \Rightarrow 4x - 3 = 9 \Rightarrow x = 3$, $y = 2$. Point $(3, 2)$.
- Practice MCQ: slope of the normal to $y = 2x^2 + 3\sin x$ at $x = 0$: $y' = 4x + 3\cos x = 3$, so the normal has slope $-\frac{1}{3}$.

**EXAMPLE** **Example 17.10: tangents with a given slope**

Find all tangents of slope 2 to $y = -\dfrac{2}{x - 3}$.

$y' = \dfrac{2}{(x-3)^2} = 2 \Rightarrow (x-3)^2 = 1 \Rightarrow x = 2$ or $4$, giving points $(2,2)$ and $(4,-2)$. Tangents: $y - 2 = 2(x-2)$, i.e. $y = 2x - 2$, and $y + 2 = 2(x-4)$, i.e. $y = 2x - 10$.

**EXAMPLE** **Example 17.11: implicit curves**

a. On $\dfrac{x^2}{4} + \dfrac{y^2}{25} = 1$: $y' = -\dfrac{25x}{4y}$. Tangent **horizontal** when $x = 0$: points $(0, \pm 5)$. Tangent **vertical** when $y = 0$: points $(\pm 2, 0)$.

b. Astroid $x^{2/3} + y^{2/3} = 2$ at $(1,1)$: $y' = -\left(\frac{y}{x}\right)^{1/3} = -1$. Tangent $x + y - 2 = 0$; normal $y = x$.

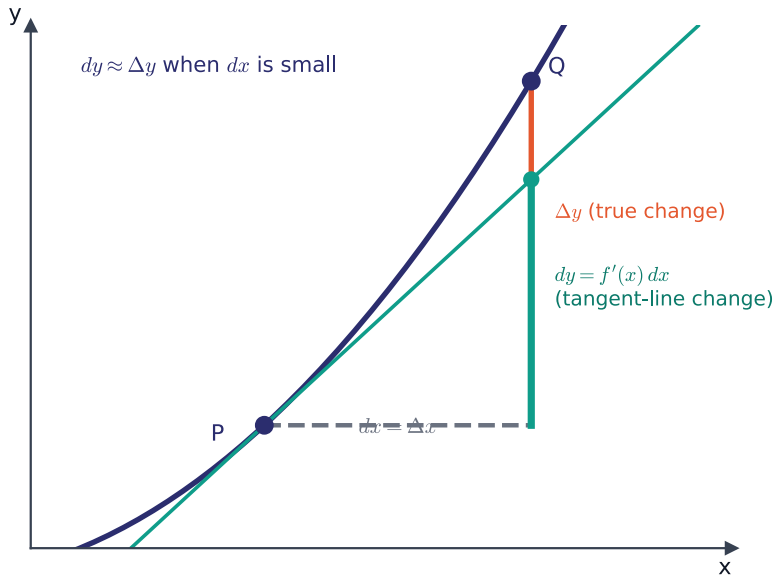
c. The line $y = x + 1$ touches $y^2 = 4x$ at $(1,2)$: $y' = \frac{2}{y} = 1$ there ✓ (practice MCQ).

**EXAMPLE** **Example 17.12: parametric curve**

$x = a\sin^3 t,\ y = b\cos^3 t$ at $t = \frac{\pi}{2}$: $\dfrac{dy}{dx} = \dfrac{-3b\cos^2 t \sin t}{3a\sin^2 t \cos t} = -\dfrac{b}{a}\cot t = 0$ at $t = \frac{\pi}{2}$; the point is $(a, 0)$. Tangent: $y = 0$.

## 17.4 Approximations using differentials

For a tiny change $\Delta x = dx$ the true change $\Delta y = f(x + \Delta x) - f(x)$ is well approximated by the **differential** $dy = f'(x)\,dx$ (the change *along the tangent line*).




$dy$ (change along the tangent) and $\Delta y$ (true change) are nearly equal when $dx$ is small.

**FORMULA** **Linear approximation**

$$\boxed{f(x + \Delta x) \approx f(x) + f'(x)\,\Delta x}$$

$$\text{absolute error} = |dy|\,, \qquad \text{relative error} = \frac{dy}{y}, \qquad \text{percentage error} = \frac{dy}{y} \times 100$$

**EXAM TIP** **How to choose $x$ and $\Delta x$**

Pick the nearest **easy** number as $x$ (a perfect square for $\sqrt{}$, a perfect cube for $\sqrt[3]{}$) and let $\Delta x$ be the small remainder, which may be negative.

**EXAMPLE** **Example 17.13: estimating roots**

$\sqrt{36.6}$: $y = \sqrt{x}$, $x = 36$, $\Delta x = 0.6$. $dy = \dfrac{1}{2\sqrt{36}}(0.6) = 0.05$. So $\sqrt{36.6} \approx 6.05$ (true: 6.0498).

$(25)^{1/3}$: $y = x^{1/3}$, $x = 27$, $\Delta x = -2$. $dy = \dfrac{1}{3 \cdot 27^{2/3}}(-2) = \dfrac{-2}{27} \approx -0.074$. So $25^{1/3} \approx 2.926$ (true: 2.924).

$\sqrt{25.3}$: $x = 25,\ \Delta x = 0.3$: $dy = \frac{0.3}{10} = 0.03$, so $\approx 5.03$.

**EXAMPLE** **Example 17.14: evaluating a polynomial nearby**

$f(x) = 3x^2 + 5x + 3$. Estimate $f(3.02)$: $f(3) = 45$, $f'(3) = 6(3) + 5 = 23$, so $f(3.02) \approx 45 + 23(0.02) = 45.46$. (Exact: 45.4612.) Likewise (practice MCQ) for $f(x) = 3x^2 + 15x + 5$: $f(3) = 77$, $f'(3) = 33$, so $f(3.02) \approx 77 + 33(0.02) = 77.66$.

**EXAMPLE** **Example 17.15: measurement errors**

- Cube side increased by 2%: $dV = 3x^2\,dx = 3x^2(0.02x) = 0.06x^3$, i.e. a 6% increase in volume (percentage errors multiply by the power).
- Sphere radius measured as 9 cm with error 0.03 cm: $dV = 4\pi r^2 dr = 4\pi(81)(0.03) = 9.72\pi$ cm$^3$.
- Diameter of a sphere has error 0.02 cm, radius 10 cm (Jain Ex. 1.2): $dr = 0.01$, $dV = 4\pi(100)(0.01) = 4\pi$, and $\dfrac{dV}{V} \times 100 = \dfrac{4\pi}{\frac{4}{3}\pi \cdot 1000} \times 100 = 0.3\%$.

UNIT II · CHAPTER 18

# Maxima and minima

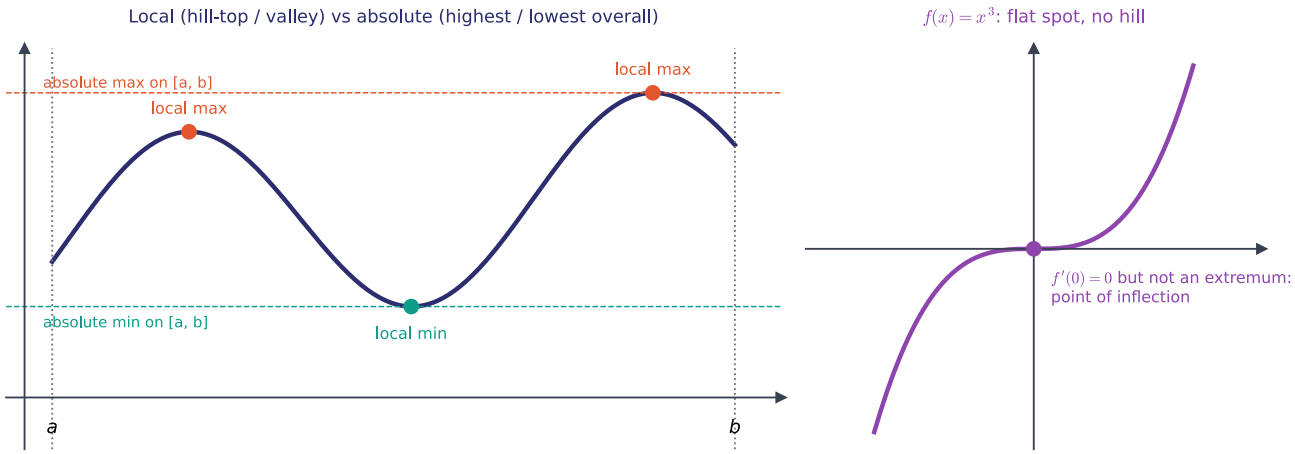
**BIG IDEA** **Big idea**

Engineers, economists and scientists constantly ask: *what is the biggest, smallest, cheapest, fastest?* On a graph, the answers sit at the **tops of hills and bottoms of valleys**, exactly the places where the tangent is flat, i.e. where $f' = 0$. Calculus turns "find the best" into "solve $f'(x) = 0$".

## 18.1 Vocabulary

**DEFINITION** **Local and absolute extrema**

- $f$ has a **local maximum** at $c$ if $f(c) \geq f(x)$ for all $x$ **near** $c$ (in some $(c-h, c+h)$); a **local minimum** if $f(c) \leq f(x)$ near $c$.
- $f$ has an **absolute (global) maximum** on an interval $I$ if $f(c) \geq f(x)$ for **all** $x \in I$; similarly an absolute minimum.
- The common word is **extremum** (plural *extrema*).
- A **critical point** of $f$ is an interior point $c$ where $f'(c) = 0$ **or** $f'(c)$ does not exist.




Local extrema (hills and valleys) versus absolute extrema on $[a, b]$ (left). Right: a flat tangent need not mean an extremum.

**THEOREM** **Fermat's necessary condition**

If $f$ has a local maximum or minimum at an **interior** point $c$, then either $f'(c) = 0$ or $f'(c)$ does not exist. *The converse is false:* $f(x) = x^3$ has $f'(0) = 0$ but 0 is neither a max nor a min (it is a **point of inflection**).

So: **extrema can only hide at critical points** (and at end points of the interval). The tests below decide which critical points are real hills or valleys.

## 18.2 The first-derivative test

**THEOREM** **First-derivative test**

Let $f$ be continuous at a critical point $c$.

- If $f'$ changes from $+$ to $-$ as $x$ increases through $c$: **local maximum**.
- If $f'$ changes from $-$ to $+$: **local minimum**.
- If $f'$ does **not** change sign: neither (a point of inflection).

**MEMORY TRICK** **Walking picture**

Going **up then down** you have crossed a peak ($+ \to -$: max). Going **down then up** you have crossed a valley ($- \to +$ : min).

**EXAMPLE** **Example 18.1**

$f(x) = x^3 - 3x + 3$: $f' = 3(x-1)(x+1)$, critical points $\pm 1$. Sign of $f'$: $+$ for $x < -1$, $-$ for $-1 < x < 1$, $+$ for $x > 1$. So **local max at** $x = -1$ with value $f(-1) = 5$, and **local min at** $x = 1$ with value $f(1) = 1$. (The Chapter 17 figure plots $x^3 - 3x$, which has the same shape, shifted down by 3.)

**EXAMPLE** **Example 18.2: no extremum**

$f(x) = 2x^3 - 6x^2 + 6x + 5$: $f' = 6(x-1)^2 \geq 0$ never changes sign, so $x = 1$ is a point of inflection and $f$ has **no extrema**.

**EXAMPLE** **Example 18.3: a corner can be an extremum**

$f(x) = 3 + |x|$ is not differentiable at 0, so the second-derivative test fails; but $f' = -1 < 0$ for $x < 0$ and $+1 > 0$ for $x > 0$ ($- \to +$), so $x = 0$ is a **local (and absolute) minimum** with value 3.

## 18.3 The second-derivative test

**THEOREM** **Second-derivative test**

Let $f'(c) = 0$ and let $f''$ exist at $c$.

- $f''(c) < 0 \Rightarrow$ **local maximum** at $c$ (curve bends like a cap $\frown$).
- $f''(c) > 0 \Rightarrow$ **local minimum** at $c$ (curve bends like a cup $\smile$).
- $f''(c) = 0 \Rightarrow$ **test fails**: use the first-derivative test (or the higher-order test below).

**EXAMPLE** **Example 18.4**

$f(x) = 3x^4 + 4x^3 - 12x^2 + 12$. $f' = 12x^3 + 12x^2 - 24x = 12x(x-1)(x+2)$, critical points $0, 1, -2$. $f'' = 36x^2 + 24x - 24$:

| $c$ | $f''(c)$ | verdict | value |
|---|---|---|---|
| $0$ | $-24 < 0$ | local **max** | $f(0) = 12$ |
| $1$ | $36 > 0$ | local **min** | $f(1) = 7$ |
| $-2$ | $72 > 0$ | local **min** | $f(-2) = -20$ |

**THEOREM** **Higher-order test (Jain, Theorem 1.6)**

If $f'(c) = f''(c) = \cdots = f^{(n-1)}(c) = 0$ but $f^{(n)}(c) \neq 0$, then: $n$ **even** $\Rightarrow$ extremum (maximum if $f^{(n)}(c) < 0$, minimum if $> 0$); $n$ **odd** $\Rightarrow$ **no** extremum (inflection).

*Example*: $f(x) = x^4$ has $f'(0) = f''(0) = f'''(0) = 0$ and $f^{(4)}(0) = 24 > 0$ ($n = 4$ even): a minimum.

## 18.4 Absolute maximum and minimum on a closed interval

**THEOREM** **Extreme Value Theorem**

A function **continuous on** $[a, b]$ attains an absolute maximum and an absolute minimum there. (On an open interval it may attain neither: $f(x) = x$ on $(0, 1)$ has no max or min.)

**BIG IDEA** **Working rule for $[a, b]$**

1. Find all critical points of $f$ **inside** $(a, b)$ ($f' = 0$ or $f'$ undefined).
2. Add the **end points** $a$ and $b$.
3. Compute $f$ at every one of these points.
4. The **largest** value is the absolute maximum; the **smallest** is the absolute minimum.

**EXAMPLE** **Example 18.5**

$f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$. $f' = 6(x-2)(x-3)$: critical points $2, 3$. Values: $f(1) = 24,\ f(2) = 29,\ f(3) = 28,\ f(5) = 56$. **Absolute max** $56$ **at** $x = 5$**; absolute min** $24$ **at** $x = 1$**.**

**EXAMPLE** **Example 18.6: a non-differentiable critical point**

$f(x) = 12x^{4/3} - 6x^{1/3}$ on $[-1, 1]$. $f'(x) = 16x^{1/3} - 2x^{-2/3} = \dfrac{2(8x-1)}{x^{2/3}}$: zero at $x = \frac{1}{8}$, undefined at $x = 0$. Both are critical points. Values: $f(-1) = 18,\ f(0) = 0,\ f(\frac{1}{8}) = -\frac{9}{4},\ f(1) = 6$. **Absolute max** $18$ **at** $x = -1$**; absolute min** $-\frac{9}{4}$ **at** $x = \frac{1}{8}$**.**

**EXAMPLE** **Example 18.7: trigonometric**

$f(x) = \sin x + \cos x$ on $[0, \pi]$: $f' = \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}$. $f(0) = 1,\ f(\frac{\pi}{4}) = \sqrt{2},\ f(\pi) = -1$. Max $\sqrt{2}$, min $-1$.

## 18.5 Optimisation: the word-problem method

**BIG IDEA** **The six-step recipe**

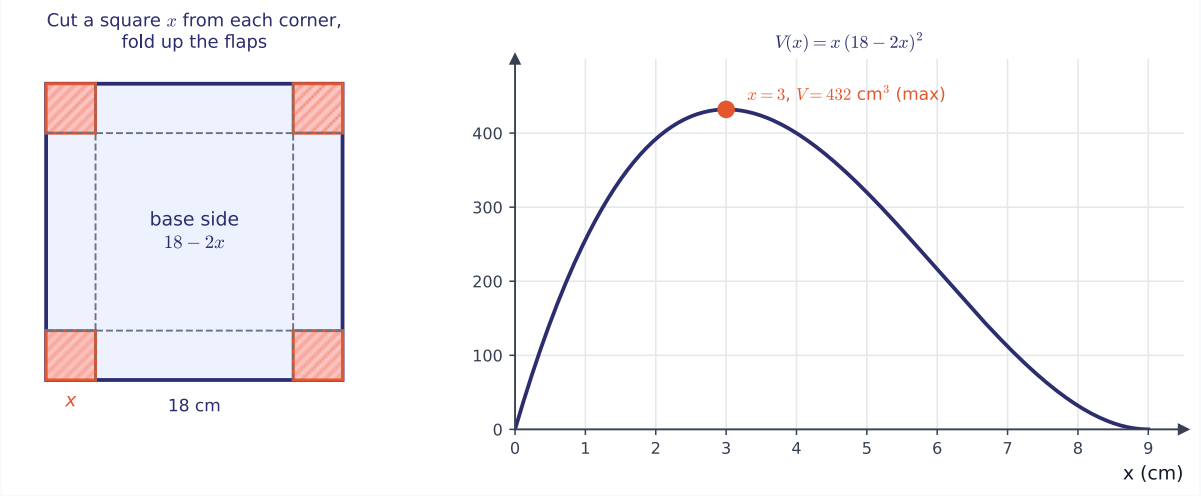
1. **Draw** a diagram; name the variables.
2. **Write the quantity** $Q$ to be maximised/minimised.
3. **Write the constraint** that links the variables.
4. **Eliminate** to get $Q$ as a function of **one** variable; state its domain.
5. **Differentiate**, set $Q' = 0$, and test the critical points (second derivative, sign change, or end points).
6. **Answer the question asked**, with units.

**EXAMPLE** **Example 18.8: minimum sum of squares**

Two positive numbers sum to 15; minimise the sum of their squares. Let them be $x$ and $15 - x$: $S = x^2 + (15-x)^2$, $S' = 4x - 30 = 0 \Rightarrow x = \frac{15}{2}$, $S'' = 4 > 0$ (min). Both numbers are $7.5$. (*In general, for a fixed sum the sum of squares is smallest when the numbers are equal.*)

**EXAMPLE** **Example 18.9: the open box**

A square tin sheet of side $18$ cm has a square of side $x$ cut from each corner and the flaps are folded up to make an open box. Maximise the volume.




Cut-and-fold box and its volume curve.

Base side $18 - 2x$, height $x$: $V = x(18-2x)^2$, $0 < x < 9$. $V' = (18-2x)^2 - 4x(18-2x) = (18-2x)(18-6x) = 0 \Rightarrow x = 3$ (not 9). $V'' = 24x - 144$ gives $V''(3) = -72 < 0$: a maximum. $V_{\max} = 3 \cdot 12^2 = \mathbf{432\ cm^3}$.

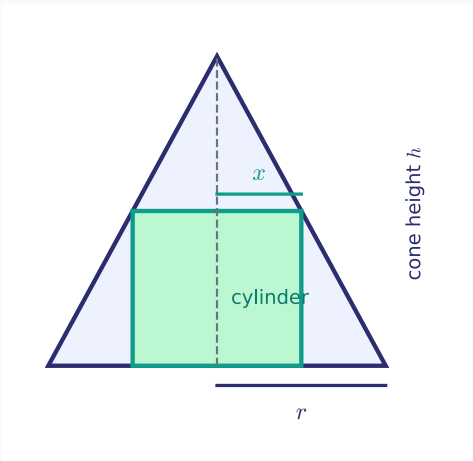
For the textbook's $45 \times 24$ sheet: $V = x(45-2x)(24-2x)$, $V' = 12(x-5)(x-18)$; only $x = 5$ is allowed ($x < 12$).

**EXAMPLE** **Example 18.10: rectangle in a circle**

Of all rectangles inscribed in a circle of radius $R$, the **square** has the greatest area. Let one side be $x$; the diagonal is $2R$, so the other side is $\sqrt{4R^2 - x^2}$ and $A = x\sqrt{4R^2 - x^2}$. Maximise $A^2 = x^2(4R^2 - x^2)$: $\frac{d}{dx} = 8R^2x - 4x^3 = 0 \Rightarrow x^2 = 2R^2$, so both sides equal $R\sqrt{2}$: a square of area $2R^2$.

**EXAMPLE** **Example 18.11: cylinder in a cone**

A right circular cylinder is inscribed in a cone of base radius $r$ and height $h$. Show the cylinder of greatest curved surface has radius $\frac{r}{2}$.




Cross-section of the cone with the cylinder of radius $x$.

By similar triangles the cylinder's height is $\dfrac{h(r-x)}{r}$. Curved surface $S = 2\pi x \cdot \dfrac{h(r-x)}{r} = \dfrac{2\pi h}{r}(rx - x^2)$. $S' = \dfrac{2\pi h}{r}(r - 2x) = 0 \Rightarrow x = \frac{r}{2}$, and $S'' = -\frac{4\pi h}{r} < 0$: a maximum. ✓

**EXAMPLE** **Example 18.12: the cheapest can**

A closed cylindrical can must hold 100 cm³. Minimise the surface area.

$V = \pi r^2 h = 100 \Rightarrow h = \dfrac{100}{\pi r^2}$. $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \dfrac{200}{r}$. $S' = 4\pi r - \dfrac{200}{r^2} = 0 \Rightarrow r^3 = \dfrac{50}{\pi}$, $r \approx 2.52$ cm, and then $h = 2r \approx 5.03$ cm. **The best can has height equal to its diameter.** $S'' = 4\pi + \frac{400}{r^3} > 0$ confirms a minimum.

**EXAMPLE** **Example 18.13: a wire cut in two**

A 28 m wire is cut; one piece forms a square, the other a circle. Minimise the total area.

Square piece $x$ (side $\frac{x}{4}$), circle piece $28 - x$ (radius $\frac{28-x}{2\pi}$): $A = \dfrac{x^2}{16} + \dfrac{(28-x)^2}{4\pi}$. $A' = \dfrac{x}{8} - \dfrac{28-x}{2\pi} = 0 \Rightarrow x = \dfrac{112}{\pi + 4}$. The pieces are $\dfrac{112}{\pi+4}$ m (square) and $\dfrac{28\pi}{\pi+4}$ m (circle).

**EXAMPLE** **Example 18.14: nearest approach**

A helicopter flies along $y = x^2 + 7$; a soldier is at $(3, 7)$. The squared distance is $D^2 = (x-3)^2 + x^4$, with derivative $2(x-3) + 4x^3 = 2(x-1)(2x^2 + 2x + 3)$. The quadratic factor has no real roots, so the only critical point is $x = 1$, where $D^2 = 4 + 1 = 5$. Check: at $x = 0$, $D^2 = 9 > 5$. **Nearest distance $\sqrt{5}$.** (*Minimise $D^2$, not $D$: same location, no square root.*)

**EXAMPLE** **Example 18.15: business and projectiles**

- **Profit** $p(x) = 41 - 24x - 18x^2$: $p' = -24 - 36x = 0 \Rightarrow x = -\frac{2}{3}$, $p = 49$.
- **Ball** thrown from a 60 m building follows $h(x) = 60 + x - \dfrac{x^2}{60}$: $h' = 1 - \frac{x}{30} = 0 \Rightarrow x = 30$, so the maximum height is $60 + 30 - 15 = \mathbf{75\ m}$.
- **Orange grove** profit $P(x) = ax + bx^2$ ($b < 0$): maximised at $x = -\dfrac{a}{2b}$ trees per acre.
- **Right triangle with hypotenuse** $h$: area $\frac{1}{2}x\sqrt{h^2 - x^2}$ is maximal when $x = \frac{h}{\sqrt{2}}$ (an isosceles right triangle), with area $\frac{h^2}{4}$ (Jain Ex. 1.14).

**EXAM TIP** **Common optimisation checks**

- Always find the **domain** (lengths positive, $x < 9$ in the box problem).
- Reject critical points outside the domain.
- If the end points make physical sense, **compare** the critical-point value with the end values.
- Minimising $D$ and $D^2$ gives the same point; use $D^2$ to avoid square roots.

## 18.6 Concavity and inflection (a short note)

$f'' > 0$ on an interval: the curve is **concave up** (holds water, like a cup). $f'' < 0$: **concave down**. A **point of inflection** is where concavity changes (e.g. $x = 0$ for $x^3$, where $f'' = 6x$ changes sign). That is why the second-derivative test works: a critical point with $f'' > 0$ sits at the bottom of a cup.

UNIT II · CHAPTER 19

# Unit II: practice and answers

## 19.1 Your professor's differentiation MCQs (Q1–Q15 of the deck)

| Q | Question | Ans | Working |
|---|---|---|---|
| 1 | $\frac{d}{dx}x^5$ | **B** | $5x^4$ |
| 2 | $\frac{d}{dx}\sin x$ | **A** | $\cos x$ |
| 3 | $\frac{d}{dx}e^{2x}$ | **C** | $2e^{2x}$ (chain rule) |
| 4 | $\frac{d}{dx}\ln x$ | **D** | $1/x$ |
| 5 | $\frac{d}{dx}(x\sin x)$ | **A** | $\sin x + x\cos x$ |
| 6 | $\frac{d}{dx}\frac{x}{x+1}$ | **B** | $\frac{(x+1)-x}{(x+1)^2} = \frac{1}{(x+1)^2}$ |
| 7 | $\frac{d}{dx}\sin(x^2)$ | **C** | $2x\cos(x^2)$ |
| 8 | $\frac{d}{dx}\tan x$ | **D** | $\sec^2 x$ |
| 9 | $\frac{d}{dx}a^x$ | **B** | $a^x \ln a$ |
| 10 | $\frac{d}{dx}\tan^{-1} x$ | **A** | $\frac{1}{1+x^2}$ |
| 11 | $\frac{d}{dx}x^x$ | **C** | $x^x(1+\ln x)$ |
| 12 | $\frac{d}{dx}\sqrt{x}$ | **D** | $\frac{1}{2\sqrt{x}}$ |
| 13 | $y = \ln\sin x$ | **B** | $\cot x$ |
| 14 | $f = x^4$, $f''$ | **C** | $12x^2$ |
| 15 | $\frac{d}{dx}\cos 3x$ | **B** | $-3\sin 3x$ |

## 19.2 The Unit II questions from the mixed practice set

| Q | Topic | Ans | Why |
|---|---|---|---|
| 1 | $(\tan x)'$ | B $\sec^2 x$ | quotient rule |
| 2 | product rule | C $uv' + vu'$ | definition |
| 5 | $x^2 + y^2 = r^2$, $y'$ | B $-\frac{x}{y}$ | implicit |
| 6 | $xy = c^2$, $y'$ | B $-\frac{y}{x}$ | implicit |
| 7 | $x = at^2, y = 2at$ | B $\frac{1}{t}$ | parametric |
| 8 | $x = a\cos\theta, y = a\sin\theta$ | D $-\cot\theta$ | parametric |
| 10 | $y = \ln x^3$ | C $\frac{3}{x}$ | $3\ln x$ |
| 12 | $y = x^x$ | C $x^x(1 + \ln x)$ | log differentiation |
| 13 | $(e^{3x})''$ | C $9e^{3x}$ | chain twice |
| 16 | $(\ln x)''$ | B $-\frac{1}{x^2}$ | $(1/x)'$ |
| 19 | $(a^x)''$ | B $a^x(\ln a)^2$ | repeat |
| 21 | not needed for Rolle | D "$f'$ strictly increasing" | only 3 conditions |
| 22 | conclusion of Rolle | B $f'(c) = 0$ | theorem |
| 24 | Rolle fails for $\lvert x\rvert$ on $[-1, 1]$ | C not differentiable at 0 | corner |
| 25 | Lagrange MVT | A $\frac{f(b)-f(a)}{b-a}$ | theorem |
| 27 | Rolle for $x^2 - 4x + 3$ on $[1, 3]$ | B $c = 2$ | $2c - 4 = 0$ |
| 28 | MVT for $x^2$ on $[0, 2]$ | C $c = 1$ | $2c = 2$ |
| 32 | Taylor coefficient of $(x - a)^n$ | B $\frac{f^{(n)}(a)}{n!}$ | derivation |
| 33 | Maclaurin of $e^x$ | C $1 + x + \frac{x^2}{2!} + \cdots$ | all derivatives $= 1$ |
| 34 | Maclaurin of $\sin x$ | B only odd powers | odd function |
| 35 | Maclaurin of $\cos x$ | A only even powers | even function |
| 36 | $(1 - x)^{-1}$, first 3 terms | B $1 + x + x^2$ | geometric series |
| 43 | $\lim \frac{\sin x}{x}$ | B 1 | standard |
| 44 | $\lim \frac{1-\cos x}{x}$ | B 0 | $\frac{\sin x}{1} \to 0$ |
| 46 | $\lim_{x\to\infty} \frac{e^x}{x}$ | D $\infty$ | growth ladder |
| 47 | $\lim_{x\to 1} \frac{\ln x}{x-1}$ | B 1 | L'Hôpital |
| 49 | $\lim \frac{\tan x}{x}$ | B 1 | standard |
| 50 | $\lim_{x\to\infty} xe^{-x}$ | C 0 | $\frac{x}{e^x} \to 0$ |
| 55 | critical points of $x^3 - 3x$ | B $x = \pm 1$ | $3x^2 - 3 = 0$ |
| 56 | max of $4x - x^2$ | C 4 | at $x = 2$ |

## 19.3 Extra practice

**TRY IT** **Questions**

1. Is $f(x) = |x-2| + |x+1|$ continuous and differentiable everywhere? Where does it fail?
2. Differentiate $y = x^{\sin x} + (\sin x)^x$.
3. Show $\dfrac{d}{dx}\tan^{-1}\dfrac{1+x}{1-x} = \dfrac{1}{1+x^2}$.
4. For $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$ find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$.
5. Verify Rolle's theorem for $\sin x$ on $[0, \pi]$ and MVT for $\ln x$ on $[1, e]$; find $c$.
6. Write the Maclaurin series of $e^x \cos x$ up to $x^4$.
7. Evaluate $\displaystyle\lim_{x\to 0}\frac{\tan x - x}{x^3}$.
8. Find the local extrema of $f(x) = x^3 - 6x^2 + 9x + 15$.
9. Find the absolute extrema of $f(x) = x^3$ on $[-2, 2]$.
10. A circle's radius grows at 3 cm/s. How fast is its area growing when $r = 10$?
11. Equation of the tangent to $y = x^3 - 3x$ at $x = 2$.
12. Intervals of increase and decrease of $f(x) = 2x^3 - 9x^2 + 12x$.
13. Of all rectangles with perimeter 40 m, which has the greatest area?
14. Differentiate $\ln\left(x + \sqrt{x^2+1}\right)$.

**Answers.** (1) Continuous everywhere (sum of continuous functions). **Not differentiable at $x = -1$ and $x = 2$** (corners). (2) $x^{\sin x}\left(\dfrac{\sin x}{x} + \cos x \ln x\right) + (\sin x)^x\,(\ln\sin x + x\cot x)$. (3) The expression equals $\frac{\pi}{4} + \tan^{-1} x$ (for $x < 1$), whose derivative is $\frac{1}{1+x^2}$. (4) $\dfrac{dx}{dt} = at\cos t$, $\dfrac{dy}{dt} = at\sin t$, so $\dfrac{dy}{dx} = \tan t$ and $\dfrac{d^2y}{dx^2} = \dfrac{\sec^2 t}{at\cos t} = \dfrac{\sec^3 t}{at}$. (5) $c = \frac{\pi}{2}$ for Rolle (since $\cos c = 0$); $c = e - 1 \approx 1.718$ for MVT. (6) $1 + x - \dfrac{x^3}{3} - \dfrac{x^4}{6}$ (the $x^2$ term cancels). (7) $\dfrac{1}{3}$ (use $\tan x = x + \frac{x^3}{3} + \cdots$ or L'Hôpital three times). (8) $f' = 3(x-1)(x-3)$: local max $f(1) = 19$ at $x = 1$; local min $f(3) = 15$ at $x = 3$. (9) Max 8 at $x = 2$; min $-8$ at $x = -2$ (no interior extremum since $f' = 3x^2 \geq 0$). (10) $\dfrac{dA}{dt} = 2\pi r\dfrac{dr}{dt} = 60\pi$ cm$^2$/s. (11) $y' = 3x^2 - 3 = 9$ at $x = 2$; point $(2, 2)$; tangent $y = 9x - 16$. (12) $f' = 6(x-1)(x-2)$: increasing on $(-\infty, 1]$ and $[2, \infty)$; decreasing on $[1, 2]$. (13) A square of side 10 m (area 100 m$^2$). (14) $\dfrac{1}{\sqrt{x^2+1}}$.

UNIT III · CHAPTER 20

# Meet the integral

**BIG IDEA** **Big idea**

Integration answers two questions that turn out to be the **same question**:

1. *Undo differentiation:* if I know the **speed**, what was the **distance**? (an **antiderivative**)
2. *Add up infinitely many tiny pieces:* how much **area** lies under a curve? (a **definite integral**)

The **Fundamental Theorem of Calculus** is the bridge: to total up a rate, find an antiderivative and subtract its values at the ends.

## 20.1 Antiderivatives and the "+C"

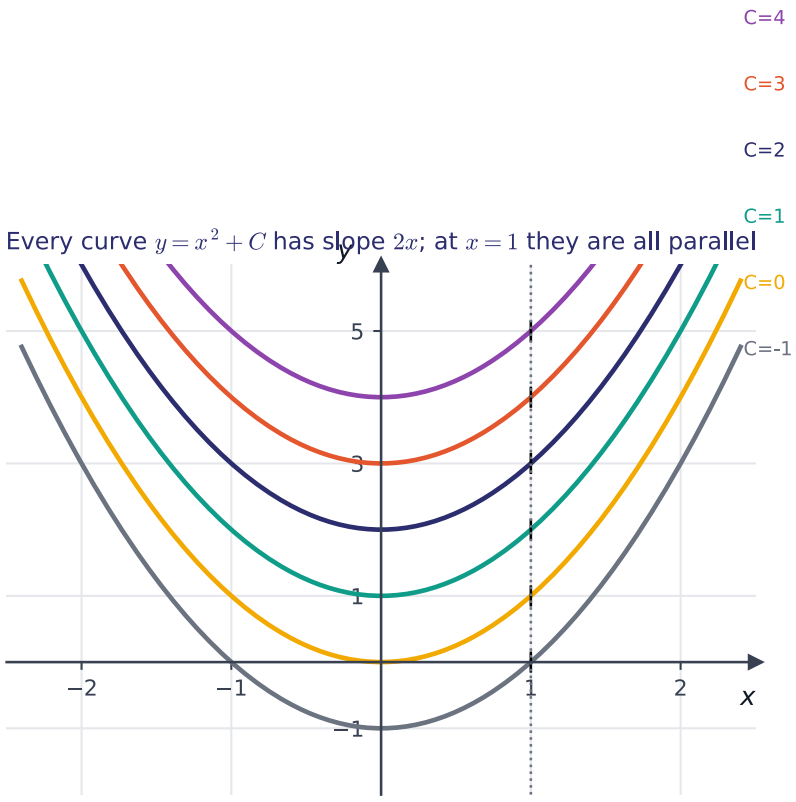
**DEFINITION** **Antiderivative and indefinite integral**

$F$ is an **antiderivative** of $f$ if $F'(x) = f(x)$. All antiderivatives of $f$ are written

$$\int f(x)\,dx = F(x) + C,$$

where $C$ is an arbitrary constant, the **constant of integration**. ($\int$ is an elongated "S" for *sum*; $f(x)$ is the **integrand**; $dx$ says which variable we integrate with respect to.)

**Why the $+C$?** Constants vanish when differentiated, so $x^2$, $x^2 + 5$ and $x^2 - 3$ all have derivative $2x$. Graphically, the antiderivatives of $2x$ are the **same parabola slid up and down**; at any given $x$ all of them have the same slope.




Antiderivatives of $2x$ are the parabolas $y = x^2 + C$: identical shape, different heights.

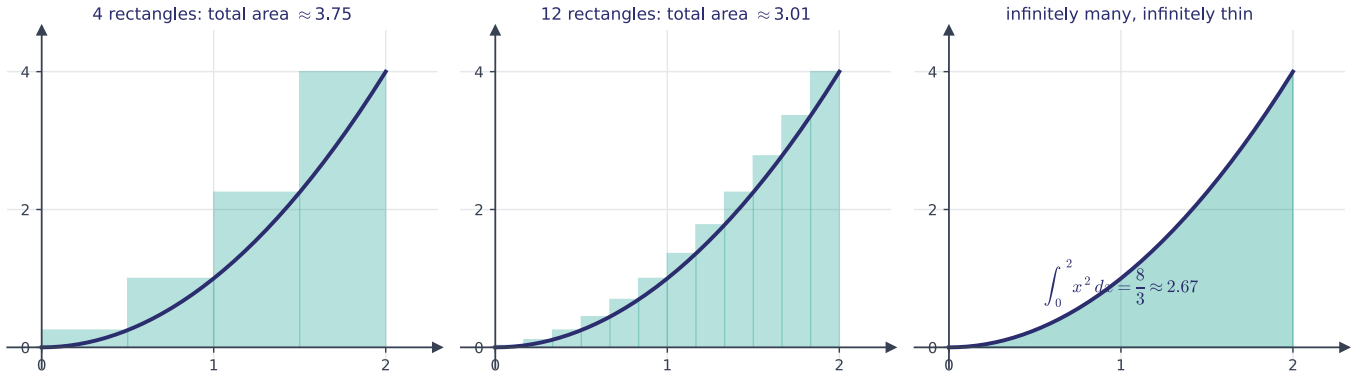
**WATCH OUT** **Never drop the $+C$**

For an indefinite integral the $+C$ is part of the answer; leaving it out loses marks. (For **definite** integrals the constant cancels, so no $+C$ there.) Different methods can give answers that *look* different, e.g. $\frac{\tan^2 x}{2}$ and $\frac{\sec^2 x}{2}$, which differ by the constant $\frac{1}{2}$.

**Check any integral by differentiating your answer.** If the derivative gives back the integrand, you are right. This is the best self-check in the whole subject.

## 20.2 Area under a curve

How do we find the area under $y = x^2$ from $x = 0$ to $x = 2$? Slice it into thin vertical strips and treat each as a rectangle: height $f(x_i)$, width $\Delta x$. Add them up. More strips give a better answer:




Approximating the area under $y = x^2$ on $[0, 2]$ by 4 and by 12 rectangles, and the exact limit.

**DEFINITION** **Definite integral (Riemann sum)**

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\,\Delta x, \qquad \Delta x = \frac{b-a}{n}.$$

It is the **signed area** between the curve and the $x$-axis from $x = a$ to $x = b$: area **above** the axis counts $+$, area **below** counts $-$.

Computing that limit directly is painful. Luckily we never have to.

## 20.3 The Fundamental Theorem of Calculus

**THEOREM** **Fundamental Theorem of Calculus**

**Part 1.** If $f$ is continuous on $[a, b]$ and $G(x) = \int_a^x f(t)\,dt$, then $G'(x) = f(x)$. (*Accumulated area grows at a rate equal to the height of the curve.*)

**Part 2.** If $F$ is **any** antiderivative of $f$, then

$$\boxed{\int_a^b f(x)\,dx = F(b) - F(a) = \Big[F(x)\Big]_a^b}$$

**EXAMPLE** **Example 20.1**

$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3} \approx 2.67$, exactly what the rectangles were closing in on. Also $\int_0^2 x^3\,dx = \left[\frac{x^4}{4}\right]_0^2 = 4$.

STORY TIME **A very old problem solved in one stroke**

*Archimedes computed the area of a parabolic segment around 250 BC by an ingenious (and exhausting) limiting argument. Two thousand years later Newton and Leibniz realised, independently, that area and slope are inverse operations. The Fundamental Theorem turned Archimedes' months of work into one line.*

**Variable limits (Leibniz rule).** $\dfrac{d}{dx}\displaystyle\int_a^{g(x)} f(t)\,dt = f(g(x))\cdot g'(x)$. For instance $\dfrac{d}{dx}\displaystyle\int_0^{x^2} \sin t\,dt = 2x\sin(x^2)$.

**Integral of a rate is net change.** If $v(t)$ is velocity, $\int_a^b v\,dt$ is the **displacement**; if $f$ is a rate of production, $\int_a^b f$ is total production.

UNIT III · CHAPTER 21

# Basic rules and standard integrals

**BIG IDEA** **Big idea**

Every derivative formula you know, read backwards, is an integral formula. Mastering this table (and the three simple rules below) lets you do a huge fraction of exam questions by **direct integration**: no clever tricks, just recognition.

## 21.1 Rules

**FORMULA** **Linearity and the linear-substitution rule**

$$\int k f(x)\,dx = k\int f(x)\,dx, \qquad \int \big(f(x)\pm g(x)\big)dx = \int f \pm \int g$$

$$\boxed{\int f(ax+b)\,dx = \frac{1}{a}\,F(ax+b)+C} \qquad \text{where } F' = f$$

The second rule is just the chain rule backwards: because $\frac{d}{dx}F(ax+b) = a\,f(ax+b)$, you must divide by $a$ to compensate. Examples: $\int \cos(3x+4)dx = \frac{1}{3}\sin(3x+4)+C$, $\int e^{3x}dx = \frac{e^{3x}}{3}+C$, $\int (2x+3)^5 dx = \frac{(2x+3)^6}{12}+C$.

**WATCH OUT** **Products and quotients do NOT split**

$\int f\,g\,dx \neq \int f\,dx \int g\,dx$ and $\int \frac{f}{g}\,dx \neq \frac{\int f}{\int g}$. There is no "product rule" shortcut; use substitution or parts (Chapters 22–23).

## 21.2 The standard table

**FORMULA** **Standard integrals (memorise all)**

| $f(x)$ | $\int f(x)\,dx$ | $f(x)$ | $\int f(x)\,dx$ |
|---|---|---|---|
| $x^n\ (n\neq -1)$ | $\frac{x^{n+1}}{n+1}+C$ | $\sin x$ | $-\cos x + C$ |
| $\frac{1}{x}$ | $\ln\lvert x\rvert + C$ | $\cos x$ | $\sin x + C$ |
| $e^x$ | $e^x + C$ | $\sec^2 x$ | $\tan x + C$ |
| $a^x$ | $\frac{a^x}{\ln a}+C$ | $\operatorname{cosec}^2 x$ | $-\cot x + C$ |
| $\frac{1}{1+x^2}$ | $\tan^{-1} x + C$ | $\sec x\tan x$ | $\sec x + C$ |
| $\frac{1}{\sqrt{1-x^2}}$ | $\sin^{-1} x + C$ | $\operatorname{cosec} x\cot x$ | $-\operatorname{cosec} x + C$ |
| $\frac{1}{x\sqrt{x^2-1}}$ | $\sec^{-1}\lvert x\rvert + C$ | $\tan x$ | $-\ln\lvert\cos x\rvert + C = \ln\lvert\sec x\rvert + C$ |
| $\cot x$ | $\ln\lvert\sin x\rvert + C$ | $\sec x$ | $\ln\lvert\sec x+\tan x\rvert + C$ |
| $\operatorname{cosec} x$ | $\ln\lvert\operatorname{cosec} x-\cot x\rvert + C$ | | |

**FORMULA** **The 'a-forms' (the constant $a > 0$ version of the inverse-trig and log integrals)**

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C, \qquad \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C, \qquad \int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln\left|x + \sqrt{x^2 \pm a^2}\right| + C, \qquad \int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$$

**FORMULA** **The 'derivative on top' patterns (huge time-savers)**

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C, \qquad \int f'(x)\big[f(x)\big]^n dx = \frac{[f(x)]^{n+1}}{n+1} + C\ (n \neq -1), \qquad \int \frac{f'(x)}{\sqrt{f(x)}}dx = 2\sqrt{f(x)} + C$$

## 21.3 Tricks before the big methods

- **Split the fraction:** $\dfrac{x^2+1}{x} = x + \dfrac{1}{x} \Rightarrow \int = \dfrac{x^2}{2} + \ln|x| + C.$
- **Use identities:** $\sin^2 x + \cos^2 x = 1$ gives $\int(\sin^2 x + \cos^2 x)dx = x + C$. Power reduction gives $\int \cos^2 x\,dx = \int \frac{1+\cos 2x}{2}dx = \frac{x}{2} + \frac{\sin 2x}{4} + C.$
- **Rewrite roots and reciprocals as powers:** $\sqrt{x} = x^{1/2}$, $\dfrac{1}{x^2} = x^{-2}$, so $\int \sqrt{x}\,dx = \frac{2}{3}x^{3/2} + C$ and $\int \frac{dx}{x^2} = -\frac{1}{x} + C.$

**EXAMPLE** **Worked set (direct integration deck, Q16–Q30)**

- $\int x^n dx = \dfrac{x^{n+1}}{n+1} + C.$ $\int \dfrac{1}{x}dx = \ln|x| + C.$ $\int e^{3x}dx = \dfrac{e^{3x}}{3} + C.$
- $\int \sin x\,dx = -\cos x + C.$ $\int \sec^2 x\,dx = \tan x + C.$ $\int \operatorname{cosec}^2 x\,dx = -\cot x + C.$
- $\int (3x^2 + 2x + 1)dx = x^3 + x^2 + x + C.$
- $\int \dfrac{dx}{1+x^2} = \tan^{-1} x + C.$ $\int \dfrac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C.$
- $\int \sqrt{x}\,dx = \frac{2}{3}x^{3/2} + C.$ $\int \dfrac{dx}{x^2} = -\dfrac{1}{x} + C.$ $\int \cos 2x\,dx = \dfrac{\sin 2x}{2} + C.$
- $\int a^x dx = \dfrac{a^x}{\ln a} + C.$ $\int(\sin^2 x + \cos^2 x)dx = x + C.$ $\int \dfrac{x^2+1}{x}dx = \dfrac{x^2}{2} + \ln|x| + C.$

**EXAMPLE** **From the mixed practice set**

$\int e^x dx = e^x + C$; $\int a^x dx = \frac{a^x}{\ln a} + C$; $\int \sec^2 x\,dx = \tan x + C$; $\int (2x+3)dx = x^2 + 3x + C$; $\int \frac{f'}{f}dx = \ln|f| + C$; $\int \tan x\,dx = -\ln|\cos x| + C$; $\int \cos(3x+4)dx = \frac{1}{3}\sin(3x+4) + C$.

**TRY IT** **Try it**

a. $\int (5x^4 - \frac{3}{x} + 2e^x)dx$ (b) $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)dx$ (c) $\int \sin^2 x\,dx$. *Answers:* (a) $x^5 - 3\ln|x| + 2e^x + C$. (b) $\frac{2}{3}x^{3/2} + 2\sqrt{x} + C$. (c) $\frac{x}{2} - \frac{\sin 2x}{4} + C$.

UNIT III · CHAPTER 22

# Integration by substitution

**BIG IDEA** **Big idea**

Substitution is the **chain rule run backwards**. If the integrand contains a function **and its derivative** (up to a constant), rename the inner function $u$. The integral collapses to something simple in $u$.

**FORMULA** **The method**

$$\int f\big(g(x)\big)\,g'(x)\,dx = \int f(u)\,du, \qquad u = g(x), \quad du = g'(x)\,dx.$$

1. Choose $u$ = the **inside** of the nastiest piece (or a denominator, or an exponent). 2. Compute $du = u'\,dx$ and rewrite the **whole** integral (including $dx$) in terms of $u$. No $x$ may remain. 3. Integrate in $u$. 4. **Substitute back** $u = g(x)$. (For definite integrals: change the limits instead, see below.)

**EXAMPLE** **Example 22.1:** $\int 2xe^{x^2}dx$

$u = x^2,\ du = 2x\,dx$. So $\int e^u du = e^u + C = e^{x^2} + C$.

**EXAMPLE** **Example 22.2:** $\int x\sqrt{x^2+1},dx$

$u = x^2+1,\ du = 2x\,dx \Rightarrow x\,dx = \frac{1}{2}du$. So $\frac{1}{2}\int u^{1/2}du = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = \frac{1}{3}(x^2+1)^{3/2} + C$.

**EXAMPLE** **Example 22.3:** $\int \tan x,dx$ **and** $\int \cot x,dx$

$\int \frac{\sin x}{\cos x}dx$ with $u = \cos x,\ du = -\sin x\,dx$: $-\int \frac{du}{u} = -\ln|\cos x| + C$. Similarly $\int \frac{\cos x}{\sin x}dx = \ln|\sin x| + C$ (with $u = \sin x$ ).

**EXAMPLE** **Example 22.4: more from the deck (Q31–Q45)**

- $\int \sin^3 x\cos x\,dx$: $u = \sin x$: $\int u^3 du = \dfrac{\sin^4 x}{4} + C$.
- $\int \dfrac{\ln x}{x}\,dx$: $u = \ln x$: $\int u\,du = \dfrac{(\ln x)^2}{2} + C$.
- $\int \dfrac{dx}{x\ln x}$: $u = \ln x$: $\int \dfrac{du}{u} = \ln|\ln x| + C$.
- $\int \dfrac{e^x}{1+e^x}dx$: $u = 1+e^x$: $\ln(1+e^x) + C$.
- $\int \cos(3x+2)dx$: $u = 3x+2$: $\frac{1}{3}\sin(3x+2) + C$.
- $\int (2x+3)^5 dx$: $u = 2x+3$: $\frac{(2x+3)^6}{12} + C$.
- $\int \dfrac{x}{x^2+1}dx$: $u = x^2+1$: $\frac{1}{2}\ln(x^2+1) + C$.
- $\int \sec^2 x\tan x\,dx$: $u = \tan x$: $\dfrac{\tan^2 x}{2} + C$ (equivalently $\frac{\sec^2 x}{2} + C'$).
- $\int \dfrac{dx}{\sqrt{1-4x^2}}$: $u = 2x$: $\frac{1}{2}\sin^{-1}(2x) + C$.
- $\int \dfrac{dx}{1+9x^2}$: $u = 3x$: $\frac{1}{3}\tan^{-1}(3x) + C$.

**EXAM TIP** **Choosing $u$: a pattern guide**

| You see… | Try |
|---|---|
| $f(ax+b)$ | $u = ax+b$ |
| a power or root of an expression whose derivative also appears | $u =$ that expression |
| $e^{g(x)}g'(x)$ | $u = g(x)$ |
| $\frac{\ln x}{x}$, $\frac{1}{x\ln x}$ | $u = \ln x$ |
| $\sin^m x\cos^n x$, $m$ **odd** | $u = \cos x$ (save one $\sin x$ for $du$) |
| $\sin^m x\cos^n x$, $n$ **odd** | $u = \sin x$ (save one $\cos x$) |
| $\sin^m x\cos^n x$, both **even** | power-reduction: $\frac{1\pm\cos 2x}{2}$ |
| $\sqrt{a^2-x^2}$ | $x = a\sin\theta$ |
| $a^2+x^2$ | $x = a\tan\theta$ |
| $\sqrt{x^2-a^2}$ | $x = a\sec\theta$ |

## 22.1 Trigonometric integrals

- **Odd power:** $\int \sin^3 x\,dx = \int(1-\cos^2 x)\sin x\,dx$, $u=\cos x$: $-\cos x + \dfrac{\cos^3 x}{3} + C$. Likewise $\int \cos^3 x\,dx = \sin x - \dfrac{\sin^3 x}{3} + C$.
- **Even powers:** $\int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$; $\int \sin^2 x\cos^2 x\,dx = \int \frac{\sin^2 2x}{4}dx = \frac{1}{8}\int(1-\cos 4x)dx = \frac{x}{8} - \frac{\sin 4x}{32} + C$.
- $\int \sec x\,dx$**:** multiply and divide by $\sec x+\tan x$: $\int \frac{\sec^2 x+\sec x\tan x}{\sec x+\tan x}dx = \ln|\sec x+\tan x| + C$ (the numerator is exactly the derivative of the denominator).

## 22.2 Completing the square and trig substitutions

**EXAMPLE** **Example 22.5: completing the square**

$\displaystyle\int \frac{dx}{x^2+2x+5}$: $x^2+2x+5 = (x+1)^2+2^2$, so the integral is $\dfrac{1}{2}\tan^{-1}\dfrac{x+1}{2} + C$.

$\displaystyle\int \frac{dx}{\sqrt{5-4x-x^2}}$: $5-4x-x^2 = 9-(x+2)^2$, so the integral is $\sin^{-1}\dfrac{x+2}{3} + C$.

**EXAMPLE** **Example 22.6:** $\int \sqrt{1-x^2}\,dx$

Let $x=\sin\theta$, $dx=\cos\theta\,d\theta$, $\sqrt{1-x^2}=\cos\theta$. So $\int\cos^2\theta\,d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + C = \frac{1}{2}\left(\sin^{-1}x + x\sqrt{1-x^2}\right) + C$.

## 22.3 Substitution in definite integrals

**Change the limits** along with the variable; then you never need to go back to $x$.

**EXAMPLE** **Example 22.7**

$\displaystyle\int_0^1 \frac{x}{1+x^2}dx$: $u = 1+x^2$; $x=0\Rightarrow u=1$, $x=1\Rightarrow u=2$. $\dfrac{1}{2}\displaystyle\int_1^2 \frac{du}{u} = \frac{1}{2}\ln 2$.

$\displaystyle\int_0^1 2xe^{x^2}dx = \int_0^1 e^u du = e-1$.

**WATCH OUT** **Typical errors**

- Leaving some $x$'s in an integral that is now in $u$.
- Forgetting the factor from $du$ (e.g. $du = 2x\,dx$ means $x\,dx = \frac{1}{2}du$).
- In a definite integral, converting the integrand but **not the limits**.

UNIT III · CHAPTER 23

# Integration by parts

**BIG IDEA** **Big idea**

Integration by parts is the **product rule run backwards**. Use it when the integrand is a **product of two different kinds of function**, e.g. $x\,e^x$, $x\cos x$, $\ln x$, $\tan^{-1} x$, where substitution fails.

**FORMULA** **The formula**

From $(uv)' = u'v + uv'$ we get

$$\boxed{\int u\,dv = uv - \int v\,du} \quad\Longleftrightarrow\quad \int u(x)\,v'(x)\,dx = u(x)v(x) - \int u'(x)v(x)\,dx.$$

**Choosing $u$: the ILATE rule.** Pick $u$ as the function that comes **first** in this list; the remaining factor is $dv$.

**MEMORY TRICK** **ILATE**

**I**nverse trig ($\sin^{-1} x, \tan^{-1} x, \ldots$) $\to$ **L**ogarithmic ($\ln x$) $\to$ **A**lgebraic ($x,\ x^2, \ldots$) $\to$ **T**rigonometric ($\sin x, \cos x$) $\to$ **E**xponential ($e^x$). *Reason:* the early ones become **simpler** when differentiated and the late ones are easy to integrate. (In ILATE, "I" stands for Inverse trigonometric functions, a practice-set question.)

**EXAMPLE** **Example 23.1:** $\int xe^x\,dx$

$u = x$ (algebraic), $dv = e^x dx$. $du = dx$, $v = e^x$. $\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C = (x-1)e^x + C$.

**EXAMPLE** **Example 23.2:** $\int x\cos x\,dx$

$u = x,\ dv = \cos x\,dx \Rightarrow v = \sin x.\ = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$. Similarly $\int x\sin x\,dx = -x\cos x + \sin x + C$.

**EXAMPLE** **Example 23.3:** $\int \ln x\,dx$ **(the sneaky one)**

There seems to be only one factor. Write $\ln x = \ln x \cdot 1$: $u = \ln x,\ dv = dx \Rightarrow du = \frac{dx}{x},\ v = x$. $\int \ln x\,dx = x\ln x - \int 1\,dx = x\ln x - x + C$.

**EXAMPLE** **Example 23.4:** $\int x\ln x\,dx$

ILATE: $u = \ln x$, $dv = x\,dx$. $= \dfrac{x^2}{2}\ln x - \int \dfrac{x^2}{2}\cdot\dfrac{1}{x}\,dx = \dfrac{x^2}{2}\ln x - \dfrac{x^2}{4} + C$. In general $\int x^n \ln x\,dx = \dfrac{x^{n+1}}{n+1}\ln x - \dfrac{x^{n+1}}{(n+1)^2} + C$.

**EXAMPLE** **Example 23.5: inverse trig functions**

$\int \tan^{-1} x\,dx$: $u = \tan^{-1} x,\ dv = dx$. $= x\tan^{-1} x - \int \dfrac{x}{1+x^2}dx = x\tan^{-1} x - \dfrac{1}{2}\ln(1+x^2) + C$. Similarly $\int \sin^{-1} x\,dx = x\sin^{-1} x + \sqrt{1-x^2} + C$.

**EXAMPLE** **Example 23.6: parts twice**

$\int x^2 e^x dx$: first $u = x^2$: $x^2e^x - 2\int xe^x dx$; then use Example 23.1: $x^2e^x - 2(x-1)e^x + C = e^x(x^2 - 2x + 2) + C$.

**EXAM TIP** **Tabular method (for polynomial × exponential/trig)**

List derivatives of the polynomial down one column until 0, and integrals of the other factor down another; multiply along the diagonals with alternating signs $+, -, +, \ldots$ For $\int x^3 e^x dx$:

| $D$ (derivatives of $x^3$) | $I$ (integrals of $e^x$) | sign |
|---|---|---|
| $x^3$ | $e^x$ | $+$ |
| $3x^2$ | $e^x$ | $-$ |
| $6x$ | $e^x$ | $+$ |
| $6$ | $e^x$ | $-$ |

Result: $e^x(x^3 - 3x^2 + 6x - 6) + C$.

## 23.1 The $e^x\left[f(x) + f'(x)\right]$ shortcut

**THEOREM** **Special integral**

$$\int e^x\left[f(x) + f'(x)\right]dx = e^x f(x) + C$$

*Proof:* by parts, $\int e^x f(x)dx = e^x f(x) - \int e^x f'(x)dx$, so the two integrals cancel after moving terms across. (Or just differentiate $e^x f$ by the product rule.)

**EXAMPLE** **Using the shortcut**

- $\int e^x(\sin x + \cos x)dx = e^x \sin x + C$ (here $f = \sin x$, $f' = \cos x$).
- $\int e^x \left(\ln x + \dfrac{1}{x}\right) dx = e^x \ln x + C$.
- $\int e^x \dfrac{x}{(x+1)^2}dx$: split $\dfrac{x}{(x+1)^2} = \dfrac{1}{x+1} - \dfrac{1}{(x+1)^2}$, which is $f + f'$ with $f = \frac{1}{x+1}$. So the integral is $\dfrac{e^x}{x+1} + C$.

## 23.2 The "boomerang" (cyclic) integrals

For $\int e^x \sin x\, dx$ apply parts twice and the original integral **reappears**; then solve for it algebraically.

**EXAMPLE** **Example 23.7**

Let $I = \int e^x \sin x\, dx$. First: $I = e^x \sin x - \int e^x \cos x\, dx$. Second: $\int e^x \cos x\, dx = e^x \cos x + \int e^x \sin x\, dx = e^x \cos x + I$.

So $I = e^x \sin x - e^x \cos x - I \Rightarrow I = \dfrac{e^x(\sin x - \cos x)}{2} + C$.

General result: $\displaystyle\int e^{ax} \sin bx\, dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} + C$.

**WATCH OUT** **Choose $u$ the wrong way and it gets worse**

For $\int x\, e^x\, dx$, taking $u = e^x$, $dv = x\, dx$ gives $\frac{x^2}{2}e^x - \int \frac{x^2}{2}e^x dx$: **harder** than before. If the new integral is nastier, swap your choice of $u$.

UNIT III · CHAPTER 24

# Integration by partial fractions

**BIG IDEA** **Big idea**

You already know how to **add** fractions: $\frac{1}{x-1}+\frac{1}{x+2}=\frac{2x+1}{(x-1)(x+2)}$. Partial fractions is doing that in **reverse**: break a complicated rational function into simple pieces, each of which integrates to a logarithm or an arctangent. It is the method for any integrand of the form $\dfrac{P(x)}{Q(x)}$ (polynomial over polynomial).

## 24.1 The four-step method

**BIG IDEA** **Algorithm**

1. **Improper?** If $\deg P \geq \deg Q$, divide first (long division). *This is the step students forget.* Then work with the proper remainder $\dfrac{R(x)}{Q(x)}$.
2. **Factorise** $Q(x)$ into linear factors and irreducible quadratics.
3. **Write the correct form** (table below) with unknown constants $A, B, C, \ldots$
4. **Find the constants**, then **integrate** each simple term.

**FORMULA** **Which form for which factor?**

| Factor in the denominator | Terms to write |
|---|---|
| distinct linear $(x-a)$ | $\dfrac{A}{x-a}$ |
| repeated linear $(x-a)^2$ | $\dfrac{A}{x-a}+\dfrac{B}{(x-a)^2}$ (one term for each power up to the highest) |
| irreducible quadratic $(x^2+px+q)$ | $\dfrac{Ax+B}{x^2+px+q}$ |
| repeated quadratic $(x^2+px+q)^2$ | $\dfrac{Ax+B}{x^2+px+q}+\dfrac{Cx+D}{(x^2+px+q)^2}$ |

**Examples of the correct form** (these are practice-set questions):

- $\dfrac{1}{(x-1)(x-2)}=\dfrac{A}{x-1}+\dfrac{B}{x-2}$
- $\dfrac{3x+1}{(x-1)^2}=\dfrac{A}{x-1}+\dfrac{B}{(x-1)^2}$
- $\dfrac{1}{(x^2+1)(x+2)}=\dfrac{Ax+B}{x^2+1}+\dfrac{C}{x+2}$

**What the integrals look like:**

$$\int \frac{A}{x-a}dx = A\ln|x-a|, \qquad \int \frac{B}{(x-a)^2}dx = -\frac{B}{x-a}, \qquad \int \frac{Ax+B}{x^2+1}dx = \frac{A}{2}\ln(x^2+1)+B\tan^{-1}x.$$

## 24.2 Finding the constants

**Method 1: substitute convenient values (cover-up).** Multiply through by the denominator, then plug in the roots of the factors. For distinct linear factors this gives each constant instantly: *cover* the factor $(x-a)$ in the original fraction and evaluate what remains at $x=a$.

*Example:* $\dfrac{x+5}{(x-1)(x+2)} = \dfrac{A}{x-1} + \dfrac{B}{x+2}$. Cover $(x-1)$ and put $x=1$: $A = \dfrac{1+5}{1+2} = 2$. Cover $(x+2)$, $x=-2$: $B = \dfrac{-2+5}{-2-1} = -1$. So the fraction is $\dfrac{2}{x-1} - \dfrac{1}{x+2}$.

**Method 2: compare coefficients.** Expand and match powers of $x$. Use it for quadratic factors or to find the leftover constants.

**Mix both:** use values for the easy constants, coefficient-comparison for the rest. Always **verify** by substituting one more value of $x$.

## 24.3 Your professor's two worked problems

**EXAMPLE** **Problem 1:** $\displaystyle\int \frac{1-x^2}{x(1-2x)}\,dx$ **(numerator and denominator have equal degree)**

**Step 1: long division.** The denominator is $x - 2x^2 = -2x^2 + x$, degree 2, same as the numerator $-x^2+1$. Dividing: $-x^2 + 1 = \frac{1}{2}(-2x^2 + x) + \left(1 - \frac{x}{2}\right)$. So

$$\frac{1-x^2}{x(1-2x)} = \frac{1}{2} + \frac{1-\frac{x}{2}}{x(1-2x)}.$$

**Step 2: partial fractions** on the proper part: $\dfrac{1-\frac{x}{2}}{x(1-2x)} = \dfrac{A}{x} + \dfrac{B}{1-2x} \Rightarrow 1 - \dfrac{x}{2} = A(1-2x) + Bx$. Put $x=0$: $1 = A$. Put $x = \frac{1}{2}$: $1 - \frac{1}{4} = \frac{B}{2} \Rightarrow B = \frac{3}{2}$.

**Step 3: integrate.**

$$\int \frac{1}{2}dx + \int \frac{1}{x}dx + \frac{3}{2}\int \frac{dx}{1-2x} = \frac{x}{2} + \ln|x| + \frac{3}{2}\cdot\frac{\ln|1-2x|}{-2} + C = \boxed{\frac{x}{2} + \ln|x| - \frac{3}{4}\ln|1-2x| + C}$$

(The $-2$ comes from the inner function $1-2x$: the linear-substitution rule.)

**EXAMPLE** **Problem 2:** $\displaystyle\int \frac{x}{(x-1)^2(x+2)}\,dx$ **(a repeated factor)**

**Form:** $\dfrac{x}{(x-1)^2(x+2)} = \dfrac{A}{x-1} + \dfrac{B}{(x-1)^2} + \dfrac{C}{x+2}$. Clear denominators:

$$x = A(x-1)(x+2) + B(x+2) + C(x-1)^2.$$

- $x=1$: $1 = 3B \Rightarrow B = \frac{1}{3}$. - $x=-2$: $-2 = 9C \Rightarrow C = -\frac{2}{9}$. - $x=0$: $0 = -2A + 2B + C = -2A + \frac{2}{3} - \frac{2}{9} \Rightarrow A = \frac{2}{9}$.

*Cross-check with the $x^2$ coefficients:* the right side has $x^2(A+C)$ and the left side has none, so $A + C = 0$ ✓.

**Integrate:** $\displaystyle\int \left(\frac{2/9}{x-1} + \frac{1/3}{(x-1)^2} - \frac{2/9}{x+2}\right)dx = \boxed{\frac{2}{9}\ln|x-1| - \frac{1}{3(x-1)} - \frac{2}{9}\ln|x+2| + C}$

## 24.4 More examples (the deck's Q37–Q55)

**EXAMPLE** **Distinct linear factors**

- $\dfrac{1}{x(x+1)} = \dfrac{1}{x} - \dfrac{1}{x+1} \Rightarrow \displaystyle\int = \ln\left|\frac{x}{x+1}\right| + C.$
- $\dfrac{1}{(x-1)(x-2)} = \dfrac{1}{x-2} - \dfrac{1}{x-1} \Rightarrow \displaystyle\int = \ln\left|\frac{x-2}{x-1}\right| + C.$
- $\dfrac{1}{x^2-1} = \dfrac{1}{2}\left[\dfrac{1}{x-1} - \dfrac{1}{x+1}\right] \Rightarrow \displaystyle\int = \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C.$
- $\dfrac{1}{x^2+3x+2} = \dfrac{1}{(x+1)(x+2)} = \dfrac{1}{x+1} - \dfrac{1}{x+2} \Rightarrow \displaystyle\int = \ln\left|\frac{x+1}{x+2}\right| + C.$
- $\dfrac{2x+3}{(x+1)(x+2)} = \dfrac{1}{x+1} + \dfrac{1}{x+2} \Rightarrow \displaystyle\int = \ln|(x+1)(x+2)| + C.$ *(Here the numerator is the derivative of the denominator $x^2+3x+2$, so $\int \frac{f'}{f} = \ln|f|$ gives it at once.)*

- $\displaystyle\int \frac{2x}{x^2-1}dx = \ln|x^2-1| + C$ (derivative on top again; partial fractions $\frac{1}{x-1} + \frac{1}{x+1}$ agree).

**EXAMPLE** **With an irreducible quadratic**

$\displaystyle\int \frac{dx}{(x^2+1)(x+2)}$: $1 = (Ax+B)(x+2) + C(x^2+1)$. $x=-2$: $C = \frac{1}{5}$. Compare $x^2$: $A + C = 0 \Rightarrow A = -\frac{1}{5}$. Compare constants: $2B + C = 1 \Rightarrow B = \frac{2}{5}$. So

$$\int \left[\frac{-\frac{1}{5}x + \frac{2}{5}}{x^2+1} + \frac{1/5}{x+2}\right]dx = -\frac{1}{10}\ln(x^2+1) + \frac{2}{5}\tan^{-1}x + \frac{1}{5}\ln|x+2| + C.$$

**EXAMPLE** **An improper fraction**

$$\int \frac{x^2+1}{x^2-1}dx = \int \left(1 + \frac{2}{x^2-1}\right)dx = \int \left(1 + \frac{1}{x-1} - \frac{1}{x+1}\right)dx = x + \ln\left|\frac{x-1}{x+1}\right| + C.$$

**WATCH OUT** **Slips to avoid**

1. Skipping long division when $\deg P \geq \deg Q$.
2. For $(x-a)^2$ writing only $\frac{A}{(x-a)^2}$ and omitting $\frac{B}{x-a}$.
3. For an irreducible quadratic writing only a constant numerator $\frac{A}{x^2+1}$ instead of $\frac{Ax+B}{x^2+1}$.
4. Forgetting the factor $\frac{1}{a}$ when integrating $\frac{1}{ax+b}$, or forgetting the absolute value in the logarithm.

UNIT III · CHAPTER 25

# Definite integrals and their properties

**BIG IDEA** **Big idea**

By the Fundamental Theorem, $\int_a^b f\,dx = F(b) - F(a)$. But many definite integrals that look impossible (e.g. $\int_0^{\pi/2} \ln \sin x\,dx$) have **no elementary antiderivative** at all. We solve them without ever finding one, using **symmetry tricks**: the properties of definite integrals. This chapter follows the handwritten sheets your professor shared.

## 25.1 The basics

$$\int f(x)\,dx = F(x) + C \quad \text{(indefinite: a family of functions)}, \qquad \int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a) \quad \text{(definite: a number)}.$$

$b$ is the **upper limit**, $a$ the **lower limit**.

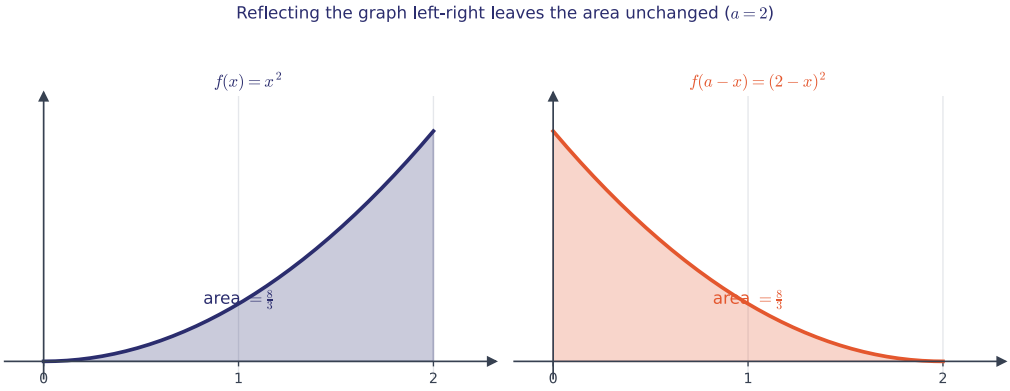
**EXAMPLE** **Warm-up from the sheet:** $\int_0^{\pi} \cos^2 x,dx$

$$= \int_0^{\pi} \frac{1+\cos 2x}{2}\,dx = \frac{1}{2}\Big[x + \frac{\sin 2x}{2}\Big]_0^{\pi} = \frac{1}{2}(\pi + 0 - 0) = \frac{\pi}{2}.$$

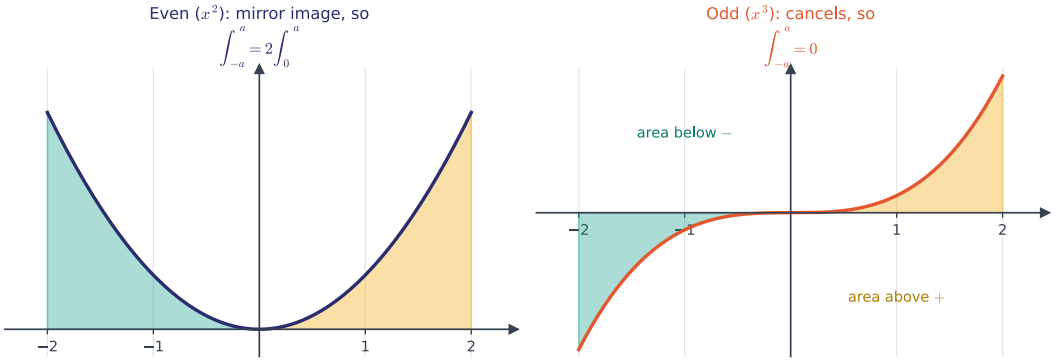
## 25.2 The eight properties

**FORMULA** **Properties $P_0$ to $P_7$ (the numbering your professor uses)**

| | Statement | In words |
|---|---|---|
| $P_0$ | $\int_a^b f(x)dx = \int_a^b f(t)dt$ | the variable name is a dummy |
| $P_1$ | $\int_a^b f\,dx = -\int_b^a f\,dx$, and $\int_a^a f = 0$ | swapping limits flips the sign |
| $P_2$ | $\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx$ | split at any point $c$ (use for $\lvert\,\rvert$ and piecewise) |
| $P_3$ | $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$ | reflect the interval about its midpoint |
| $P_4$ | $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ | special case of $P_3$ with lower limit 0 |
| $P_5$ | $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$ | split $[0, 2a]$ at $a$ |
| $P_6$ | $\int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & f(2a-x) = f(x) \\ 0, & f(2a-x) = -f(x) \end{cases}$ | halve the work, or get 0 |
| $P_7$ | $\int_{-a}^a f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & f \text{ even} \\ 0, & f \text{ odd} \end{cases}$ | symmetric limits |




$P_4$: replacing $x$ by $a - x$ reflects the graph left-right and leaves the area unchanged.




$P_7$: even functions double; odd functions cancel.

**Even and odd test.** Compute $f(-x)$. If it equals $f(x)$, $f$ is **even** (e.g. $x^2,\ \cos x,\ |x|$). If it equals $-f(x)$, it is **odd** (e.g. $x,\ x^3,\ \sin x,\ \tan x$).

**BIG IDEA** **Your professor's decision rule for $\int_0^a f(x)\,dx$**

Compute $f(a-x)$ and compare:

- **Same function back** ($f(a-x) = f(x)$): apply $P_6$: $I = 2\displaystyle\int_0^{a/2} f(x)\,dx$.
- **Negative of it** ($f(a-x) = -f(x)$): $I = 0$.
- **A *new* function** (e.g. sin → cos): apply $P_4$ to get a second expression for $I$, **add the two**, and the integrand usually collapses: $2I = \int(\text{something simple})$.

**EXAM TIP** **Order of attack for any definite integral**

| Look at the limits and integrand | Use |
|---|---|
| limits $-a$ to $a$ | test even/odd first ($P_7$) |
| contains $\lvert\text{expression}\rvert$ | find where the expression is 0, split there ($P_2$), drop the bars piece by piece |
| limits 0 to $a$, trig or $x \leftrightarrow a - x$ pairing | $P_4$ then add; or $P_6$ |
| limits $a$ to $b$ | $P_3$ ($x \to a + b - x$) |
| limits 0 to $2a$ | $P_6$ |
| a messy antiderivative | change variable first and *change the limits* |

## 25.3 Worked examples (the professor's sheets)

**EXAMPLE** **Example 25.1:** $\displaystyle\int_0^1 x(1-x)^{10}\,dx$

Here $f(1-x)=(1-x)x^{10}$, a *new* function, but expanding it is easy. By $P_4$: $I=\displaystyle\int_0^1(1-x)x^{10}dx=\int_0^1(x^{10}-x^{11})dx=\frac{1}{11}-\frac{1}{12}=\boxed{\frac{1}{132}}$.

**EXAMPLE** **Example 25.2:** $\displaystyle\int_0^{\pi/2}\frac{\sin x}{\sin x+\cos x}\,dx$

$f(\frac{\pi}{2}-x)=\dfrac{\cos x}{\cos x+\sin x}$ (a new function). By $P_4$, $I=\displaystyle\int_0^{\pi/2}\frac{\cos x}{\cos x+\sin x}dx$. **Add:** $2I=\displaystyle\int_0^{\pi/2}\frac{\sin x+\cos x}{\sin x+\cos x}dx=\int_0^{\pi/2}1\,dx=\frac{\pi}{2}$. So $I=\boxed{\frac{\pi}{4}}$.

**EXAMPLE** **Example 25.3:** $\displaystyle\int_0^{\pi}\frac{2^{|\sin x|}}{2^{|\sin x|}+2^{|\cos x|}}\,dx$

First, $f(\pi-x)=f(x)$, because $|\sin(\pi-x)|=|\sin x|$ and $|\cos(\pi-x)|=|\cos x|$. By $P_6$ (with $2a=\pi$): $I=2J$, where $J=\displaystyle\int_0^{\pi/2}\frac{2^{\sin x}}{2^{\sin x}+2^{\cos x}}dx$ (on $[0,\frac{\pi}{2}]$ the bars are unnecessary).

Now $f(\frac{\pi}{2}-x)$ swaps $\sin\leftrightarrow\cos$, a new function, so use $P_4$: $J=\displaystyle\int_0^{\pi/2}\frac{2^{\cos x}}{2^{\cos x}+2^{\sin x}}dx$. **Add** the two forms of $J$: $2J=\displaystyle\int_0^{\pi/2}1\,dx=\frac{\pi}{2}$, so $J=\frac{\pi}{4}$ and $I=2J=\boxed{\frac{\pi}{2}}$.

**EXAMPLE** **Example 25.4: odd + even pieces,** $\displaystyle\int_{-3}^{3}\left(4x^3-\tan x+x^2\sin x+1\right)dx$

$4x^3$, $\tan x$ and $x^2\sin x$ (even $\times$ odd) are **odd**; the constant 1 is **even**. So the first three cancel over $[-3,3]$ and $I=0+2\displaystyle\int_0^3 1\,dx=\boxed{6}$.

**EXAMPLE** **Example 25.5: absolute value,** $\displaystyle\int_{-\pi}^{\pi}|\cos x|\,,dx$

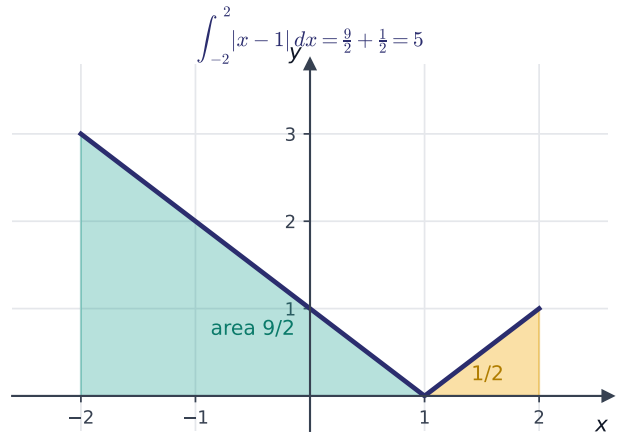
$|\cos(-x)|=|\cos x|$: even. So $I=2\displaystyle\int_0^{\pi}|\cos x|\,dx$. $\cos x\ge 0$ on $[0,\frac{\pi}{2}]$ and $\le 0$ on $[\frac{\pi}{2},\pi]$ ($P_2$): $I=2\Big[\displaystyle\int_0^{\pi/2}\cos x\,dx-\int_{\pi/2}^{\pi}\cos x\,dx\Big]=2\big[(1-0)-(0-1)\big]=\boxed{4}$.

**EXAMPLE** **Example 25.6:** $\displaystyle\int_0^{2\pi}\cos^5 x\,,dx$

$f(2\pi-x)=\cos^5 x=f(x)$, so $I=2\displaystyle\int_0^{\pi}\cos^5 x\,dx$. On $[0,\pi]$: $f(\pi-x)=\cos^5(\pi-x)=(-\cos x)^5=-\cos^5 x=-f(x)$, so $\displaystyle\int_0^{\pi}\cos^5 x\,dx=0$ by $P_6$ (with $2a=\pi$). Hence $I=\boxed{0}$.

**EXAMPLE** **Example 25.7:** $\displaystyle\int_{-2}^{2}|x-1|\,,dx$

$x-1$ changes sign at $x=1$. $I=\displaystyle\int_{-2}^{1}(1-x)dx+\int_1^2(x-1)dx=\Big[x-\frac{x^2}{2}\Big]_{-2}^{1}+\Big[\frac{x^2}{2}-x\Big]_1^2=\frac{9}{2}+\frac{1}{2}=\boxed{5}$. (Two triangles: areas $\frac{9}{2}$ and $\frac{1}{2}$.)




The two triangles under $|x-1|$.

**EXAMPLE** **Example 25.8: the logarithm integral, $I=\displaystyle\int_0^{\pi/2}\ln\sin x\,dx$**

By $P_4$: $I=\displaystyle\int_0^{\pi/2}\ln\cos x\,dx$. Add: $2I=\displaystyle\int_0^{\pi/2}\ln(\sin x\cos x)dx=\int_0^{\pi/2}\ln\frac{\sin 2x}{2}dx=\int_0^{\pi/2}\ln\sin 2x\,dx-\frac{\pi}{2}\ln 2$.

For the first integral put $t=2x$: $\displaystyle\int_0^{\pi/2}\ln\sin 2x\,dx=\frac{1}{2}\int_0^{\pi}\ln\sin t\,dt$. Since $\sin(\pi-t)=\sin t$, $P_6$ gives $\int_0^{\pi}\ln\sin t\,dt=2I$, so that integral equals $\frac{1}{2}\cdot 2I=I$. Therefore $2I=I-\frac{\pi}{2}\ln 2$, i.e.

$$\boxed{I=-\frac{\pi}{2}\ln 2}.$$

**EXAMPLE** **Example 25.9: your professor's challenge question**

Evaluate $I=\displaystyle\int_{-\pi/16}^{3\pi/16}\ln(1+\tan 2x)\,dx$.

The limits sum to $a+b=-\frac{\pi}{16}+\frac{3\pi}{16}=\frac{\pi}{8}$, so use $P_3$: replace $x$ by $\frac{\pi}{8}-x$:

$$1+\tan\Big(\frac{\pi}{4}-2x\Big)=1+\frac{1-\tan 2x}{1+\tan 2x}=\frac{2}{1+\tan 2x}.$$

So $I=\displaystyle\int\big[\ln 2-\ln(1+\tan 2x)\big]dx=\ln 2\cdot\Big(\frac{3\pi}{16}+\frac{\pi}{16}\Big)-I$. Hence $2I=\frac{\pi}{4}\ln 2$ and

$$\boxed{I=\frac{\pi}{8}\ln 2}.$$

## 25.4 Further reminders

- $\displaystyle\int_0^1 xe^x dx=\Big[(x-1)e^x\Big]_0^1=0-(-1)=1$ (parts, then limits).
- **Bounds:** if $m\le f(x)\le M$ on $[a,b]$ then $m(b-a)\le\displaystyle\int_a^b f\,dx\le M(b-a)$.
- $\displaystyle\int_a^b f\,dx\le\int_a^b|f|\,dx$.
- $P_3$ **is the heart of every "add the two forms" trick:** the integrand and its reflection add up to something easy.

**WATCH OUT** **Frequent errors**

- Using $P_7$ without checking that the **limits are symmetric** ($-a$ to $a$).
- Applying $P_4$ to limits that do not start at 0 (use $P_3$ instead).
- Removing absolute value bars without splitting at the sign change.
- After a substitution, forgetting to **change the limits**.

UNIT III · CHAPTER 26

# Unit III: practice and answers

## 26.1 Your professor's MCQ deck (Q16–Q80) with results

| Q | Ans | Result | Q | Ans | Result |
|---|---|---|---|---|---|
| 16 | D | $\int x^n dx = \frac{x^{n+1}}{n+1}$ | 49 | A | $\frac{Ax+B}{x^2+1} + \frac{C}{x+2}$ |
| 17 | A | $\ln \lvert x \rvert$ | 50 | C | do long division |
| 18 | B | $\frac{e^{3x}}{3}$ | 51 | C | $\frac{1}{2}\ln\left\lvert\frac{x-1}{x+1}\right\rvert$ |
| 19 | C | $-\cos x$ | 52 | B | $\frac{1}{x} - \frac{1}{x+1}$ |
| 20 | D | $\tan x$ | 53 | D | $\ln \lvert (x+1)(x+2) \rvert$ |
| 21 | B | $x^3 + x^2 + x$ | 54 | C | $A = 2$ |
| 22 | C | $\tan^{-1} x$ | 55 | A | $\ln\left\lvert\frac{x+1}{x+2}\right\rvert$ |
| 23 | A | $\sin^{-1} x$ | 56 | B | $uv - \int v\, du$ |
| 24 | B | $\frac{2}{3}x^{3/2}$ | 57 | B | I = Inverse trig |
| 25 | D | $-\frac{1}{x}$ | 58 | D | $(x-1)e^x$ |
| 26 | C | $\frac{\sin 2x}{2}$ | 59 | A | $x\sin x + \cos x$ |
| 27 | B | $\frac{a^x}{\ln a}$ | 60 | C | $x\ln x - x$ |
| 28 | A | $-\cot x$ | 61 | D | $\frac{x^2}{2}\ln x - \frac{x^2}{4}$ |
| 29 | D | $x$ | 62 | C | $x\tan^{-1} x - \frac{1}{2}\ln(1+x^2)$ |
| 30 | C | $\frac{x^2}{2} + \ln \lvert x \rvert$ | 63 | B | $e^x f(x)$ |
| 31 | B | $e^{x^2}$ | 64 | D | $e^x \sin x$ |
| 32 | C | $u = x^2 + 1$ | 65 | B | $e^x(x^2 - 2x + 2)$ |
| 33 | B | $\frac{1}{3}(x^2+1)^{3/2}$ | 66 | D | $-x\cos x + \sin x$ |
| 34 | D | $-\ln \lvert \cos x \rvert$ | 67 | B | $u = \ln x$ |
| 35 | C | $\ln \lvert \sin x \rvert$ | 68 | C | $-\int_b^a f$ |
| 36 | A | $\frac{\sin^4 x}{4}$ | 69 | B | $\frac{1}{3}$ |
| 37 | C | $\frac{(\ln x)^2}{2}$ | 70 | A | $5 - 2 = 3$ |
| 38 | D | $\ln \lvert \ln x \rvert$ | 71 | D | $f(a+b-x)$ |
| 39 | B | $\ln(1+e^x)$ | 72 | B | $2\int_0^a f$ |
| 40 | C | $\frac{1}{3}\sin(3x+2)$ | 73 | D | 0 |
| 41 | D | $\frac{(2x+3)^6}{12}$ | 74 | C | 0 (odd) |
| 42 | C | $\frac{1}{2}\ln(x^2+1)$ | 75 | A | 1 |
| 43 | B | $\frac{\tan^2 x}{2}$ | 76 | B | $\frac{\pi}{4}$ |
| 44 | A | $\frac{1}{2}\sin^{-1} 2x$ | 77 | C | 1 |
| 45 | C | $\frac{1}{3}\tan^{-1} 3x$ | 78 | D | $e - 1$ |

| Q | Ans | Result | Q | Ans | Result |
|---|---|---|---|---|---|
| 46 | B | $\frac{A}{x-1}+\frac{B}{x-2}$ | 79 | B | $m(b-a)$ to $M(b-a)$ |
| 47 | C | $\ln\left\lvert\frac{x-2}{x-1}\right\rvert$ | 80 | D | 1 |
| 48 | D | $\frac{A}{x-1}+\frac{B}{(x-1)^2}$ | | | |

## 26.2 The Unit III questions from the mixed practice set

| Q | Question | Answer |
|---|---|---|
| 4 | $\int e^x dx$ | **(C)** $e^x + C$ |
| 5 | $\int a^x dx$ | **(B)** $\frac{a^x}{\ln a} + C$ |
| 7 | $\int \sec^2 x\, dx$ | **(A)** $\tan x + C$ |
| 8 | $\int (2x+3)dx$ | **(B)** $x^2 + 3x + C$ |
| 12 | $\int \frac{f'}{f} dx$ | **(C)** $\ln\lvert f(x)\rvert + C$ |
| 13 | $\int \tan x\, dx$ | **(C)** $-\ln\lvert\cos x\rvert + C$ |
| 15 | $\int \cos(3x+4)dx$ | **(C)** $\frac{1}{3}\sin(3x+4) + C$ |
| 16 | $\int \frac{e^x}{1+e^x} dx$ | **(A)** $\ln(1+e^x) + C$ |
| 24 | $\int xe^x dx$ | **(B)** $e^x(x-1) + C$ |
| 26 | $\int \ln x\, dx$ | **(A)** $x\ln x - x + C$ |
| 27 | $\int e^x[f + f']dx$ | **(C)** $e^x f(x) + C$ |
| 28 | $\int e^x(\sin x + \cos x)dx$ | **(B)** $e^x \sin x + C$ |
| 37 | $\int \frac{1}{x(x+1)} dx$ | **(B)** $\ln\left\lvert\frac{x}{x+1}\right\rvert + C$ |
| 40 | $\int \frac{2x}{x^2-1} dx$ | $\ln\lvert x^2 - 1\rvert + C$ |
| 41 | Fundamental Theorem | **(B)** $F(b) - F(a)$ |
| 42 | $\int_0^2 x^3 dx$ | **(B)** 4 |
| 43 | $\int_0^{\pi/2} \cos x\, dx$ | **(B)** 1 |
| 44 | $\int_1^e \frac{1}{x} dx$ | **(B)** 1 |
| 45 | $\int_0^1 e^x dx$ | **(B)** $e - 1$ |

## 26.3 Extra practice

**TRY IT** **Questions**

1. $\int \sin^2 x\, dx$ 2. $\int \dfrac{x}{\sqrt{1+x^2}} dx$ 3. $\int \cos^3 x\, dx$
2. $\int \dfrac{2x+1}{x^2+x+5} dx$ 5. $\int x^2 e^{-x} dx$ 6. $\int x\sec^2 x\, dx$
3. $\int \dfrac{x+1}{(x-1)(x-2)} dx$ 8. $\int \dfrac{dx}{x^2+2x+5}$ 9. $\int x^3 \ln x\, dx$
4. $\int \dfrac{dx}{x(x^2+1)}$ 11. $\int_0^{\pi/2} \dfrac{\cos x}{\sin x + \cos x} dx$ 12. $\int_{-1}^{1} (x^2 + x)\, dx$
5. $\int_0^{\pi/4} \ln(1+\tan x)\, dx$ 14. $\int_0^3 \lvert x-2\rvert\, dx$ 15. $\int_0^1 x(1-x)^5 dx$

**Answers.** (1) $\frac{x}{2} - \frac{\sin 2x}{4} + C$. (2) $\sqrt{1+x^2} + C$. (3) $\sin x - \frac{\sin^3 x}{3} + C$. (4) $\ln(x^2+x+5) + C$ (derivative on top). (5) $-e^{-x}(x^2+2x+2) + C$. (6) $x\tan x + \ln|\cos x| + C$. (7) $-2\ln|x-1| + 3\ln|x-2| + C$. (8) $\frac{1}{2}\tan^{-1}\frac{x+1}{2} + C$. (9) $\frac{x^4}{4}\ln x - \frac{x^4}{16} + C$. (10) $\ln|x| - \frac{1}{2}\ln(x^2+1) + C$ (write $\frac{1}{x} - \frac{x}{x^2+1}$). (11) $\frac{\pi}{4}$ (same as Example 25.2). (12) $\frac{2}{3}$ ($x^2$ even gives $2\cdot\frac{1}{3}$; $x$ is odd, so 0). (13) $\frac{\pi}{8}\ln 2$ ($P_4$ with $a=\frac{\pi}{4}$: $1+\tan(\frac{\pi}{4}-x) = \frac{2}{1+\tan x}$). (14) $\int_0^2 (2-x)dx + \int_2^3 (x-2)dx = 2 + \frac{1}{2} = \frac{5}{2}$. (15) $\int_0^1 (1-x)x^5 dx = \frac{1}{6} - \frac{1}{7} = \frac{1}{42}$.

APPENDIX A

# Appendix A: The one-stop formula sheet

**QUICK RECALL** **How to use this**

Print this section. Read it the night before the exam and again just before you walk in. Every formula here appears, with its reasoning, in a chapter of this book.

## A.1 Unit I: matrices

**FORMULA** **Matrices and determinants**

$(AB)^T = B^T A^T \quad (AB)^{-1} = B^{-1}A^{-1} \quad A^{-1} = \dfrac{\operatorname{adj} A}{|A|} \quad |AB| = |A|\,|B| \quad \left|A^{-1}\right| = \dfrac{1}{|A|} \quad |kA| = k^n\,|A|$

$|\operatorname{adj} A| = |A|^{n-1}$

$2 \times 2$ inverse: $\dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Invertible $\iff |A| \neq 0 \iff \operatorname{rank} A = n$.

**FORMULA** **Rank, systems, eigenvalues**

**Rouché–Capelli** ($r = \operatorname{rank} A,\ r' = \operatorname{rank}[A \mid B],\ n$ unknowns): $r \neq r'$: no solution; $r = r' = n$: unique; $r = r' < n$: infinitely many ($n - r$ parameters). Homogeneous: non-trivial solution $\iff r < n$ (square: $|A| = 0$).

**Eigenvalues:** $|A - \lambda I| = 0$. $2 \times 2$: $\lambda^2 - T\lambda + D = 0$. $3 \times 3$: $\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$. Sum $= \operatorname{tr} A$, product $= |A|$. $A^{-1} \to \frac{1}{\lambda}$, $A^m \to \lambda^m$, $kA \to k\lambda$, $A + kI \to \lambda + k$.

**Cayley–Hamilton:** $A$ satisfies its own characteristic equation; $2 \times 2$: $A^2 - TA + DI = O$.

## A.2 Unit II: differentiation

**FORMULA** **Rules and standard derivatives**

$(uv)' = u'v + uv' \quad \left(\dfrac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2} \quad \dfrac{d}{dx}f(g(x)) = f'(g(x))g'(x) \quad \text{parametric: } \dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$

$(x^n)' = nx^{n-1}\ \ (e^x)' = e^x\ \ (a^x)' = a^x \ln a\ \ (\ln x)' = \frac{1}{x}\ \ (\sin x)' = \cos x\ \ (\cos x)' = -\sin x\ \ (\tan x)' = \sec^2 x\ \ (\cot x)' = -\operatorname{cosec}^2 x\ \ (\sec x)' = \sec x \tan x\ \ (\operatorname{cosec} x)' = -\operatorname{cosec} x \cot x$

$(\sin^{-1} x)' = \dfrac{1}{\sqrt{1-x^2}}\ \ (\cos^{-1} x)' = -\dfrac{1}{\sqrt{1-x^2}}\ \ (\tan^{-1} x)' = \dfrac{1}{1+x^2}\ \ (\cot^{-1} x)' = -\dfrac{1}{1+x^2}\ \ (\sec^{-1} x)' = \dfrac{1}{|x|\sqrt{x^2-1}}$

Log differentiation: $y = u^v \Rightarrow y' = y\left[\dfrac{vu'}{u} + v' \ln u\right]$.

**FORMULA** **Theorems**

**Rolle:** continuous on $[a,b]$, differentiable on $(a,b)$, $f(a)=f(b) \Rightarrow f'(c)=0$ for some $c \in (a,b)$.

**Lagrange MVT:** $f'(c)=\dfrac{f(b)-f(a)}{b-a}$. **Cauchy MVT:** $\dfrac{f(b)-f(a)}{g(b)-g(a)}=\dfrac{f'(c)}{g'(c)}$.

**Taylor:** $f(x)=\displaystyle\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k+\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$. **Maclaurin:** $a=0$.

**L'Hôpital:** $\dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$: $\lim \dfrac{f}{g}=\lim \dfrac{f'}{g'}$. Other forms: convert; for $1^\infty, 0^0, \infty^0$ take ln.

**Leibniz:** $(uv)^{(n)}=\displaystyle\sum_{k=0}^{n}\binom{n}{k}u^{(n-k)}v^{(k)}$.

**FORMULA** **$n$-th derivatives**

$D^n e^{ax}=a^n e^{ax}$ $D^n \sin(ax+b)=a^n \sin(ax+b+\frac{n\pi}{2})$ $D^n \cos(ax+b)=a^n \cos(ax+b+\frac{n\pi}{2})$ $D^n \dfrac{1}{ax+b}=\dfrac{(-1)^n n! a^n}{(ax+b)^{n+1}}$ $D^n \ln(ax+b)=\dfrac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n}$ $D^n x^n=n!$

**FORMULA** **Standard limits and series**

$\dfrac{\sin x}{x}, \dfrac{\tan x}{x}, \dfrac{\sin^{-1}x}{x}, \dfrac{\tan^{-1}x}{x}, \dfrac{e^x-1}{x}, \dfrac{\ln(1+x)}{x} \to 1$ $\dfrac{1-\cos x}{x^2}\to\frac{1}{2}$ $\dfrac{a^x-1}{x}\to \ln a$ $(1+\frac{1}{x})^x \to e$

$e^x=\sum \frac{x^n}{n!}$ $\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots$ $\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots$ $\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots$ $\tan^{-1}x=x-\frac{x^3}{3}+\frac{x^5}{5}-\cdots$ $\frac{1}{1-x}=1+x+x^2+\cdots$ $(1+x)^n=1+nx+\frac{n(n-1)}{2!}x^2+\cdots$

**FORMULA** **Applications**

Tangent: $y-y_0=f'(x_0)(x-x_0)$. Normal: slope $-\dfrac{1}{f'(x_0)}$. Approximation: $f(x+\Delta x)\approx f(x)+f'(x)\Delta x$. $MC=\dfrac{dC}{dx}$.

**Extrema:** critical points $f'=0$ or undefined. First-derivative test ($+\to-$ max, $-\to+$ min). Second-derivative test ($f''<0$ max, $f''>0$ min, $=0$ inconclusive). On $[a,b]$: compare $f$ at critical points and end points.

## A.3 Unit III: integration

**FORMULA** **Standard integrals (add $+C$)**

$\int x^n=\frac{x^{n+1}}{n+1}$ $\int \frac{1}{x}=\ln|x|$ $\int e^x=e^x$ $\int a^x=\frac{a^x}{\ln a}$ $\int \sin x=-\cos x$ $\int \cos x=\sin x$ $\int \sec^2 x=\tan x$ $\int \operatorname{cosec}^2 x=-\cot x$ $\int \sec x\tan x=\sec x$ $\int \operatorname{cosec} x\cot x=-\operatorname{cosec} x$

$\int \tan x=-\ln|\cos x|$ $\int \cot x=\ln|\sin x|$ $\int \sec x=\ln|\sec x+\tan x|$ $\int \operatorname{cosec} x=\ln|\operatorname{cosec} x-\cot x|$

$\int \frac{dx}{1+x^2}=\tan^{-1}x$ $\int \frac{dx}{\sqrt{1-x^2}}=\sin^{-1}x$ $\int \frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}$ $\int \frac{dx}{x^2-a^2}=\frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|$ $\int \frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}$

$\int \frac{f'}{f}=\ln|f|$ $\int f(ax+b)dx=\frac{1}{a}F(ax+b)$ $\int e^x[f+f']=e^x f$

**FORMULA** **Methods**

**Substitution:** $u=g(x)$. **By parts:** $\int u\,dv=uv-\int v\,du$, ILATE. **Partial fractions:** divide first if improper; forms $\frac{A}{x-a}$, $\frac{A}{x-a}+\frac{B}{(x-a)^2}$, $\frac{Ax+B}{x^2+px+q}$.

**Definite:** $\int_a^b f=F(b)-F(a)$. $P_2$ split; $P_3$ $f(a+b-x)$; $P_4$ $f(a-x)$; $P_6$ $\int_0^{2a}=2\int_0^a$ or 0; $P_7$ even $\to 2\int_0^a$, odd $\to 0$. If $m\le f\le M$: $m(b-a)\le\int_a^b f\le M(b-a)$.

## A.4 Trigonometry

**FORMULA** **Identities**

$\sin^2+\cos^2=1,\ 1+\tan^2=\sec^2,\ 1+\cot^2=\operatorname{cosec}^2.\ \sin 2\theta=2\sin\theta\cos\theta,\ \cos 2\theta=\cos^2\theta-\sin^2\theta=2\cos^2\theta-1=1-2\sin^2\theta.\ \cos^2\theta=\frac{1+\cos 2\theta}{2},\ \sin^2\theta=\frac{1-\cos 2\theta}{2}.\ \sin(A\pm B)=\sin A\cos B\pm\cos A\sin B.\ \tan(A\pm B)=\frac{\tan A\pm\tan B}{1\mp\tan A\tan B}.\ \sin 2\theta=\frac{2\tan\theta}{1+\tan^2\theta},\ \cos 2\theta=\frac{1-\tan^2\theta}{1+\tan^2\theta}.\ \sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}.$

APPENDIX B

# Appendix B: The 30 most expensive mistakes

Every item below is a real, common way to lose marks. Skim the left column; if you recognise yourself, read the right.

| # | The slip | The fix |
|---|---|---|
| 1 | $(a+b)^2 = a^2 + b^2$ | $(a+b)^2 = a^2 + 2ab + b^2$ |
| 2 | $\sqrt{x^2} = x$ | $\sqrt{x^2} = \lvert x \rvert$ |
| 3 | $\sin^{-1} x = \dfrac{1}{\sin x}$ | $\sin^{-1} x$ is the inverse function (an angle); $\frac{1}{\sin x} = \operatorname{cosec} x$ |
| 4 | $(fg)' = f'g'$ | product rule: $f'g + fg'$ |
| 5 | $\frac{d}{dx} x^x = x \cdot x^{x-1}$ | log differentiation: $x^x(1 + \ln x)$ |
| 6 | $\frac{d}{dx} \sin(x^2) = \cos(x^2)$ | multiply by the inside: $2x \cos x^2$ |
| 7 | $\frac{d}{dx} e^{2x} = 2xe^{2x-1}$ | $x^n$ and $a^x$ are different: $2e^{2x}$ |
| 8 | quotient rule with the subtraction reversed | low d-high minus high d-low |
| 9 | "$\frac{0}{0} = 0$" or "$\frac{0}{0} = 1$" | $\frac{0}{0}$ is **indeterminate**; simplify or use L'Hôpital |
| 10 | treating $\infty$ as a number | it is a direction; $\infty - \infty$ is indeterminate |
| 11 | using the quotient rule on $f/g$ inside L'Hôpital | L'Hôpital differentiates numerator and denominator **separately** |
| 12 | L'Hôpital on a limit that is not $\frac{0}{0}$ or $\frac{\infty}{\infty}$ | check the form first |
| 13 | $\lim_{x \to c} f(x) = f(c)$ always | only when $f$ is continuous at $c$ |
| 14 | continuous $\Rightarrow$ differentiable | false ($\lvert x \rvert$ at 0); the converse is true |
| 15 | $f'(c) = 0 \Rightarrow$ extremum | false ($x^3$ at 0); check the sign change |
| 16 | Rolle/MVT without verifying conditions | check continuity, differentiability, $f(a) = f(b)$ |
| 17 | accepting a $c$ outside $(a, b)$ | $c$ must be in the **open** interval |
| 18 | second-derivative test "proves" something when $f''(c) = 0$ | inconclusive; use the first-derivative test |
| 19 | absolute extrema on $[a, b]$ but forgetting the end points | always evaluate $f(a)$ and $f(b)$ |
| 20 | related rates: plugging numbers in **before** differentiating | differentiate first, substitute after |
| 21 | no domain check in optimisation | restrict to the physically allowed domain |
| 22 | forgetting $+C$ | every indefinite integral needs it |
| 23 | $\int fg = \int f \int g$ | there is no such rule; use substitution or parts |
| 24 | $\int \cos 3x \, dx = \sin 3x$ | divide by the inner derivative: $\frac{1}{3} \sin 3x$ |

| # | The slip | The fix |
|---|---|---|
| 25 | $\int \frac{1}{x} = \ln x$ for all $x$ | $\ln \lvert x \rvert$ |
| 26 | substitution: leaving $x$'s behind, or forgetting $du$'s factor | rewrite **everything** in $u$ |
| 27 | definite substitution without changing limits | change them (or go back to $x$) |
| 28 | wrong choice of $u$ in parts | ILATE; switch if it gets harder |
| 29 | partial fractions on an improper fraction | long division first |
| 30 | matrices: $AB = BA$, $(AB)^{-1} = A^{-1}B^{-1}$, $\lvert kA \rvert = k \lvert A \rvert$ | $AB \neq BA$ in general; $(AB)^{-1} = B^{-1}A^{-1}$; $\lvert kA \rvert = k^n \lvert A \rvert$ |

APPENDIX C

# Appendix C: Map of your professor's resources

Use this to practise on exactly the material you were given.

| Resource | Covers | Where in this book |
|---|---|---|
| **Syllabus** (Units I – III) | the three units | the whole book; see the table of contents |
| **Practice problems from Units 1, 2, 3** (mixed MCQs) | matrices, derivatives, Rolle/MVT, Taylor, limits, critical points, integrals | Chapters 8, 19, 26 |
| **Practice problems Units 2 & 3** (80-question deck) | differentiation, integration (direct, substitution, partial fractions, parts), definite integrals | Chapters 19, 26 and Appendix D |
| **Chapter 5: Continuity and Differentiability** (textbook) | continuity, chain rule, implicit, inverse trig, exp/log, log differentiation, parametric, second order, Rolle, MVT | Chapters 10 – 14 |
| **Chapter 6: Applications of Derivatives** (textbook) | rate of change, increasing/decreasing, tangents/normals, approximations, maxima/minima | Chapters 17 – 18 |
| **Unit 2 (R. K. Jain, Engineering Mathematics)** | differentials, Rolle, Lagrange and Cauchy MVT, L'Hôpital, monotonicity, extrema, Leibniz, Taylor | Chapters 13 – 18 |
| **Taylor and Maclaurin sheet** | the two theorems with remainder, the 10 standard series | Chapter 15 |
| **Partial fractions and properties (handwritten)** | integrals of rational functions; $P_4$, $P_6$; log and trig integrals | Chapters 24 – 25 |
| **Properties of definite integrals (handwritten)** | $P_0$–$P_7$, even/odd, worked examples Q1–Q6 | Chapter 25 |

## Exercise map for the textbook chapters

| Exercise | Topic | Chapter |
|---|---|---|
| 5.1 (Q1–34) | continuity, finding constants | 10 |
| 5.2 (Q1–10) | chain rule; $\|x-1\|$ and $[x]$ not differentiable | 11, 12.1 |
| 5.3 (Q1–15) | implicit differentiation (and inverse trig, Examples 40–45) | 12.2, 12.3 |
| 5.4 | exponential and log derivatives | 12.4 |
| 5.5 | logarithmic differentiation | 12.5 |
| 5.6 | parametric derivatives | 12.6 |
| 5.7 | second-order derivatives | 13 |
| 5.8 | Rolle's theorem and MVT | 14 |
| 6.1 | rates of change, marginal cost/revenue | 17.1 |
| 6.2 | increasing and decreasing | 17.2 |
| 6.3 | tangents and normals | 17.3 |
| 6.4 | approximations | 17.4 |
| 6.5 | maxima, minima, optimisation | 18 |
| Jain Ex. 1.1 | Q1–5 approximations, Q6–14 MVT/Rolle, Q15–28 limits, Q29–37 monotonicity, Q38–45 extrema, Q46–50 higher derivatives, Q51–75 Taylor | 17, 14, 16, 17, 18, 13, 15 |

APPENDIX D

# Appendix D: Complete MCQ answer key

The 80-question deck on differentiation and integration (Units II & III). Check your attempts here, then use the chapter tables to see *why*.

| | | |
|---|---|---|
| **1.** B | **2.** A | **3.** C |
| **4.** D | **5.** A | **6.** B |
| **7.** C | **8.** D | **9.** B |
| **10.** A | **11.** C | **12.** D |
| **13.** B | **14.** C | **15.** B |
| **16.** D | **17.** A | **18.** B |
| **19.** C | **20.** D | **21.** B |
| **22.** C | **23.** A | **24.** B |
| **25.** D | **26.** C | **27.** B |
| **28.** A | **29.** D | **30.** C |
| **31.** B | **32.** C | **33.** B |
| **34.** D | **35.** C | **36.** A |
| **37.** C | **38.** D | **39.** B |
| **40.** C | **41.** D | **42.** C |
| **43.** B | **44.** A | **45.** C |
| **46.** B | **47.** C | **48.** D |
| **49.** A | **50.** C | **51.** C |
| **52.** B | **53.** D | **54.** C |
| **55.** A | **56.** B | **57.** B |
| **58.** D | **59.** A | **60.** C |
| **61.** D | **62.** C | **63.** B |
| **64.** D | **65.** B | **66.** D |
| **67.** B | **68.** C | **69.** B |
| **70.** A | **71.** D | **72.** B |
| **73.** D | **74.** C | **75.** A |
| **76.** B | **77.** C | **78.** D |
| **79.** B | **80.** D | |

APPENDIX E

# Appendix E: Plain-English glossary

| Term | In plain English |
|---|---|
| **Function** | a rule giving exactly one output for each input |
| **Domain / range** | the allowed inputs / the outputs actually produced |
| **Limit** | the value a function is heading towards as $x$ nears a point |
| **Continuous** | no breaks; can draw it without lifting the pen |
| **Derivative** | instantaneous rate of change; slope of the tangent |
| **Tangent / normal** | the line that just touches the curve / the line perpendicular to it |
| **Chain rule** | derivative of a function of a function: multiply the rates |
| **Implicit** | $y$ is defined by an equation mixing $x$ and $y$ |
| **Critical point** | where $f' = 0$ or $f'$ does not exist; candidates for max/min |
| **Local vs absolute extremum** | highest/lowest nearby vs highest/lowest on the whole interval |
| **Point of inflection** | where the curve changes from cup-shaped to cap-shaped (or vice versa) |
| **Indeterminate form** | an expression like $\frac{0}{0}$ whose value is not decided by the form alone |
| **Taylor/Maclaurin series** | rewriting a function as an infinite polynomial around a point (around 0) |
| **Antiderivative** | a function whose derivative is the given function |
| **Indefinite / definite integral** | a family of antiderivatives / a number (signed area) |
| **Integrand** | the function being integrated |
| **Substitution** | renaming an inner function $u$ to simplify an integral |
| **Integration by parts** | the product rule in reverse |
| **Partial fractions** | splitting a rational function into simple fractions |
| **Matrix / order** | a rectangular table of numbers / its number of rows $\times$ columns |
| **Determinant** | a number measuring whether a square matrix can be undone |
| **Singular / non-singular** | determinant zero (no inverse) / non-zero (invertible) |
| **Rank** | the number of independent rows (non-zero rows in echelon form) |
| **Linearly independent** | none of the vectors can be built from the others |
| **Consistent system** | a system of equations that has at least one solution |
| **Eigenvalue / eigenvector** | a stretch factor / a direction that is only stretched by the matrix |
| **Characteristic equation** | $\lvert A - \lambda I \rvert = 0$, whose roots are the eigenvalues |